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Q&A with Matthew Barnard

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As equity research adapts to rapid shifts in technology, markets and client behaviour, Matt Barnard, Head of US Equity Research, is helping steer how Deutsche Bank evolves its research offering. In this edition of our Q & A series, Matt shares his perspective on the new ways clients are consuming research, his outlook for US markets, and what it's like leading research teams through a period of significant change.

Q: You've spoken about the equity research landscape undergoing a fundamental shift. What has changed most in how clients discover, engage with and use research today?

AI is fundamentally changing how clients consume sell-side research. As information becomes more easily accessible through AI tools, the value of differentiated insights from trusted sources has moved higher up the value chain. Investors are increasingly using AI to discover new ideas and challenge consensus views more quickly and efficiently. This allows them to filter out noise and spend more time engaging in deeper, more meaningful conversations with our Deutsche Bank research analysts.

Q: Against a complex backdrop for US geopolitics and policy, how do you currently assess the outlook for US markets – and which themes matter most for clients as we move through 2026?

Geopolitical developments in the US and globally continue to be a significant driver of volatility across risk assets. Deutsche Bank research's macro and strategy teams have consistently taken bold, often counter-consensus views, which are especially valuable in this environment. While conflicts in the Middle East will likely continue to create ripple effects across markets, the underlying fundamentals for US equities remain relatively strong. The rapid acceleration of AI adoption – and the associated capital investment – remains a key thematic focus for clients, both in the US and globally.

Q: You now lead teams through a period of rapid change. What do you find most rewarding about working in research management at this point in the cycle?

At its core, the role of an equity research analyst is to form forward-looking views with imperfect information. The strongest analysts thrive in periods of uncertainty and rapid change, identifying emerging trends and uncovering new investment opportunities early. From a management perspective, the most rewarding part of my role is seeing some of the brightest minds in research stay ahead of these shifts and deliver true alpha for our clients.

Q&A with Matthew Barnard

Deutsche Bank
Research

Q2 Corporate Bank Quarterly

A guide to Macro, Markets, Geopolitics & Tech for CFOs and Corporates

April 2026

Marion Laboure | Camilla Siazon | Luke Templeman,
Adrian Cox | Helen Belopolsky | Miha Hribernik | Jim Reid

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Introduction: Macro, Markets, Geopolitics & Tech

A guide for CFOs & Corporates

We're pleased to share our April 2026 Guide for CFOs and Corporates, providing essential insights into macroeconomics, markets, geopolitics and technology. This report offers critical analysis and perspectives to help corporate clients navigate today's complex economic environment.

Key highlights:

- Macroeconomic outlook: The Iran War is driving an energy price shock that threatens a global inflation shock, despite growth remaining resilient outside the euro area.
- Market analysis: Higher energy costs and rates are pulling on corporate earnings growth. Brent crude averages $84.80/bbl in our baseline. Dollar weakness narrative resumes.
- Geopolitics: A long-term solution between the US and Iran is likely, but post-conflict normalization will take months. Key Q2 events to watch: Xi-Trump summit (14-15 May) and Ukraine ceasefire talks.
- Artificial Intelligence: AI disruption concerns have waned somewhat, however, the software sector remains under intense scrutiny and Q1 outlook announcements will be critical.
- The future of monetary policy: The energy shock has materially shifted central bank trajectories. Fed on hold through 2026; ECB hiking twice. The US continues to lead stablecoin adoption.

Upcoming key macro events

04 April

27-28	JN	Bank of Japan decision
28-29	US	Federal Reserve meeting
29-30	EZ	ECB decision
30	UK	Bank of England decision

05 May

06 June

10-11	EZ	ECB decision
14-16	G7	G7 summit in France
15-16	JN	Bank of Japan decision
16-17*	US	Federal Reserve meeting
18	UK	Bank of England decision
18-19	EU	European Council meeting

07 July

01	IR	Ireland holds EU Council presidency
22-23	EZ	ECB decision
28-29	US	Federal Reserve meeting
30	UK	Bank of England decision
30-31	JN	Bank of Japan decision

08 August

09 September

-	FR	France Senate election
09-10	EZ	ECB decision
15-16*	US	Federal Reserve meeting
17	UK	Bank of England decision
17-18	JN	Bank of Japan decision
30	US	Fiscal-year ends- multiple agencies' funding expires

10 October

04-25	BR	Brazilian general election
27-28	US	Federal reserve meeting
28-29	EZ	ECB decision
29-30	JN	Bank of Japan decision

11 November

03	US	US midterm elections (House & Senate)
05	UK	Bank of England decision
09-20	COP31	COP31 in Türkiye

12 December

-	CH	China Central Economic Work Conference
08-09*	US	Federal reserve meeting
14-15	G20	G20 summit in US
16-17	EZ	ECB decision
17	UK	Bank of England decision
17-18	JN	Bank of Japan decision

Source: The House View: Infographic - 2026 Global Macro Events Calendar – 06-Jan-2026

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DB forecasts

GDP Growth (%)	2024	2025	2026F	2027F
Global	3.4	3.4	3.3	3.3
US	2.8	2.1	2.4	2.2
Eurozone	0.9	1.5	0.5	1.1
Germany	-0.5	0.2	0.5	1.3
France	1.1	0.9	1.0	1.1
Italy	0.6	0.7	0.8	1.0
Spain	3.5	2.8	2.2	2.1
Japan	-0.2	1.2	0.8	1.0
UK	1.1	1.4	0.9	1.2
China	5.0	5.0	4.9	4.5
India	7.2	7.5	6.7	7.2
Russia	4.9	1.0	0.8	1.3
Asia (ex Japan)	5.4	5.5	5.2	5.1
EEMEA	3.4	3.0	2.9	3.3
LatAm	2.5	1.9	2.0	1.8
EM	4.6	4.5	4.3	4.4

- *CPI (%) forecasts are period averages. **Germany National CPI taken.
  EEMEA: Czech Rep., Israel, Egypt, Hungary, Poland, Russia, Saudi Arabia, South Africa, Turkey, UAE
- LATAM (ex Venezuela): Argentina, Brazil, Chile, Colombia, Mexico, Peru
- ASIA (ex Japan): China, HK, India, Indonesia, Korea, Malaysia, Philippines, Singapore, Sri Lanka, Taiwan, Thailand, Vietnam
- Current prices as of April 28, 2026.
- *China rates are 7d OMO rates.
- **Gold, Brent and WTI forecasts are for Q2 / Q3 / Q4 2026.

CPI Inflation, YoY* (%)	2025	2026F	2027F	2028F
US	2.7	3.3	2.5	2.5
Eurozone	2.1	2.8	2.2	2.2
Germany**	2.2	2.9	2.4	2.9
Japan	3.2	1.8	2.2	2.2
UK	3.4	3.1	2.6	2.0
China	0.1	1.5	2.0	2.0
India	2.2	4.6	4.2	
Russia	8.7	5.7	5.4	

Central Bank policy rate (%)	Current	Jun 26F	Sep 26F	Dec 26F
US	3.625	3.625	3.625	3.625
Eurozone	2.00	2.25	2.50	2.50
UK	3.75	3.75	3.75	3.75
Japan	0.75	0.75	1.00	1.00
China*	1.40	1.40	1.40	1.40
India	5.25	5.25	5.25	5.25

Key market metrics	Current	Jun 26F	Sep 26F	Dec 26F
US 10Y Yield (%)	4.34	4.30	4.45	4.45
EUR 10Y Yield (%)	3.03	2.90	3.00	3.10
EUR/USD	1.17	1.20	1.22	1.25
USD/JPY	159	155	152	150
USD/CNY	6.83	6.80	6.80	6.70
S&P 500	7174	7450	7675	8000
**Oil WTI (USD/bbl)	97.51	90	82	73
**Oil Brent (USD/bbl)	109.56	96	87	78
**Gold	4669	5,100	5,500	5,800

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Source: Deutsche Bank Research, Bloomberg Finance LP.

01.

Macroeconomics

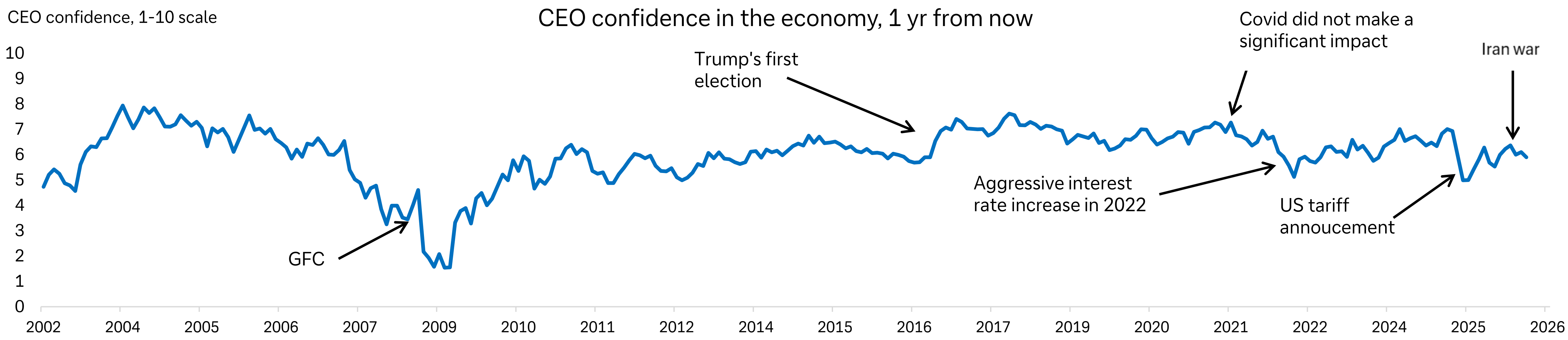
2026 Global growth

Energy shock weighs on growth, especially in the Euro Area

	View	Rationale
(Globe)	Growth at 3.3% in 2026 and 2027.	– The energy price shock from the Iran war is reverberating through the global economy, although the impact varies across regions.
(US)	Growth revised to 2.4% in 2026 (2.3% q4/q4) and 2.2% in 2027 (2.2% q4/q4).	– The downgrade reflects the drag from higher oil prices and softer consumer spending, counterbalanced by continued tailwinds from supportive fiscal policy, and strong AI investments.
(EU)	Euro area growth decreased to 0.5% in 2026 and 1.1% in 2027. German GDP downgraded to 0.5% in 2026 and 1.3% in 2027.	– The energy shock is impacting the region at a time when growth was already softening, lowering the bar to stagnation or even recession.

	View	Rationale
(China)	Growth increased to 4.9% in 2026, 4.5% in 2027.	– Strong performance driven by a +15% y/y export surge in Q1, rebound in domestic investment, and some stabilization in the property sector.
EM	Notable downward growth revisions due to Iran war, including India (FY27 growth 6.7% from 7.5%).	– In Asia, fuel price controls have limited immediate inflation rise, but full effects will emerge as domestic prices adjust over time to higher imported input prices. – CEEMEA has largest upward inflation revisions, but more limited drag on activity and sentiment compared to Asia. LatAM impact less pronounced, electoral cycles remain in focus.

The drop in corporate sentiment from the Iran war is not as steep as one may think



Read: Global Economics: Monthly Flyover (April 2026) - Deutsche Bank Research

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3 key risks for the global economy

The growing economic effects of the war in the Middle East are raising further questions about the government response, not just in the US but also in Europe.

1. Iran conflict

- Oil-driven energy crisis risks global stagflation, leading to higher prices and slower growth for corporates.
- Despite volatile environment, the probability of a long-term solution is strong.

2. Private credit

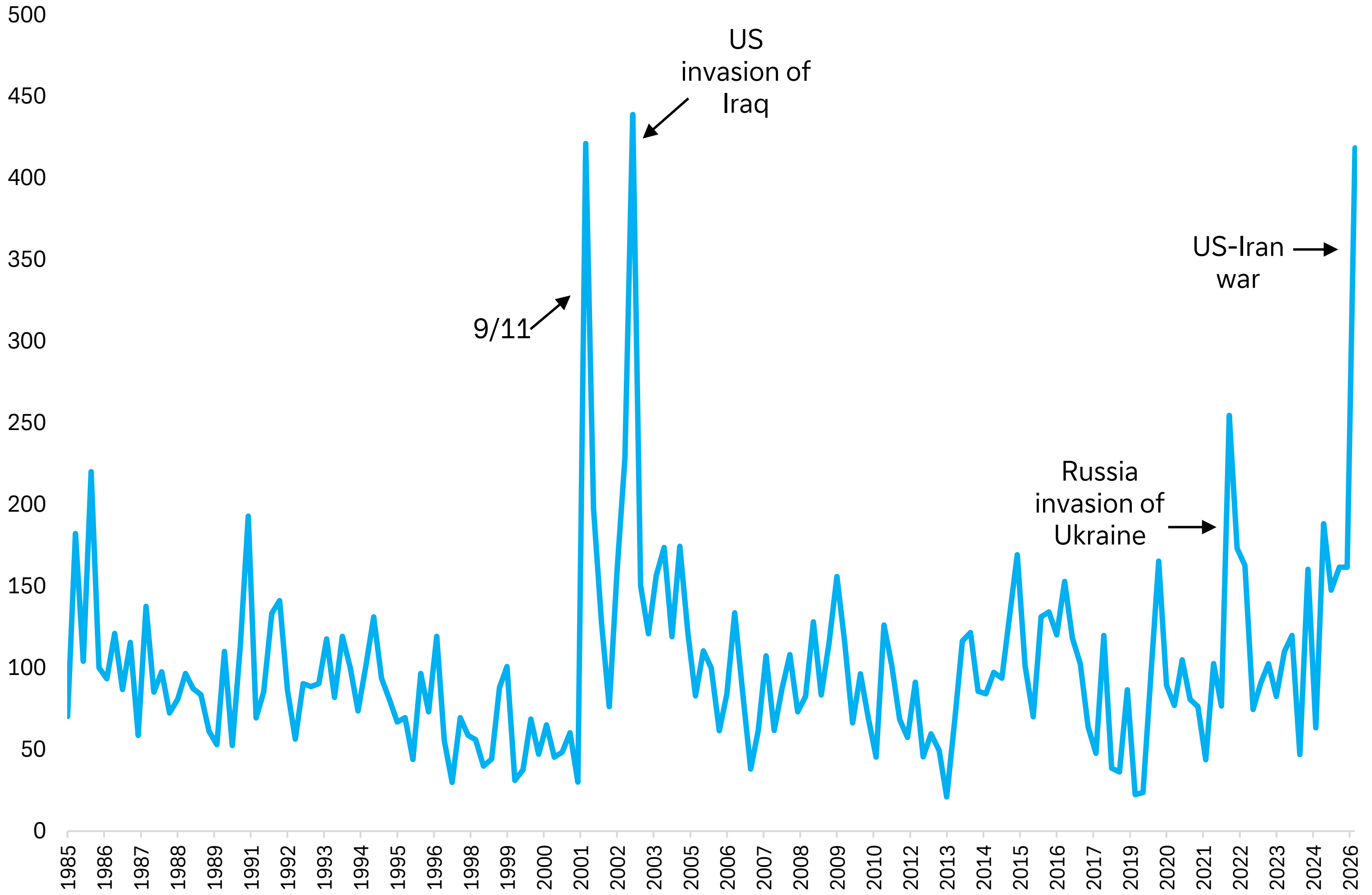
- Liquidity pressures are building but default rates remain low while the most material exposures are with larger managers with more buffer.
- There are expectations of a severe shock (BDCs' discount to NAV is about 20%) however, this fear has been relatively muted in broader public credit markets.

3. AI disruption

- Concerns that AI will disrupt the labour force are, so far, unfounded in aggregate. The most likely scenario remains that job losses will be limited to specific roles and most will use AI to amplify existing roles.
- The Nasdaq has recovered from another AI-related sell-off.

Geopolitical uncertainty is now at its highest level since the US invasion of Iraq

Caldara Iacoviello Geopolitical Risk Index, quarterly uncertainty



Source: Deutsche Bank Research, Caldara, Iacoviello Geopolitical Risk Index. Takes automated text-search results of the electronic archives of newspapers

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Central banks

The energy shock has shifted central bank trajectories

Fed

- Fed to remain on hold near neutral through 2026.
- Core PCE (2.9% q4/q4) set to remain above target, coupled with labour market stabilization, makes chances of easing unlikely.

ECB

- 25bp hikes in Jun and Sep to 2.50% year-end, previously on hold.
- 2.5% at the upper end of neutral; ECB is sending signal on their commitment to price stability.

BoE

- On hold at 3.75% through 2026. Upside risks stem from energy prices trickling down into core goods and food.

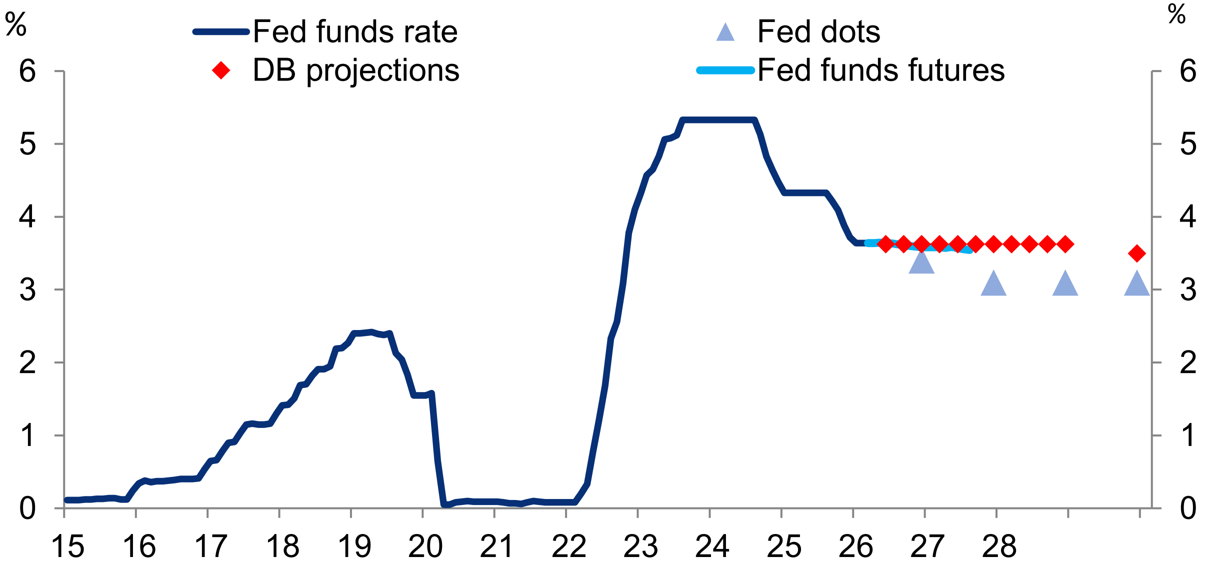
BoJ

- 25bps hikes in Jul and Jan 27, 1.25% by Jan 2027.

PBoC

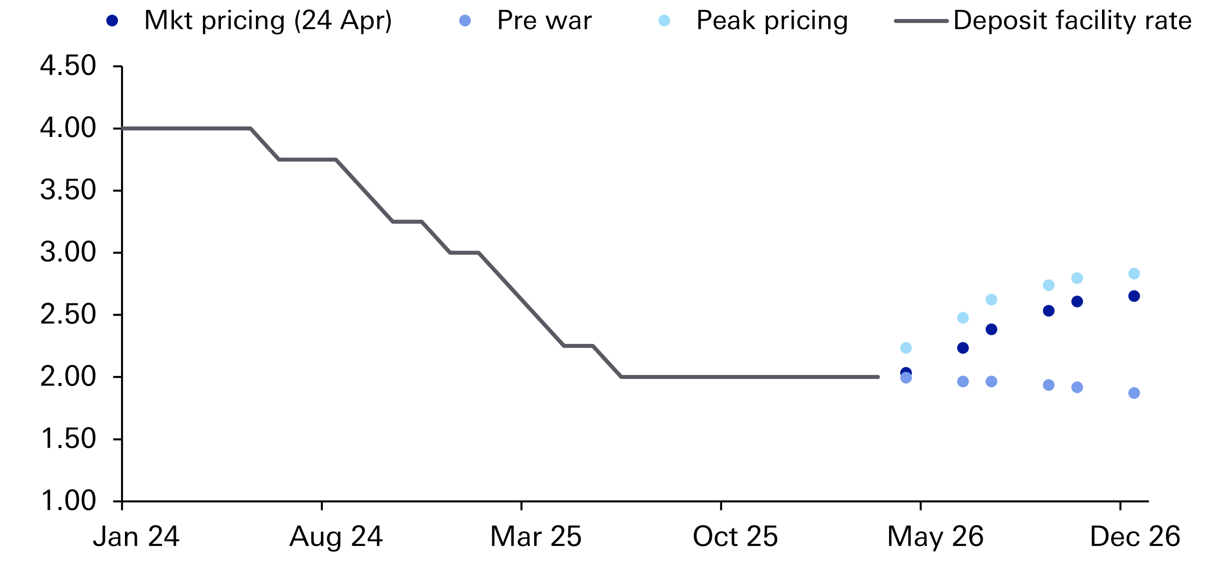
- Hold at 1.40% through 2026.

FED: Central bank is done cutting interest rates from here



Source: FRB, Bloomberg Finance LP, Haver Analytics, Deutsche Bank Research. Updated April 24, 2026

ECB: 2.50% YE, previously 2.0%



Source: Deutsche Bank Research, ECB. Updated April 24, 2026

Read: Focus Europe: ECB Preview: Tough talk, tough decisions - Deutsche Bank Research and Fed Notes: FOMC preview: Powell's last dance? - Deutsche Bank Research

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02.

Geopolitics

Iran conflict

An end in sight?

While diplomacy is continuing, the situation remains extremely fragile. The two sides' postures continue to diverge on Iran's nuclear programme, the pace of sanctions relief, potential reparations, Iranian support for its proxies, and long-term control of the Strait of Hormuz. While the ceasefire has been extended, the crisis remains unresolved and we should expect continued escalations and bellicose rhetoric by all parties to the conflict.

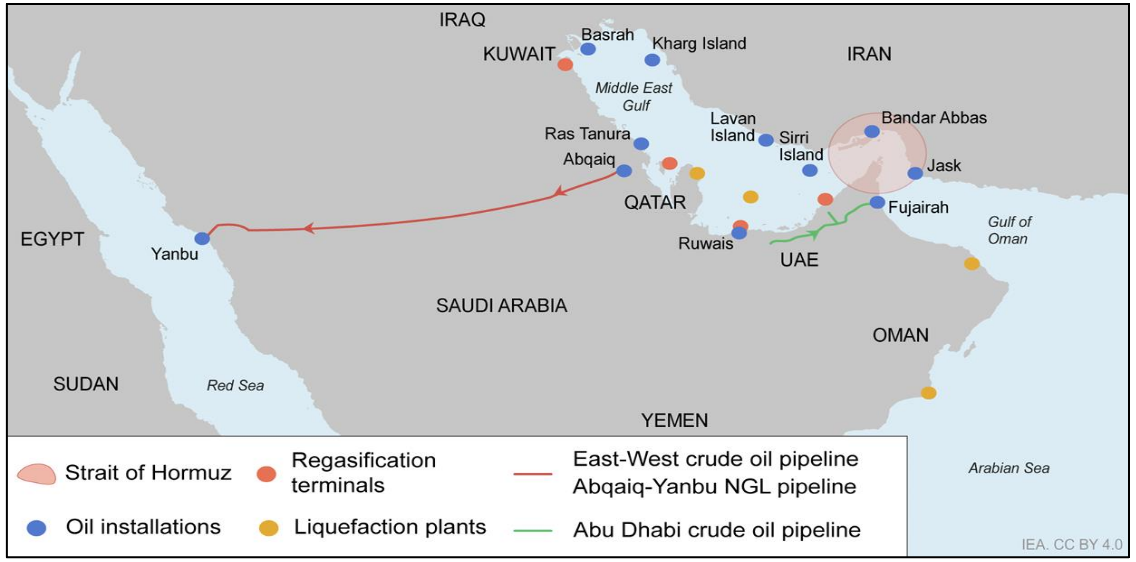
Despite the precarious nature of the ceasefire, and the added uncertainty due to the US blockade of the Strait of Hormuz, we assess that the probability of a longer-term solution is strong. The US is facing diminishing returns and growing costs. We have already witnessed increasing domestic and international pressure on Trump, which we expect will only increase as the economic fallout of crisis persists and expands.

However this ends, it will be a gamechanger for the Middle East. The long-standing Middle East geopolitical equilibrium has been shattered. Iran will likely walk away from this conflict with enhanced influence over the Strait of Hormuz, potential sanctions relief, and with greater political leverage having withstood the US and Israeli attacks. Meanwhile, the conflict illustrates that the Gulf is no longer buffered from conflict. Consequently, its relationship with the US will inevitably be impacted. We expect to see a diversification in Gulf relations in the aftermath of the acute crisis.

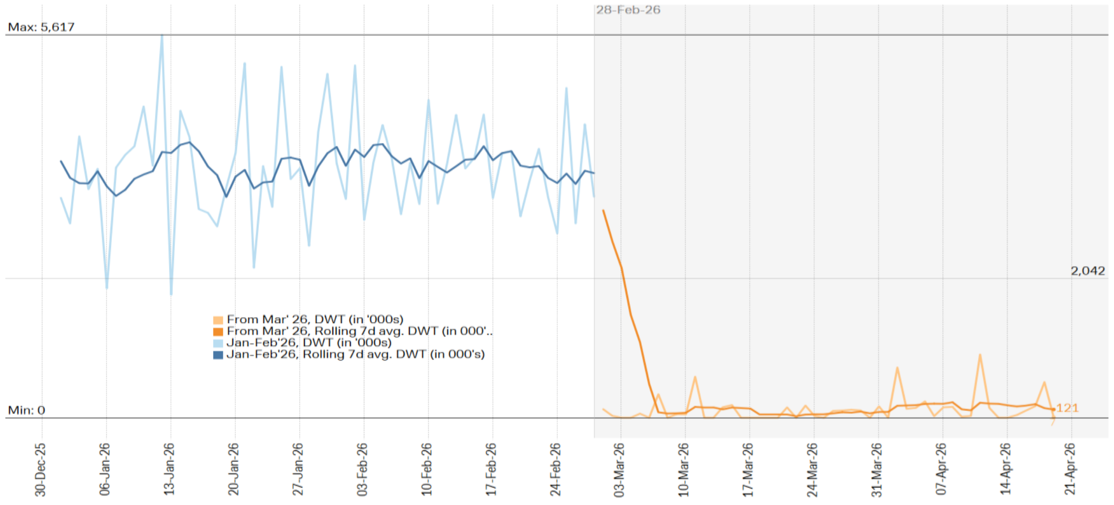
Post-conflict normalization could take months. Even if the war concludes in the near term, rebuilding critical infrastructure and restoring shipping and energy flows will take significant time. For example, repairs to Qatar's Ras Laffan natural gas field – the world's largest – could take months, inevitably delaying a rebound in global supply.

A rebound in shipping (see chart) will also be contingent on confidence – both by shippers and insurers – that the Strait of Hormuz is safe and clear of mines. This could take weeks beyond an end to the conflict. Significant knock-on effects, including for inflation, could continue well into 2026-H2 even with a near-term peace deal, including in the form of jet fuel and fertilizer shortages.

Oil and gas infrastructure in the Gulf



Tanker activity exiting the Strait of Hormuz (in DWT'000s)



Source: Deutsche Bank Research, International Energy Agency, dbDataInsights.

Other geopolitical flashpoints

Key topics to monitor during 2026-Q2

- Trump-Xi Summit: Barring a significant escalation in the Iran war, the summit between presidents Xi and Trump on 14-15 May will likely go ahead as planned – and deliver a package of tactical economic agreements. Despite recent signs of friction over Iran, both sides appear to be prioritizing bilateral stability for 2026, and the threshold for another summit deferral is likely high. However, a breakdown of the US-Iran ceasefire and a return to fighting would be a notable downside risk, which would increase the probability of another summit delay, or even a cancellation.

- Putin's visit to China: It will likely aim to cement the China-Russia partnership (including more energy supplies for Beijing), and possibly to coordinate US policies after the Xi-Trump summit. Unconfirmed Russian media reports suggest Putin could visit during the week of 18 May. Putin and Xi could also seek to deepen diplomatic and security ties, and coordinate on the Iran conflict, given Tehran's status as both countries' close partner.

- Ukraine ceasefire talks: Negotiations for a ceasefire in Ukraine are largely stalled. While there is ongoing diplomatic activity, no new rounds of ceasefire talks are publicly scheduled for May or June. Ukrainian President Zelenskyy has stated that the US wants Russia and Ukraine to reach a peace agreement by June. However, the specifics of how this would manifest in terms of scheduled meetings are unclear.

- While a ceasefire over the coming months is far from guaranteed, even credible steps toward a deal would have implications for markets. These could include a possible easing of some sanctions on Russia, the potential for increased Ukrainian agriculture exports, and for enabling reconstruction – with implications for investments in minerals, gas infrastructure, technology, city reconstruction, data centres and artificial intelligence.

For more on geopolitical flashpoints to watch out for, read our recent publications here:
Iran Ceasefire: An End in Sight? ; This Month in Geopolitics: April 2026 ; Geopolitical Crossroads: India's Strategic Balancing Act ; High-Stakes Diplomacy: Navigating the Xi-Trump Summit in the Shadow of Iran

Source: Weichao Deng on Unsplash

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Selected geopolitical events in April - June

- **7 May:** UK local elections and Scottish and Welsh parliamentary elections.
- **8-9 May:** The Philippines will host the ASEAN Leaders' Summit.
- **11-12 May:** EU foreign and defence ministers meetings.
- **14-15 May:** Presidents Xi and Trump are expected to meet in Beijing.
- **Mid-May:** President Putin is expected to visit Beijing.
- **Mid-May:** Indian Prime Minister Narendra Modi is expected to visit the Netherlands, Norway and Italy.
- **22 May:** Nuclear Non-Proliferation Review Conference concludes in New York.
- **29-31 May:** Singapore will host the annual Shangri-La Dialogue defence summit.
- **31 May:** Colombia Presidential election first round
- **Late May:** US Secretary of State Marco Rubio will visit India for talks.
- **Late May/Early June:** The OECD Ministerial Council Meeting is expected to take place in Paris.
- **11 June:** FIFA World Cup begins in US, Canada and Mexico
- **15-17 June:** France will host the G7 Leaders' Summit in Évian.
- **19 June:** European Council meeting
- **1 July:** The first review of the US-Mexico-Canada (USMCA) trade agreement begins.

Source: aboodi vesakaran on Unsplash

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SAUDI ARABIA
EGYPT
KAZAKSTAN
SOMALIA
ETHIOPIA
KENYA
Tropic of Cancer
GULF OF ADEN
MALDIVES
SEYCHELLES
Direction of the Earths Rotation

03.

Markets

Commodity markets

The key question: When do the stresses in financial assets directly affected by the war begin to catalyse confidence problems in broader markets?

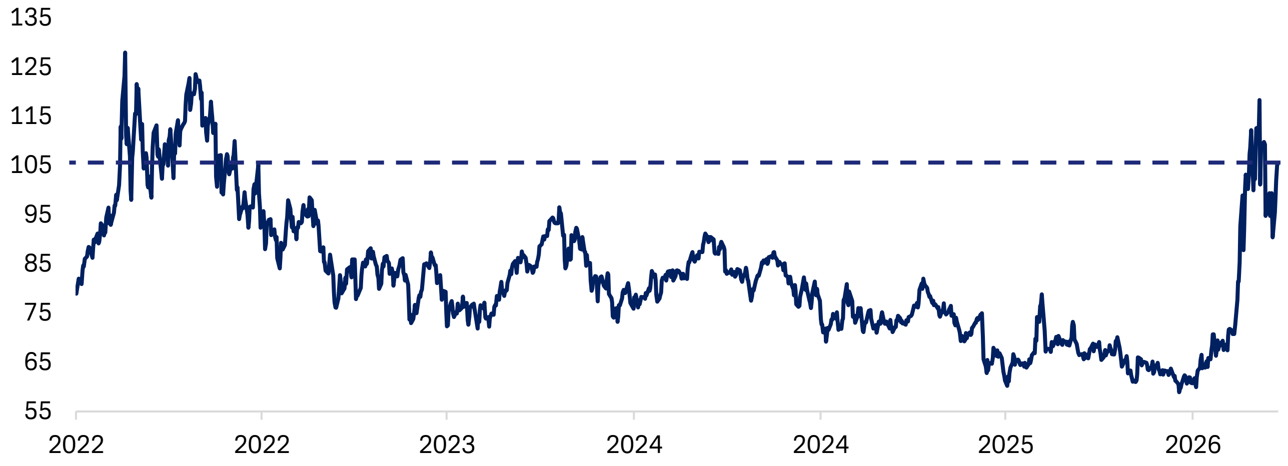
Oil: Brent $96/bbl (Q2'26), $84.8/bbl (2026) average, assuming the Strait of Hormuz will re-open in May.

- Improving but uncertain backdrop, thanks to (i) supply diversion and remaining flows limiting the magnitude of oil supply disruption; and (ii) significant demand reduction rising to -2.3mmb/d y/y in April, according to the IEA.
- Risks: Longer suspension of the SoH (reopening beyond May) could accelerate the run-up in Brent well above $100/bbl near term. This would translate into higher-for-longer inflation.
- Central banks' ability to look-through supply shocks is tested.

Gold: moderate YE target to $5,000/oz (prevs. $6,000/oz)

- Forecast based on belief that dollar weakness will resume.

Brent crude oil prices ($/bbl) have risen but are clearly lower today than they were in 2022



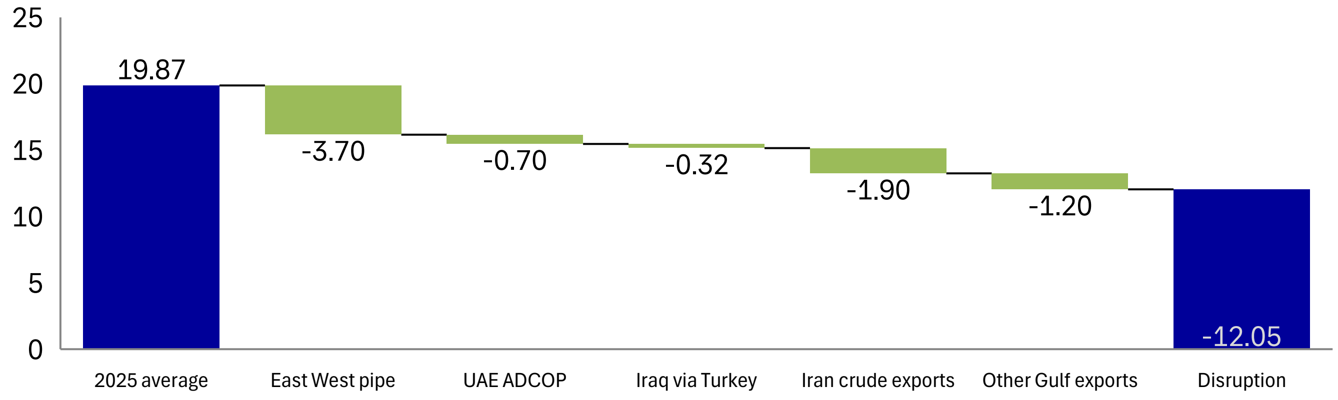
Source: Deutsche Bank Research, Bloomberg Finance LP

Latest oil price forecasts

USD	2025	Q1 2026	Q2 2026	Q3 2026	Q4 2026	2026	2027
WTI (bbl)	64.9	72.7	90.0	82.0	73.0	79.4	71.0
% change		-0.5%	12.5%	9.3%	4.3%	6.6%	4.4%
Brent (bbl)	68.20	78.10	96.00	87.00	78.00	84.80	75.00
% change		0.1%	12.9%	8.7%	4.0%	6.6%	4.2%

Reports: Hsueh On Oil: Problem solved? - Deutsche Bank Research and Precious Special Report: Gold investors to emerge from war bunker - Deutsche Bank Research

Strait of Hormuz April supply disruption down to 12mmb/d



Source: Deutsche Bank Research, Bloomberg Finance LP, IEA, Kpler

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Rates & Credit markets

Dual shock of higher rates and energy costs introduces some uncertainty whether strong corporate growth forecasts will continue into year-end

Private credit outflows have had some effect on broader leveraged loans and some larger US money centres/regional banks, tightening financial conditions somewhat.

- Private credits remain centred on AI disruption, particularly in the software sector, rather than the ongoing energy shock.
- We see credit spreads to widen throughout 2026, with €HY 420bps, $HY at 410bps YE and €IG at 110bps, $IG at 110bps YE.
- In Europe, weaker macro fundamentals plus energy shock driven restrictive policy rates also account for wider spreads. Political risks in Europe are growing.

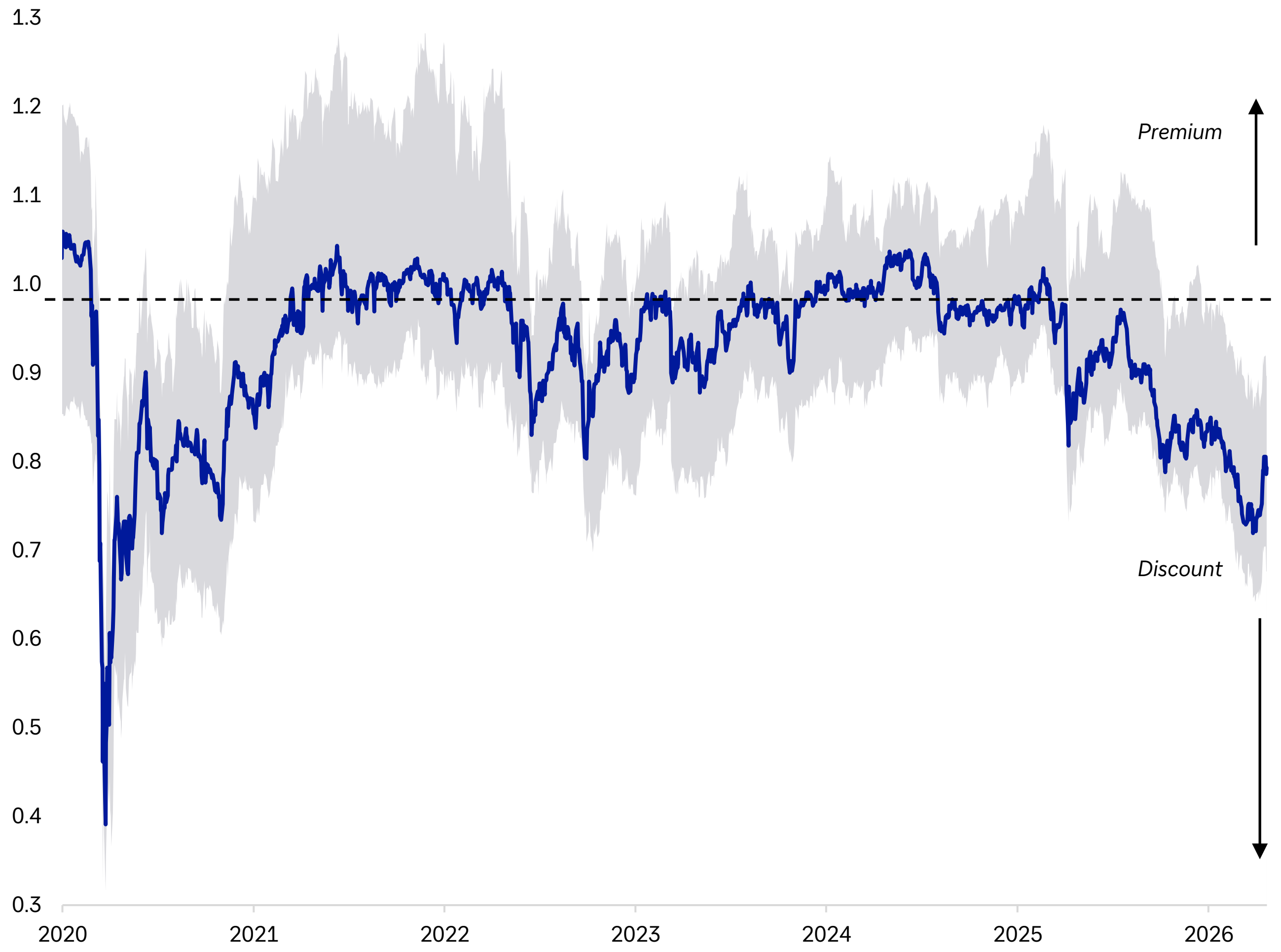
We retain our conviction that the 10Y US Treasury Yield will continue to fall, impacting corporate funding costs.

- Difficult to justify further Fed rate cuts given (i) unemployment rate is broadly stable; (ii) inflation is sticky above 2% and; (iii) policy rates are unlikely to be restrictive.
- AI tailwinds (either bubble or labour market impact) potentially challenge our bearish view on 10Y UST (4.45% by YE).

Reports: IG & HY Strategy: The Narratives Are Turning Upside Down - Deutsche Bank Research and IG & HY Strategy: Private Credit: The gates are up, but the problem is within - Deutsche Bank Research and Global Fixed Income Weekly: Strategy Update - Deutsche Bank Research

US business development companies are a key proxy for the market's private credit concerns. The current discount to NAV of about 20% suggests investors expect severe stress events.

Median price-to-NAV ratio of funds in the S&P BDC index



Source: Deutsche Bank Research, S&P Global. See: Thematic Research: Private capital monitor: Fear and loathing - Deutsche Bank Research

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Foreign exchange markets

The dollar bullish narrative has eroded, staying dollar bearish

Despite some reprieve during war, big picture remains dollar bearish

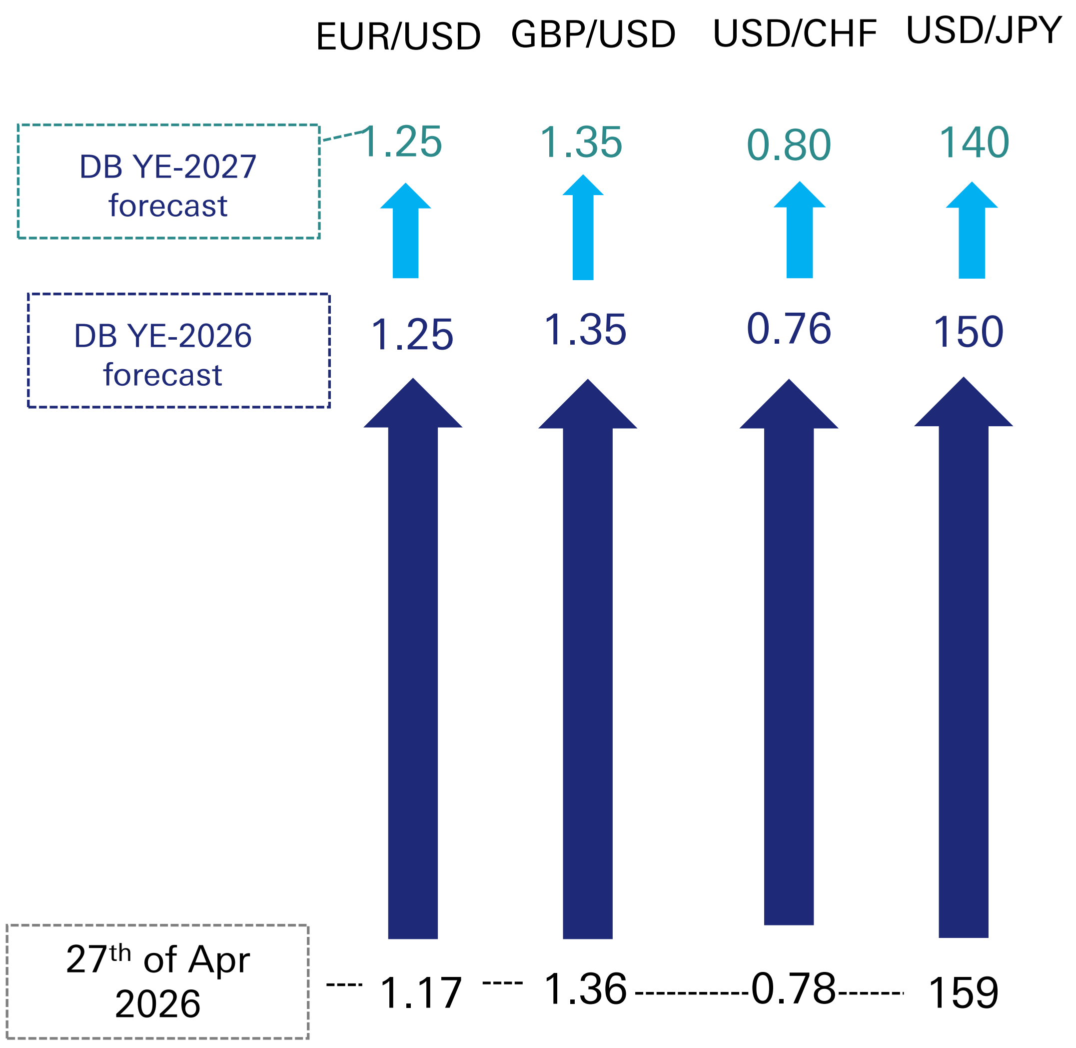
- Usually, we would expect the dollar (as a haven currency) to have rallied more than it did during the Iran war. That highlights some weakness.
- Strengthens the argument that the USD is still no longer as exceptional as a high-yielder and as a global growth outlier.
- The Iran war exacerbated dollar weakness due to Asia and the Middle East running down FX reserves and excess savings to finance higher energy costs. Countries were already moving away from the USD into smaller, unspecified reserve currencies (up from 3% to 6% of these assets in the last 3 years).

Currency outlook

- EUR/USD: Rise to $1.25 by year-end. While the energy price spike complicates the Euro Area's growth picture, it has also brought forward the ECB rate hike cycle, leading to a stronger Euro.
- USD/JPY: Raised end-2026 to 150. JPY remains very cheap as the BOJ remains unlikely to tighten policy rates this year.
- RMB: Supported by de-escalation in US-China tensions and the government's continued aims to internationalise the yuan. RMB remains cheap but a stronger RMB would increase the appeal for global corporates to transition to using and saving in the yuan, especially if most Middle East energy continues to travel to Asia. Corporates continue to have scope to increase conversion of their large stock and flow of FX into RMB.

Read: FX Blog: Time to sell USD - Deutsche Bank Research and The Big FX View: War and Peace - Deutsche Bank Research

Currency movements and DB forecasts for YE-2026 and 2027



Source: Deutsche Bank Research, Bloomberg Finance LP.

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04.

Technology

Artificial intelligence

AI disruption concerns are rising in 2026, but there is no conclusive relationship between modest job declines and substitution with AI

Job displacement from AI remains a concern, but the aggregate evidence so far is encouraging

– A Dec 2025 NBER study suggests that AI, by substituting primarily cognitive tasks, can shift relative demand away from high-skilled roles toward jobs with lower education, lower wages and more male workers.

– The evidence is still inconclusive, as noted by our US analysts. A survey by Indeed found no noticeable deterioration amongst roles of recent market concerns, such as legal, marketing, accounting and management.

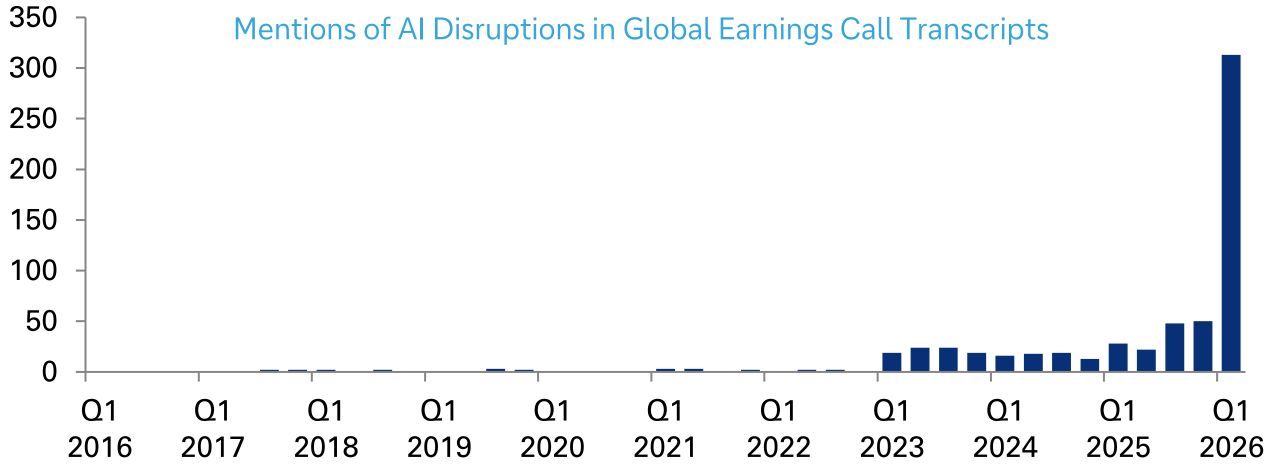
The case for AI's positive benefits is growing

– According to an Atlanta Fed survey, business executives are already seeing labour productivity gains from AI and anticipate it to double in 2026. Firms in high-skill services are seeing the greatest gains. Finance, for example, saw productivity growth of 0.8% last year, implied by revenues and employment, a figure which is set to exceed 2% in 2026.

– Firms in manufacturing and construction saw smaller but still positive implied gains of roughly 0.4 percent in 2025.

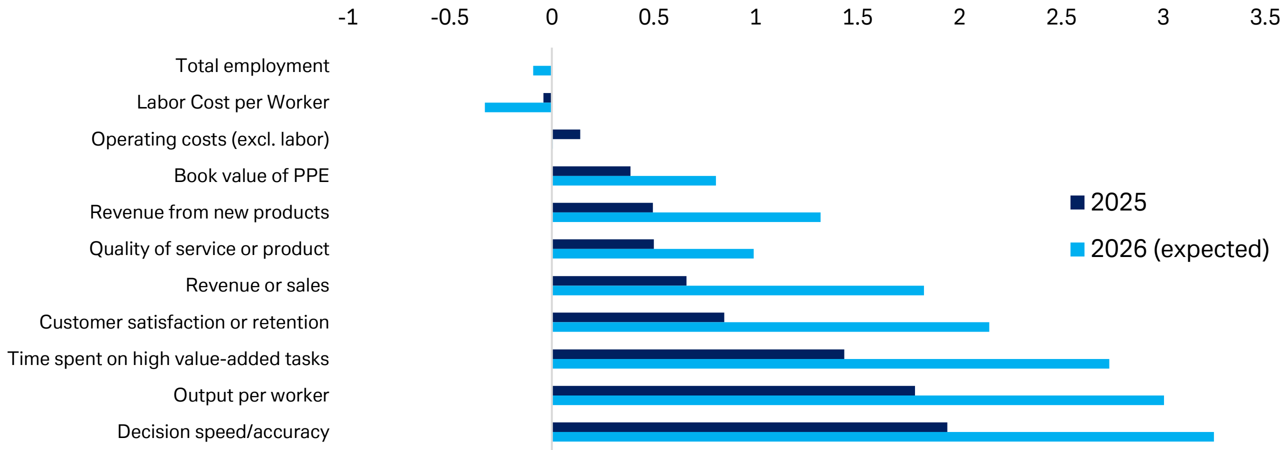
– The study found that productivity gains were associated with innovation and demand-orientated channels rather than cost reduction alone.

In Q1 26, mentions increased more than sixfold from Q4 2025 in earnings calls.



Source: Deutsche Bank Research, Bloomberg Finance LP.

Impact of AI usage on firm outcomes



Source: Deutsche Bank Research, The CFO Survey, Federal Reserve Bank of Richmond, Federal Reserve Bank of Atlanta. Question asks for likert scale options that are mapped to percentage outcomes (i.e. increased moderately, 5.1 to 10%. Above chart uses midpoint of ranges provided and +/- 5% above/below highest/lowest options to calculate numeric averages across respondents.

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Artificial intelligence

Companies will continue to invest in AI to varying degrees, but lack of trust remains a key barrier

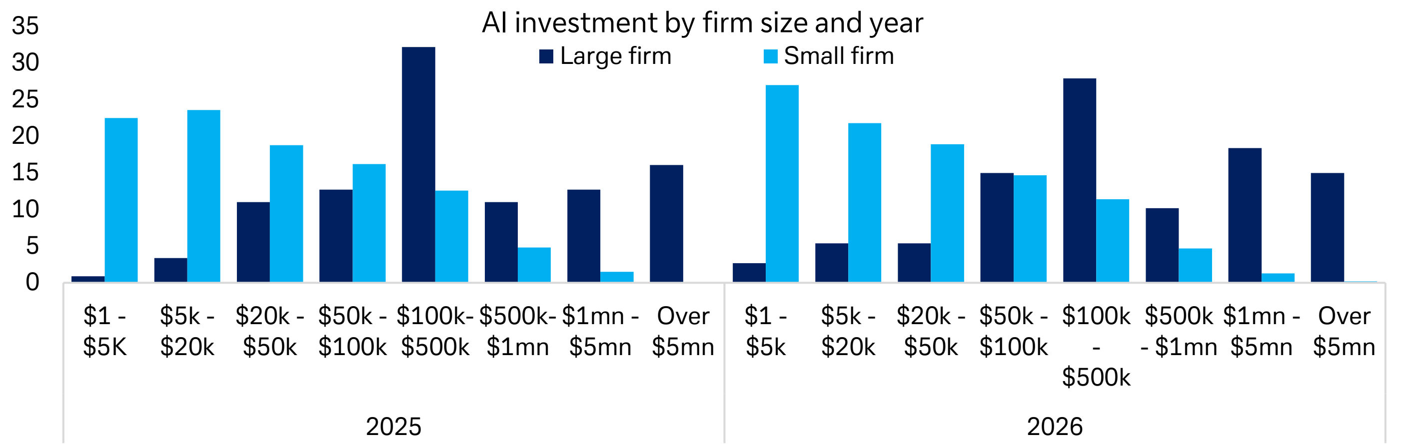
Company adoption continues to be underway, with more pickup in 2026

- Large firms led corporate AI adoption in 2025; smaller firms are only beginning to invest in AI now, according to a survey led by the Atlanta Fed and Richmond Fed.
- Around 30% of large firms expect to invest more than $1mn in AI in 2026 (vs 1% of small firms). But nearly 60% of smaller firms plan to invest less than $20,000 in AI in 2026 (vs 14% of larger firms).

Western economies lag in AI trust and adoption

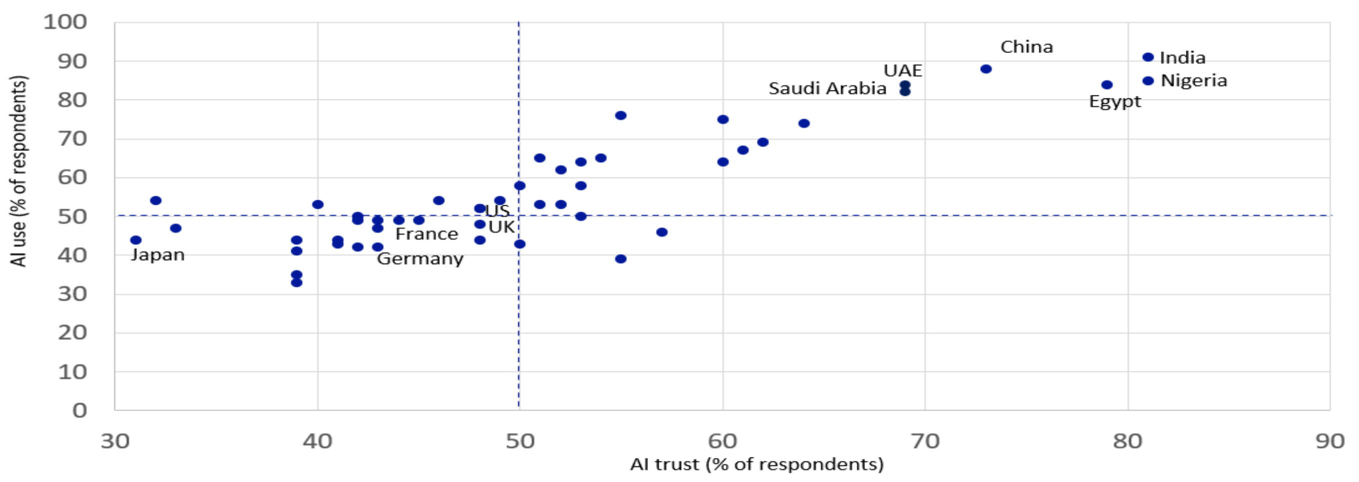
- Barriers to adoption continue to be lack of workforce training, belief AI is not yet sufficiently mature and privacy concerns (see here)
- While the US leads in the most advanced models and chips, and accounts for half of the world's data centres, US consumers are more suspicious of AI than their peers in markets such as India, the UAE, China and Saudi Arabia. Western consumers also lag behind in adoption.
- 48% of US consumers say they trust AI for work purposes, and only 52% say they use it on a regular basis, according to a survey.
- That compares with well over two-thirds trusting AI for work in the leading Asian and Gulf markets and around 85 to 90 percent using it.

80% of firms will invest in AI in 2026, with large firms to invest the most, according to a CFO survey led by the Atlanta Fed and Richmond Fed



Source: Deutsche Bank Research, The CFO Survey. Large firms have > 500 employees. What do you expect your company's total expenditure and financial investment in AI technology/solutions will be over the next 12 months? Federal Reserve Bank Richmond, Federal Reserve Bank Atlanta.

Western economies lag behind their peers in trust in AI and intentional use at work



Source: Deutsche Bank Research, University of Melbourne, KPMG.

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Digital currencies

The digital euro is coming faster than most CFOs realise; Stablecoins are becoming a corporate treasury tool

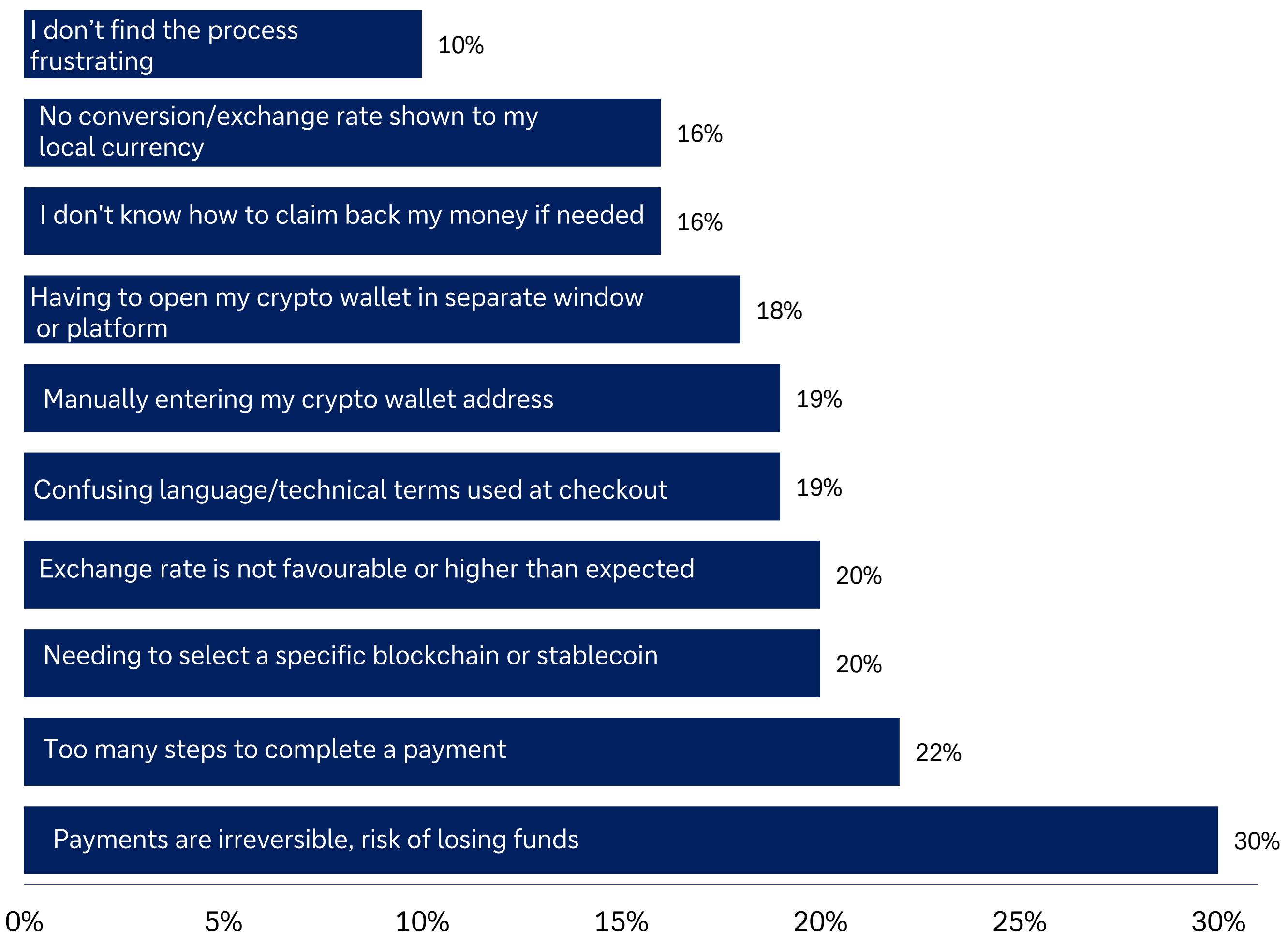
Stablecoins

- Over 80% of transitions occurred outside the US in 2025. Use cases expanding beyond crypto trading into treasury management, payments & settlements, banking & fintech, e-commerce & services.
- Top CFO risks: irreversible payments (30%), too many steps to complete a transaction (22%), fragmented wallet infrastructure.
- In Europe, political debate on permissibility of multi-jurisdictional stablecoins – agreement pending between ECB and European Commission.
- US CLARITY Act still pending. Key debate includes stablecoin yield and bank lending impact.
- March 2026: The OCC proposed stricter stablecoins rules, including a minimum capital floor of $5mn and a ban on interest payments to holders. This favours established payment giants and raises barriers for new entrants. See here.

CBDCS

- Wholesale: ECB's Project Pontes – a settlement solution for DLT-based large-value transaction – launches Q3 2026.
- Retail: European Parliament vote scheduled 23 June 2026. Pilot expected to begin in 2027; retail issuance targeted for 2029.

Challenges with stablecoin payments experience (BVNK Survey)



Source: Deutsche Bank Research, BVNK Survey (March 2026).

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Appendix 1

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Deutsche Bank
Research Institute

The return of history: gold, the dollar, and the monetary future

April 27, 2026

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In 1989, Francis Fukuyama argued that humanity had reached "the end of history". In the years that followed, the US became the uncontested hegemon, global trade exploded in a US-defined liberal order, developed market central banks sold gold, while emerging markets accumulated vast amounts of US dollar FX reserves. We argue that the end of history has come to an end. The world is back in a superpower struggle; the US is retreating from free trade, alliances, and security provision; the Great Economic Moderation is behind us; and the dollar banking system has been weaponized. The "return of history" has big implications for gold and the dollar.

Contrary to conventional thinking, we argue that the share of gold in central bank reserves is not driven by the global monetary system, but by the global geopolitical environment. Gold's decline as a share of reserves did not happen with the fall of Bretton Woods in the 1970s, but the fall of the Berlin Wall and the assertion of US hegemony in the 1990s. As tectonic geopolitical plates shift again, the share of US dollars in central bank reserves is once more in decline. It has fallen from over 60% to just 40%, while gold's share has tripled from its lows to 30% today.

We create a framework for the share of gold in central bank reserves as a function of: (1) the volume of gold held by central banks; (2) the price of gold; and (3) the amount of global FX reserves. We see all three pillars on the move, driven by EM. EM central banks have been actively buying gold and driving upward pressure on prices; crucially - their FX reserves may also now begin to structurally decline.

A "return of history" would be consistent with gold getting to at least a 40% share of global reserves. There is significant scope for EM to add towards this. We find that EM countries with closer non-Western defence ties hold more gold. If the world diversifies trade and security dependence away from the US, this would be consistent with less USD and more gold in reserves.

We simulate a range of different outcomes for gold prices depending on the level of FX reserves EM central banks end up with, and the share of gold they target. Even in an environment where EM FX reserves decline to USD5tn, gold prices could still rise to $8000 over the next five years, if EM countries all target a 40% gold share.

For now, EM central bank gold buying likely has to do with preserving the value and accessibility of foreign savings in a changing geopolitical climate. But in the long-run, we consider how gold may one day play a role in anchoring a monetary order that builds independence from the dollar.

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All that glitters

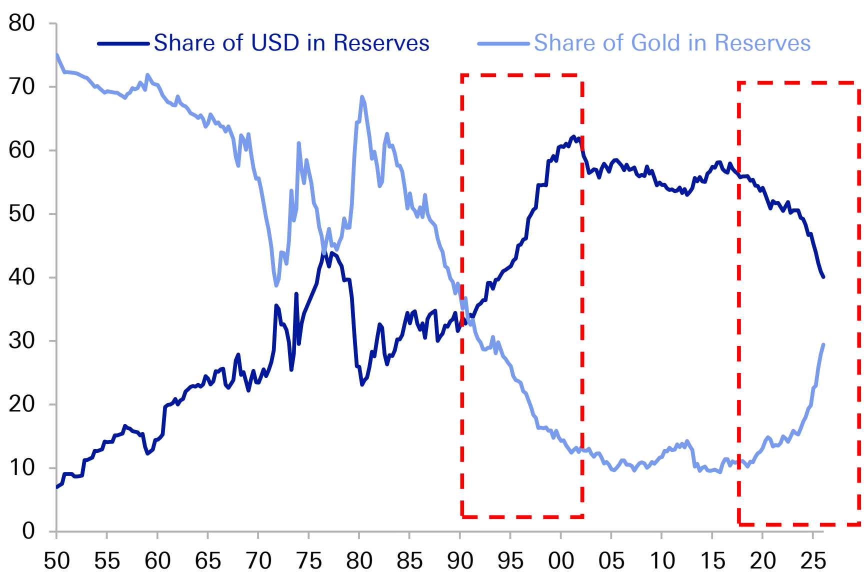
The reserves baton is passing back to gold from the dollar

The share of the USD in global central bank reserves has dropped sharply from around 60% at its peak to just 40% today, with attrition accelerating in the past few years. As Figure 1 illustrates, the USD's share of reserves peaked at the start of this century and sustained those levels for the next two decades before recent losses Importantly, the dollar's losses as a share of central bank reserves have not gone to other fiat currencies, but to gold.

Gold's share in global central bank reserves has doubled in the past four years to nearly 30% today. The fact that the gap between the dollar and gold as a share of reserves is now just 10% is extremely notable. As Figure 1 shows, there appears to be a marked reversal underway of the 1990s trend when central banks moved away from gold and towards the USD in their reserves. Before the 1990s, gold had consistently been a larger share of central bank reserves than the fiat dollar. But by the end of the 1990s, the dollar was over four times the share of gold. This seems to now be going in reverse with gold clawing back its share rapidly. What happened in the 1990s and why is this unwinding today? How far can it go and to what end? These are the questions which motivate this paper.

The 1990s began with a declaration by historian Francis Fukuyama that humanity had reached "The End of History." We argue in this paper that history has returned, but the contest and its leaders will be different. This is the lens through which we think about the resurgence of gold, the decline of the dollar, and the international monetary architecture that may await us this century.

Figure 1: The "end of history" in the 1990s saw a sharp rise in the USD's share of global central bank reserves, with gold's role declining sharply. We argue that history has returned, with gold clawing back its share from the dollar.

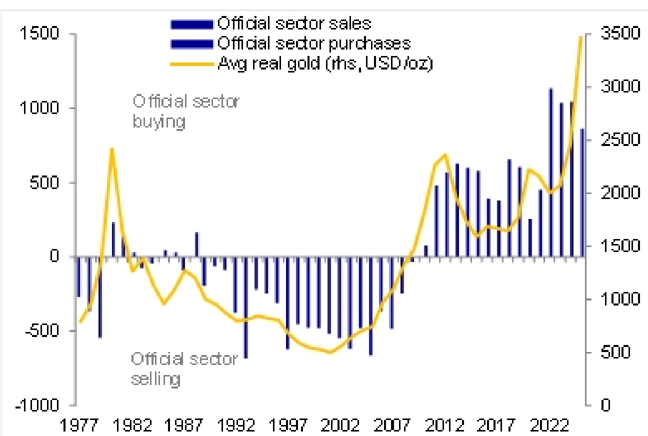


Source :Deutsche Bank, Bloomberg Finance LP, Eichengreen, Chitu and Mehl (2014)

Let's address the skepticism. Skeptics may say the rising share of gold in central bank reserves is simply reflective of gold price increases. Indeed, around 80% of the rise in gold's share has been on account of prices. But there is a genuine volume driver underlying this: central bank purchases have arguably themselves been behind significant price momentum. There is indeed a close relationship between official purchases and sales of gold and the change in the real gold price (Figure 2). Volume and prices are thus endogenously related and are both doing the legwork of gold's rising share.

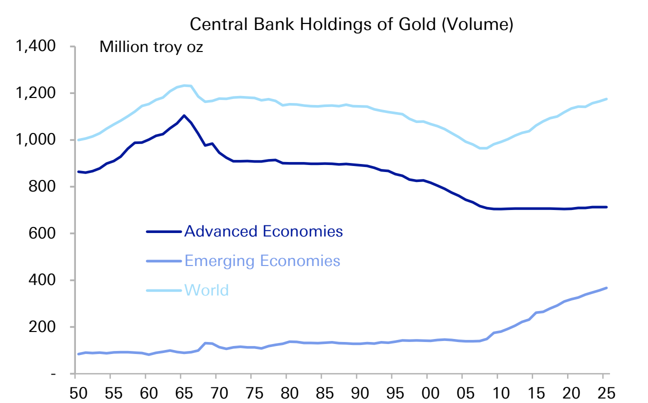
Crucially, all central bank purchases are occurring in emerging markets. As Figure 3 illustrates, it is EM central banks that have been steadily purchasing gold since the 2008 GFC, adding over 225mn troy oz over the past 17 years. Importantly, this is more than advanced economy central banks sold in the 1990s. As we discuss in depth in this paper, we think there could be a long way to go in the trend of EM central bank buying of gold.

Figure 2: Official sector purchases and sales have been closely associated with real price movements in gold



Source : Deutsche Bank, Bloomberg Finance LP, WGC

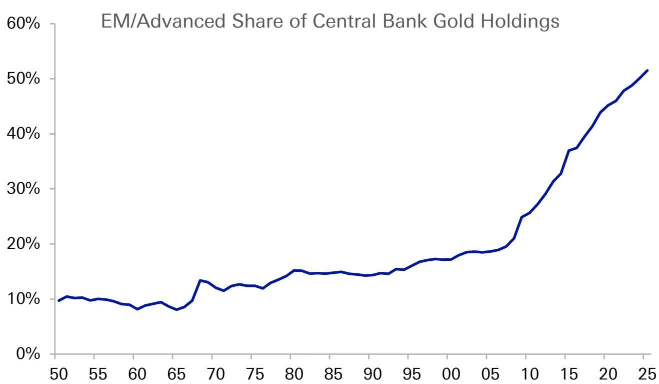
Figure 3: All central bank gold purchases since 2008 have been by emerging markets, reversing the gold sales by DM central banks in the 1990s



Source : Deutsche Bank, IMF International Liquidity Statistics

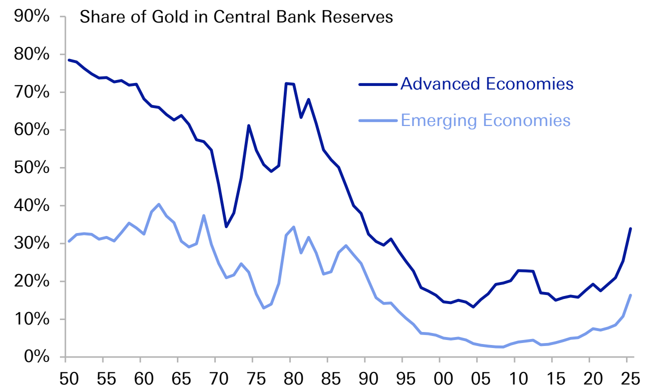
EM central banks still only hold half the amount of physical gold of developed markets. In stock terms, EM central banks held 367mn troy oz at the end of 2025 compared to 712mn troy oz by Advanced Economy central banks, according to IMF classification. This has however been on a steadily rising trend. The ratio of EM/DM central bank gold holding is at around 52%, having risen from just 20% before the 2008 GFC (Figure 4). As a share of total reserves (including foreign exchange holdings), which is where our focus in this paper lies, EM central banks had just 16% of total reserves in gold compared to 34% for DM central banks by end-2025. There thus remains a significant gap to close, if not ultimately exceed.

Figure 4: EM central banks still only own half the gold of DM central banks, but their share has been steadily increasing



Source : Deutsche Bank, IMF

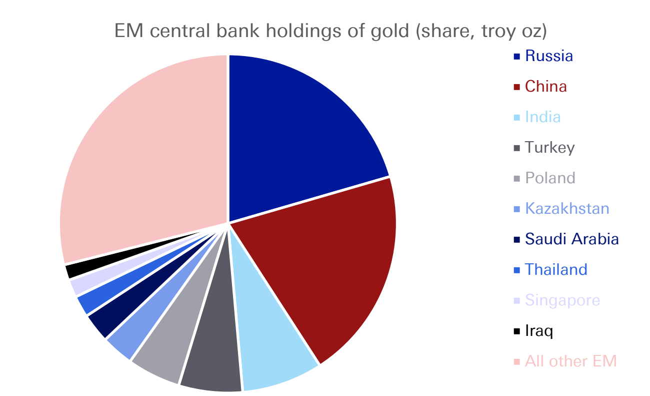
Figure 5: Gold was around 16% of EM reserves compared to 34% of DM reserves at the end of 2025



Source : Deutsche Bank, IMF

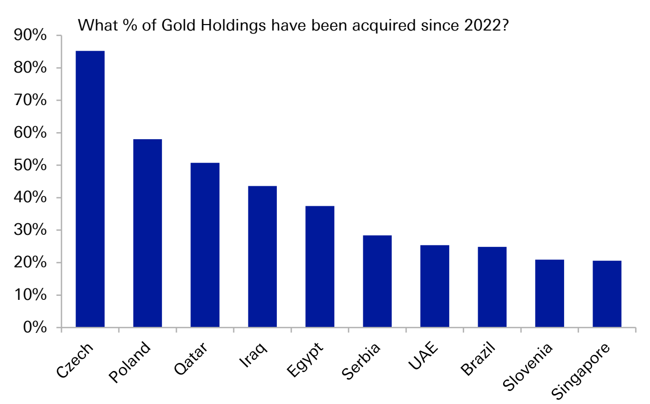
This is not just about the BRICS. Almost half of EM central bank holdings are accounted for by just China, Russia and India. But many middle powers like Turkiye, Kazakhstan, and Saudi Arabia are also significant holders (Figure 6). Perhaps most notable in recent years is the momentum of purchases in certain regions. Strikingly, in Eastern Europe, more than half of the gold holdings of Czechia and Poland have been acquired in the past four years alone, after Russia's invasion of Ukraine (Figure 7). Many MENA states like Qatar, Egypt and the UAE have acquired between 25-50% of their total gold holdings in the last few years alone. We will be exploring the motivations for these purchases in more detail later in the paper.

Figure 6: China, Russia, India and Turkey are the biggest holders but there is a long and growing tail



Source : Deutsche Bank, IMF

Figure 7: Many central banks in Eastern Europe and MENA have been adding quickly to their gold holdings in recent years



Source : Deutsche Bank, IMF, Bloomberg Finance LP

A brief history of gold

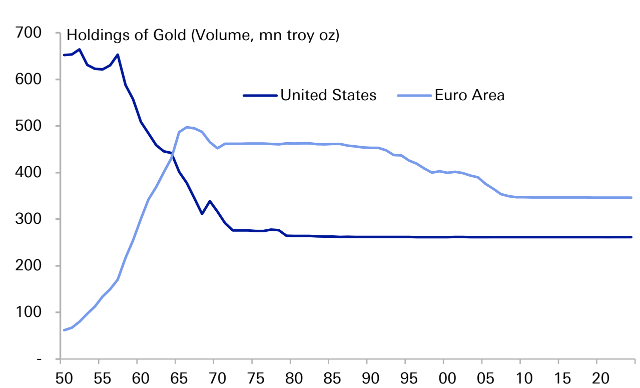
Gold was the heart of Bretton Woods but did not fall with it

The US had over 70% of the world's central bank gold reserves after World War 2, and in constructing the Bretton Woods monetary architecture, it used gold to back a system that centered on the dollar.

As Monnet and Puy (2019) explain, the Bretton Woods system - which followed WW2 and lasted till 1971 - was not like the gold standard that preceded World War 1, or that was tried unsuccessfully in the interwar period. Under Bretton Woods, gold was not redeemable against bank notes at a fixed rate everywhere in the world. Nor were central banks strictly required to back currency in circulation with gold reserves. *Only* the US dollar was convertible into gold at $35/oz by *other central banks* via the Fed. Gold was essentially only a means of settlement between monetary authorities. Foreign countries could choose to hold their claims on the US in US dollars, or exchange them for gold. As the US began to run balance of payments deficits in the 1950s and 60s, some European countries began to make claims on the Fed's gold. By the mid-1960s, more gold was held at European central banks than in the US (Figure 8). By 1971, the system collapsed with Nixon ending the US dollar's convertibility to gold.

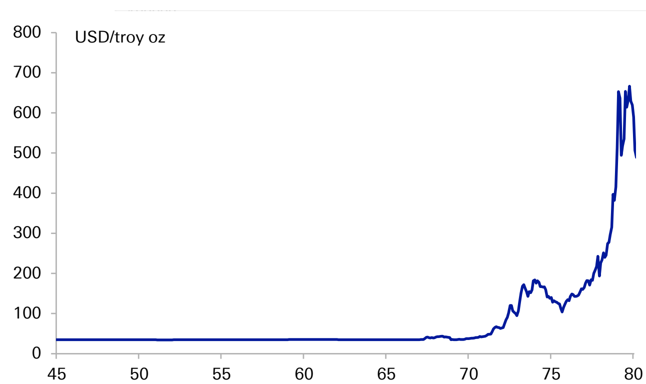
Gold did not however drop as a share of central bank reserves with the fall of Bretton Woods. Even as gold stopped serving a purpose as an official means of international settlement, gold averaged around a 50% share of central bank reserves through the 1970s, albeit with significant gyrations. Notably, it remained more important than the fiat dollar throughout the decade as the earlier Figure 1 depicts. Indeed, the 1970s were a period of enormous price gains for gold which rose more than 1000% in nominal terms through the decade driven by high inflation and geopolitical shocks from the 1973 Arab oil embargo, to the Iranian revolution (Figure 9).

Figure 8: In the 1960s, significant amounts of gold left the US for Europe to settle US deficits. Brettons Woods ultimately collapsed in 1971.



Source : Deutsche Bank, IMF

Figure 9: Gold prices rose more than 1000% in the 1970s - a period of high inflation and geopolitical uncertainty



Source : Deutsche Bank, Bloomberg Finance LP

Therefore, even as the global exchange rate system changed in the 1970s, gold's share in global reserves did not. This points to deeper and different drivers of gold's role. If central bank gold holdings in reserves were not ultimately only about convertibility of the USD, what were they about? And what brought about the change? We turn to this next.

The end of history

The ultimate decline in gold's share in global central bank reserves came in the 1990s. As Figure 1 earlier showed, gold fell below the dollar's share at the start of the 1990s, with the rest of the decade featuring a consistent rise in the dollar and decline of gold. What was behind this crossover?

It was not a transition in the monetary system - which had occurred two decades prior - but a shift in the geopolitical environment that changed the role of gold.

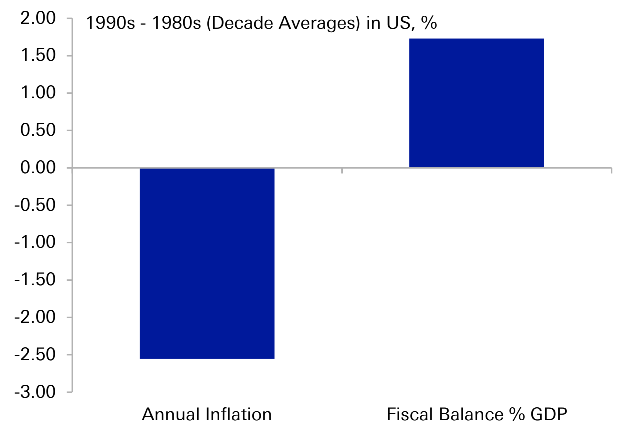
In 1989, Francis Fukuyama questioned whether humanity had reached "the end of history?". A lot of the 20th century had featured ideological violence, but the ending of the Cold War he believed brought a "triumph of the West, of the Western idea," "the endpoint of mankind's ideological evolution" and the "unabashed victory of economic and political liberalism." The Berlin Wall fell the year of his thesis, and the Soviet Union had fully dissolved by 1991. The US thus became an uncontested hegemon in what appeared to be a geopolitically unipolar world. Japan, which had been the US' closest economic competitor, was well within the US security and dollar system, and China was still a decade from joining the WTO.

In the economic realm, a lot was changing in the 1990s from the turbulent decades that preceded it. In the US, a "Great Moderation" was underway. Inflation was falling with tight monetary policy having tamed the price instability of the 1970s and 80s. Indeed, average inflation in the 1990s was 2.5% below 1980 averages. The US was running growing fiscal surpluses. The US fiscal position was 1.5% GDP better in the 1990s than in the 1980s on average (Figure 10). The combination of better inflation and fiscal data, and independent central banking raised trust in monetary and fiscal systems, making US Treasuries a more attractive safe asset. Unlike gold, Treasuries paid a positive yield, had a deep and liquid market which made them easy to hold and transact in, and had no storage costs.

Developed world central banks led by Europe thus began to sell gold reserves in the 1990s (see again Figure 2), which came to be seen as a "barbarous relic" (Arslanalp et al, 2023). Countries like Switzerland, UK, Belgium, Netherlands, Austria, Australia, Canada all sold gold. Gold was seen as a vestige of the past, a zero-yielding asset with little role to play, especially in a world where European countries were moving towards a common currency. As gold sales began to impact the price of gold, central banks formalized a Central Bank Gold Agreement to coordinate sales in 1999, a framework that lasted for the next two decades.

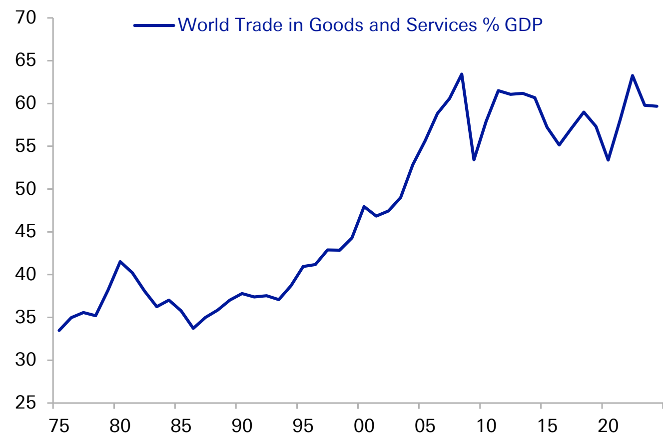
Stabilizing geopolitics, combined with large improvements in technology, communications and transport, led to an explosion in trade and globalization in the 1990s. Trade in goods and services close to doubled as a share of global GDP from 1990 to the 2008 GFC (Figure 11). A rise in global trade contributed to the deepening in dollarization via the channels of invoicing, finance, and the accumulation of dollar surpluses in exporting EM economies. This brings us to arguably the biggest driver of gold's relative decline against the dollar.

Figure 10: The Great Moderation of the 1990s: US inflation was falling and the fiscal position improving



Source : Deutsche Bank, Haver Analytics

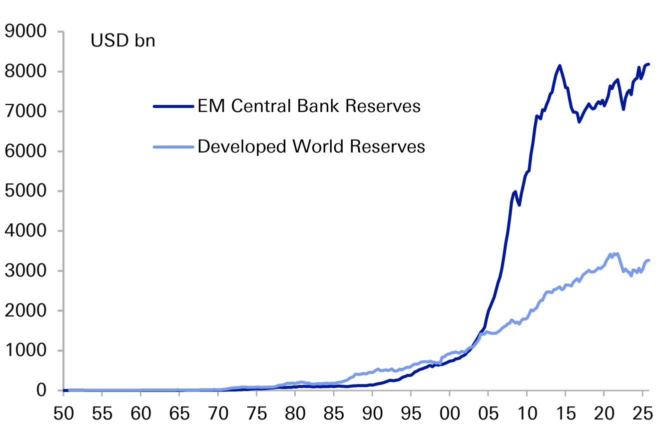
Figure 11: The 1990s were also a period of dramatic growth in trade which was very dollarizing



Source : Deutsche Bank, World Bank

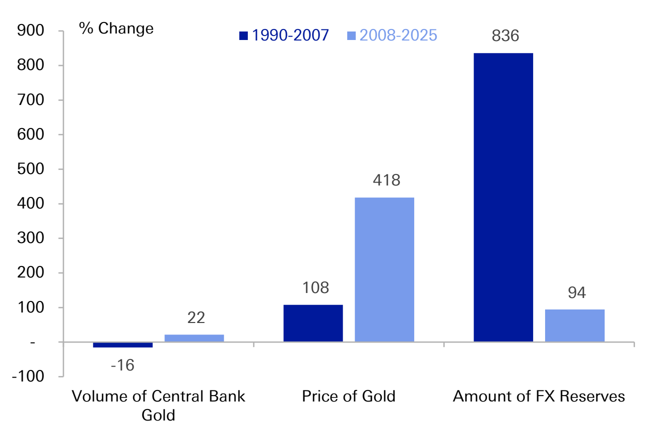
The biggest driver of the decline of gold's share in global central bank reserves in the 1990s was the meteoric rise of emerging market FX reserves, held in dollars. As Figure 12 illustrates, global reserves began to rise in the 1990s. Japan was the biggest driver in the 1990s, with the baton passing to China in the 2000s alongside a wide swathe of EM countries from Russia, to Saudi Arabia, India, Korea, Taiwan, Brazil and others. The total amount of global FX reserves went up 9 times between 1990-2007. A majority of these reserves were built in USD and saved in US Treasuries given the predominance of the US driven economic and trade order. The Asian Financial Crisis in the late 1990 also led to a deeper preference for self insurance by EM countries and a dramatic shift in favour of building reserves, after the failures of capital account liberalization policies. The IMF itself shifted to encouraging reserves adequacy frameworks that stressed savings in USD liquidity.

Figure 12: EM central bank reserves began rising in the 1990s and turned parabolic in the 2000s



Source : Deutsche Bank, IMF

Figure 13: The rise in USD reserves between 1990-GFC were a much bigger driver of gold's relative decline than DM central bank gold sales



Source : Deutsche Bank, IMF, Bloomberg Finance LP

Bringing it all together: one can think of the share of gold in central bank reserves as a function of three things: (1) the *volume* of gold held by central banks; (2) the

***price* of gold; and (3) the amount of *foreign exchange* reserves.** The former two influence the numerator, while the latter influences the denominator. The decline of gold as a share of global central bank reserves from around 40% on the eve of the 1990s to just 10% by the eve of the GFC was predominantly about the denominator. While developed world central banks were selling gold in the 1990s, this was in fact largely offset by the price increase of gold (Figure 13).

In sum, the biggest driver of gold's decline in global reserves in the 1990s was the rise of EM FX reserves accumulated in USD. This was in turn a function of dramatic globalization, in a US-driven neo-liberal unipolar order, amidst sound and improving economic fundamentals in the US. By extension, the dollar's fate as the world's reserve currency will have a lot to do with how these same countries treat their USD holdings in a de-globalizing world in which the US driven order is fraying: do EM countries still add to USD reserves, diversify away from them into gold, or actively draw them down? The future of the dollar as a reserve currency could well be determined in emerging markets, consistent with our thesis that the Global South will be a much bigger economic and geopolitical force to understand.

The return of history

The drivers of the "end of history" have almost all been going in reverse. The world is back in an ideological struggle between competing economic and political models, this time led by the US and China. Both countries are engaged in growing competition across technology, energy, resources, and influence. The world is no longer unipolar. China is on many metrics a bigger industrial, trade and naval power than the US.

The US is stepping back from free trade and fracturing traditional alliances, with China having positioned itself to step into the fray with years of building trade and investment relationships across the Global South. The provision of global public goods guaranteed by the US - namely freedom of navigation and security for key allied regions - has come into question with the closure of the Straits of Hormuz and the vulnerability of the Gulf in the latest conflict.

If the 1990s was a world where the US was happy to outsource manufacturing and labour to EM, with many EM countries happy to outsource security and savings to the US, this is now reversing. The US is keen to onshore more critical manufacturing, while many EM regions like Asia and the Gulf will be reconsidering their need for strategic autonomy in areas like energy and defence. They may well need the savings they have been investing in the USD to build these capabilities.

On the economic front, the Great Moderation is behind us. US inflation has been above target for over five years, independent central banking has come into question, monetary balance sheets have expanded dramatically under QE, and the US fiscal trajectory is on a worrying path.

Finally, the weaponization of the US dominated banking system with the freezing of Russia's USD and EUR FX reserves in 2022 has increased the appeal of saving in gold, which can be held physically and locally, away from the arm of sanctions or asset seizures. Indeed, Russia and China hold 100% of their gold locally.

The end of history has itself come to an end, with significant implications for gold and the dollar, which are becoming increasingly apparent.

What the future may hold

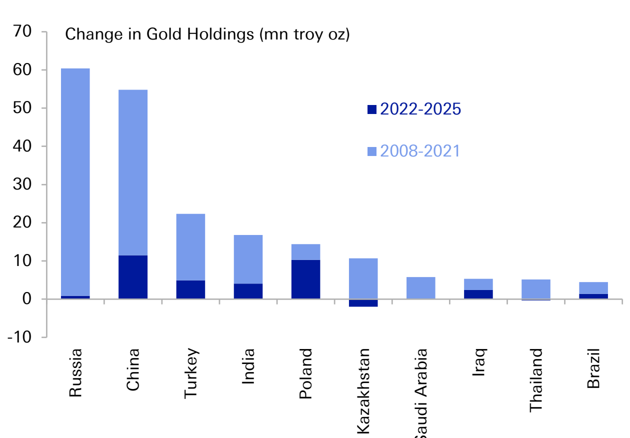
EM is not only buying gold, but could soon be selling the dollar too

As discussed earlier, gold's share in central bank reserves is a function of three main drivers: the volume of gold held, the price of gold, and the stock of FX reserves. For emerging market central banks, all three are on the move.

First, EM central banks have been actively buying gold. EM purchases since 2008 exceed the sales made by developed world central bank sales in the 1990s. Figure 14 illustrates the biggest buyers in volume terms since 2008, split between purchases before and after Russia's invasion of Ukraine (2008-2021 and 2022-25). It is notable that Russia had actively diversified into gold ahead of 2022 (Figure 14). China has continued to purchase gold at roughly the same pace since 2022, while countries in Eastern Europe, India, and MENA have accelerated gold purchases since then.

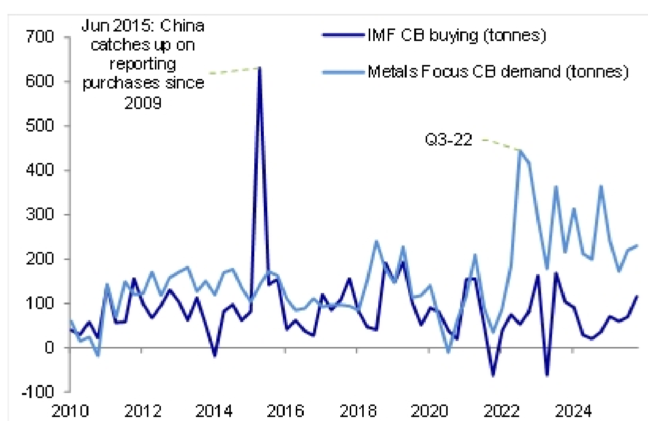
It is important to note that our analysis in this paper focuses on IMF data on central bank reserves, which understates the true accumulation of EM official savings in foreign assets. We discussed this phenomenon in depth **here**. We have chosen to focus on IMF data for a range of reasons: longer time-series history of the data, comparability to COFER estimates of central bank FX holdings, and availability of country level data series. But there are important caveats to this data. While IMF data shows EM central bank reserves have been stagnant in recent years, EM countries have in fact continued to build and recycle savings abroad via large sovereign wealth funds, state banks, and public pension funds. A significant amount of these savings are held in USD, but often with a riskier, less liquid profile across equities and private markets rather than necessarily US Treasuries. Global SWF pegged sovereign wealth fund holdings in Asia and MENA at over USD12tn in March 2026, greater than the size of their central bank holdings.

Figure 14: Russia, China, Turkey, India and Poland have been the biggest buyers in this wave of EM central bank demand



Source : Deutsche Bank, IMF, Bloomberg Finance LP

Figure 15: IMF central bank data understates the true extent of official diversification into gold, which could be three times as large



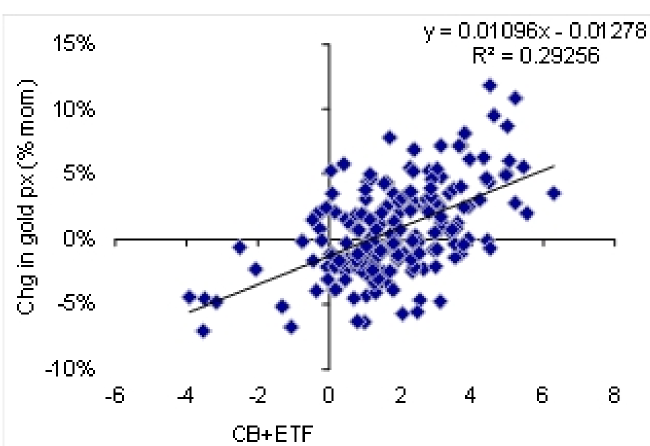
Source : Deutsche Bank, IMF, Bloomberg Finance LP, World Gold Council

If IMF data on central bank reserves is understating the true extent of official foreign savings in EM, it is also understating the full extent of official diversification into gold. Indeed, a series by the World Gold Council (WGC) that tracks quarterly demand by "Central Banks and Other Institutions" and is believed to include sovereign wealth funds and other state-directed flows, shows gold purchases at three times the pace of the IMF series since 2022 (Figure 15). IMF data shows roughly 10 mn troy oz of annual purchases in the past four years by central banks, while the WGC series pegs this at over 30mn troy oz or over 1000 tonnes per year. While this does not change the trend or conclusions of this paper, it does suggest scope to magnify already significant implications.

Second, central bank purchases are driving upward pressure on gold prices. Central banks and official institutions as defined by the WGC have accounted for over 40% of the investment demand for gold since 2022, excluding demand for jewelry, technology and industrial purposes. ETF purchases which picked up meaningfully last year may themselves be piggy-backing off the consistent underlying bid from central banks. A simple regression that looks at central bank and ETF demand suggests that every 1 mn troy oz of purchases leads to a 1% increase in the gold price (Figure 16). We adopt this as a simple back-of-the-envelope heuristic, to be used in analysis below.

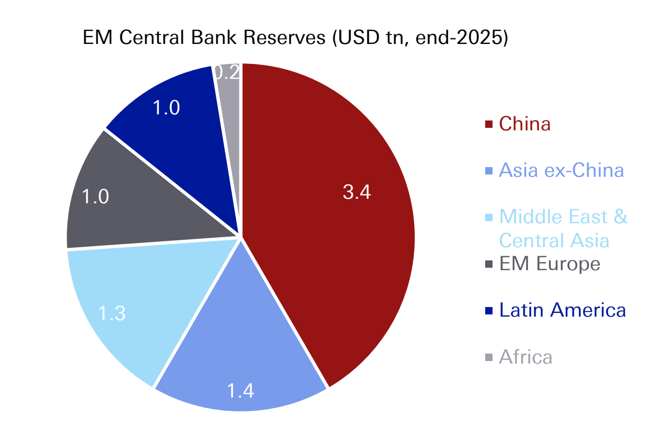
Third, the enormous build of foreign exchange reserves in emerging markets may now go into reverse. As discussed above, the dramatic rise of the dollar's share in global central bank reserves in the 1990s had almost everything to do with the sharp rise in EM reserves. There is a possibility that EM reserves may begin to decline from here. This would be motivated less by foreign investor capital exits or the need to defend currencies, but as countries in Asia and the Gulf draw on their savings to build strategic autonomy in defence and energy, which will require capital, imports and investment. While a majority of the roughly USD8tn in EM central bank reserves are held in China, a significant share are also in Asia ex-China and Middle East & Central Asia (Figure 17). Recent reports that the UAE has asked the US Treasury for a currency swap, suggests the need for USD liquidity. Indeed, the Gulf may turn to their savings not just to tide over the effects of the ongoing war, but for rebuilding efforts, to address economic scarring and diversification needs, and ultimately to build greater domestic defence resilience.

Figure 16: Every 1mn troy oz of buying by central banks and ETFs leads to a 1% rise in gold prices



Source : Deutsche Bank, World Gold Council, IMF, Bloomberg Finance LP

Figure 17: The roughly USD 8tn in EM reserves is held mostly in Asia and the Middle East



Source : Deutsche Bank, IMF

The first two forces are already underway. EM central banks have been actively buying gold and prices have been rising. The third force of active reduction in US holdings is yet to begin, but could be very significant. All three drivers could be at play together, suggesting there is more to go in gold's rise and the dollar's decline as a share of global reserves. Where might they end up?

What share of reserves should gold be?

The combination of gold purchases and price rises has already pushed gold's share of global central bank reserves to around 30%. Where might we go from here?

There is likely still a long way to go. We begin with the key observation that before the 1990s, gold's share of reserves fluctuated between 40-70% of total reserves as illustrated in Figure 18. "The return of history" we described above should make getting to the lower bound of this range - or 40% for gold's share - a reasonable initial target. If the world is going back to looking like it did before the 1990s - with geopolitical competition between superpowers, high inflation, and less universal support for free trade - then it may make sense to expect gold's share of reserves to also go back to similar levels. While before the 1990s, emerging market central banks held very little total reserves, today they are the dominant holders of reserves. And whilst the share of gold in advanced economies reached 34% at the end of 2025, or fairly close to the 40% level, it was only 16% in emerging markets. If the dominant trends of central bank purchases and USD reserve sales are taking place in emerging markets, then there could still be a long way to go: from 16% to 40%.

Figure 18: A complete "return of history" should take us back to at least a 40% share for gold in global reserves - EM has relatively more room to catch up

Source : Deutsche Bank, Eichengreen et al (2014), Bloomberg Finance LP, IMF

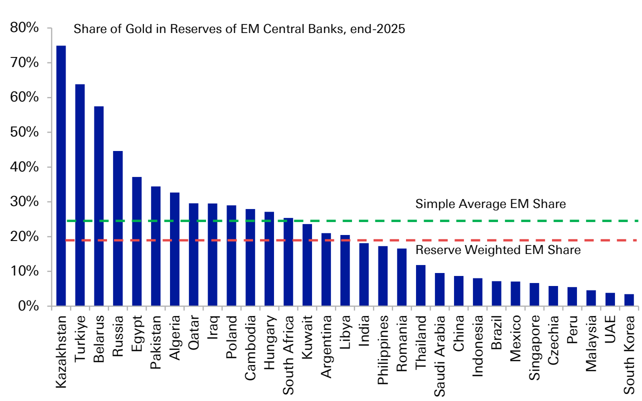
There is a lot of heterogeneity across EM in gold's share in reserves. While EM central banks as a whole held 16% of reserves in gold (end-2025), this masks large variation. The average has a downward bias driven by the largest reserves holder - China: PBOC holds only 9% of reserves in gold. Clearly, the scope for increase driven by China is huge. Kazakhstan and Turkiye had over 60% of their reserves in gold, Russia and Egypt near 40%, Poland and Hungary had reached nearly 30%, while traditional US allies like South Korea and UAE had under 5% in gold.

Academic literature explains this heterogeneity through both an economic and geopolitical lens. Arslanalp, Eichengreen and Simpson-Bell (2025) argue that "central banks operating floating exchange rates hold more gold, consistent with the presumption that they have less need to use their reserves in foreign exchange market intervention." Indeed, gold holdings are less effective in helping central banks defend currencies against large scale capital flows. Thus, an increasing share of gold may reflect less concern amongst EM central banks about sudden stops or withdrawals of capital, especially with foreign ownership at lower levels than in the past. More interestingly, they find that "geopolitical alignment with the United States, proxied by the existence of a defence pact, increases dollar reserves. An interpretation is that governments grateful for US military presence encourage their central banks to hold dollars as a show of good faith."

We also find that EM countries with closer defence ties to the Western bloc hold less gold in reserves, while EM countries with closer ties to China and Russia hold more gold. We draw on an analytical metric first used in our Global South framework to measure military alignment. We look at SIPRI data on arms imports for all emerging market countries over the last 10 years and calculate the proportion of arms imports from the "Eastern bloc", which we define as including China and Russia, versus the "Western bloc" which includes the US, Israel, Europe, and South Korea. We split EM countries into two groups: those that import more than a third of arms from the "Eastern bloc", and those that have limited defence integration with China and Russia. As Figure 20 illustrates, EM countries in the former group have double the share of gold in reserves than the latter group.

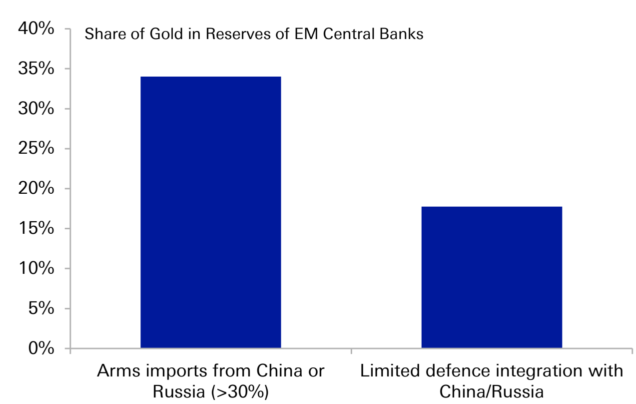
The implication of this analysis is that should more countries diversify defence dependence away from the US, this would be consistent with a reduced share of USD and a greater share of gold in reserves. The US has actively pressured major allies like NATO, South Korea and Japan to take on more ownership for their defence. And we have written about how the US-Iran conflict has challenged the US security umbrella over the Gulf and thereby support for the petrodollar and dollar savings. And while the ultimate geopolitical equilibrium remains to be seen, we think it could lead to some diversification of Gulf ties away from the US, and an acceleration in the localization of defence capabilities.

Figure 19: There is significant heterogeneity across EM in the share of gold in reserves



Source : Deutsche Bank, Bloomberg Finance LP

Figure 20: EM countries with more defence integration with China & Russia hold more gold in reserves



Source : Deutsche Bank, Bloomberg Finance LP, SIPRI

What might this mean for gold prices?

We simulate a range of different outcomes for gold prices depending on the level of FX reserves EM central banks end up with, and the share of gold in reserves they target.

- For total FX reserves, we consider four potential scenarios: a steady state for EM FX reserves of USD8tn (similar to current levels); and then three alternatives: a drastic decline in EM FX reserves to USD2.5tn, a decline to USD5tn, or an increase in EM FX reserves to USD10tn.
- For the share of gold in reserves, we look at a range of different shares rising from 15% to 40%. As discussed earlier, a 40% gold share could be a reasonable target for a "return to history" scenario, while 15-20% is where EM central banks are as a whole today.
- We use our earlier heuristic that every 1mn troy oz of purchases (sales) drives a 1% increase (decrease) in the gold price for our analysis. Our simulation is therefore dynamic, as we assume gold purchases by central banks influence the gold share through their impact on the prices as well. The rise in gold's share captures the blended impact of both prices and volumes.

The table in Figure 21 shares our simulation results. Prices shaded in green are above current gold prices (at the time of writing), while prices shaded in red are below. The numbers are not intended to be forecasts (found here), but to illustrate the range of potential price impacts from different combinations of EM central bank behaviour.

Figure 21: We simulate a range of different gold price outcomes depending on where EM central bank FX reserves end up, and the share of gold they target

EM FX Reserves (USD tn)	Share of Gold in EM Total Reserves (%)					
	15%	20%	25%	30%	35%	40%
Big Decline in EM Reserves (USD 2.5bn)	1700	2200	2700	3300	3900	4600
Decline in EM FX Reserves (USD 5tn)	2800	3700	4600	5600	6700	8000
Stable EM FX Reserves (USD 8tn)	4000	5300	6700	8100	9800	11600
EM FX Reserves Rise (USD10tn)	4800	6300	8000	9800	11700	14000

Source : Deutsche Bank, Bloomberg Finance LP

As intuition would suggest, if EM central banks are targeting a rising share of gold in a steady or growing FX reserves environment, this would be the most bullish outcome for gold. But even in an environment where EM FX reserves decline to USD5tn, if central banks target an increase of gold's share to 40%, this could still be consistent with gold prices rising to near $8000 over the next five years. We walk through the calculation in the simulation to illustrate. At USD5tn in FX reserves, gold would need to be worth USD3.3tn for gold to be a 40% share. Getting to USD3.3tn in gold would be a function of central banks buying more gold, and that buying driving up prices. Assuming every 1mn troy oz of purchases drives up gold prices by 1%, if EM central banks build gold holdings to 417mn oz, or an additional

52mn oz, this would drive up prices to around $7977, which would put total gold valuations at around USD3.3tn. At the current pace of gold purchases of roughly 10mn troy oz per year by central banks (based on IMF central bank only data series), this would be consistent with five more years of gold buying. In other words, EM central banks could push gold prices to $8000 over the next five years even in a declining FX reserves environment.

In the extreme case of EM FX reserves falling to USD2.5tn, the current volume of gold held by EM central banks at current prices (USD1.7tn), would already give gold a 40% share. There is thus no upside to gold projected in the first row of the table.

This simulation also helps us understand the price action of gold in March 2026 during the Iran war. IMF reserves data for the month of March showed overall EM central banks *sold* gold, led by Turkiye. These gold sales are likely a function of the fact that Turkiye has a high share of gold in reserves at over 60% which were therefore leveraged for liquidity. As our colleagues note, net FX intervention by the CBT reached USD23bn in the last two weeks of March, and the CBT mobilised around USD 20bn worth of gold — USD 11.1bn via gold-FX swaps and USD 8.2bn through outright gold sales. While acute, this is not likely to be representative of a broader trend in EM. Countries from Poland to Kazakhstan were still buying gold in March according to IMF data, and China's PBOC data also shows buying at the fastest pace in a year. Most EM countries have gold shares at far lower levels than Turkiye with significant scope to raise them.

If more of the rise of gold's share for EM central banks is achieved through the drawdown of FX reserves, this will be less bullish for gold. If, however, EM FX reserves are stable or fall more gradually, and central banks engage in active buying of gold to raise their gold shares, this will be more bullish for gold.

What does this suggest about the monetary order to come?

As this paper has discussed, central banks in emerging markets have been actively adding to their gold holdings. The rationale for this is mostly seen as being about diversifying official savings into a physical long-lived asset, that can be held beyond the reach of sanctions, and which is likely to hold value better than fiat currencies amidst greater fiscal and inflation risks.

But it is worth considering whether the build up in physical gold in emerging markets might be a precursor to a potential return of gold as an anchor for an alternative future monetary system. Since the collapse of the Bretton Woods, gold has not had a formal role in international monetary architecture. But history has long alternated between periods of fiat and physical-backed money. It would be consistent with - not counter to - history, to expect gold to return at some stage.

The US backed the dollar with gold when it created the post-war monetary architecture of Bretton Woods. It would thus be intuitive to expect any effort by other countries looking to create a bigger role for their currencies in payments and savings to also turn to gold. Gold has been part of monetary orders for over 2500 years and is not anyone's liability. And while production does expand supply - with above-ground gold stocks growing at around 2% a year this century - this is less than the growth in most countries' fiscal deficits. For countries in the Global South, where economic regimes, rule of law, and capital account openness, may be less well understood by global corporates and investors, backing payment currencies with a share of physical gold could be an important trust-building mechanism.

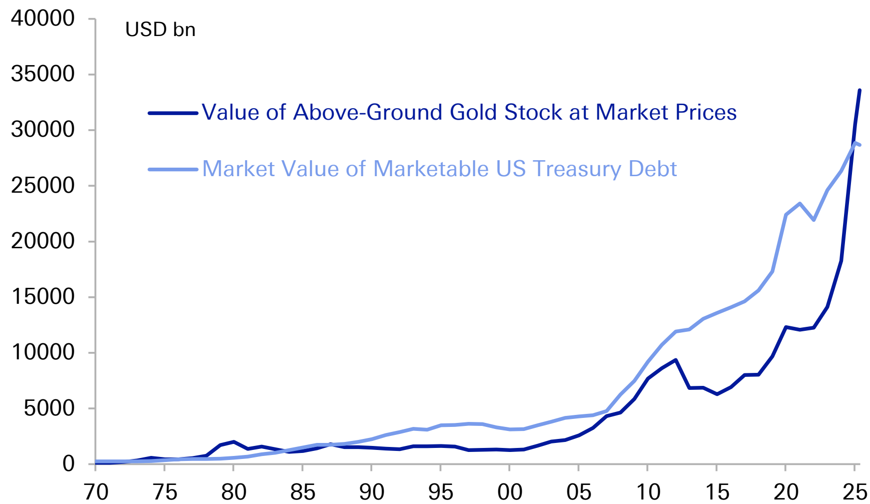
While far from formalized as policy, there are pockets of discussion around gold playing a role in backing a future BRICS currency. A paper from OMFIF, an independent think tank for central banking, economic policy and public investment, noted that the “BRICS are exploring the creation of a common currency that would be pegged partly to gold and partly to a basket of their own currencies.” They noted that "by tokenizing gold reserves, each digital unit would be backed by tangible assets stored in secure vaults, with regular audits ensuring accountability.” They note that a "gold linked alternative BRICS payment system" could again give gold a "distinct role in payments." Media reports in late 2025 indeed noted the pilot release of a BRICS Unit backed 40% by physical gold and 60% by an equal split of CNY, RUB, INR, ZAR, and BRL fiat currencies. We would stress that the "Unit" does not appear to be formal policy by the BRICS and thus should be considered more a thought experiment at this stage. But the momentum for creating alternative payment rails that facilitate local currency payments in the Global South are well underway. For instance, Project mBridge has already reached minimum viability for payments and involves central banks in China, Saudi Arabia, UAE, Hong Kong, and Thailand settling cross-border trade via wholesale CBDCs over the blockchain.

The economic clout of China and the Global South in PPP-adjusted GDP already exceeds 50% of the world today, and a majority of global trade flow growth is happening down these corridors. Payment innovations in this part of the world should thus be paid heed. We have written in great detail (see here) how competition around payments is heating up and how this is deeply linked with reserve currency status through the linkages between invoicing and savings. The US has opted for stablecoins as its solution to ensuring dollar dominance, namely in the Global South where traditional USD payments mechanisms were at risk. We note that while US stablecoins would be backed by US T-bills, a BRICS currency of the future could well be backed in part by physical gold.

In sum, while EM central bank diversification into gold likely has much to do with preserving the value and accessibility of their foreign savings in a changing geopolitical climate, it may also - in the long-run - play a role in anchoring a monetary order that builds independence from the dollar. There is of course a very long way to go. EM central banks as a whole still only hold half the physical gold of advanced economy central banks. But there is a world where gold returns to the centre of a future monetary system with different leaders.

To conclude, we find it notable that the value of above-ground gold exceeded the total value of marketable US Treasury debt last year for the first time in 40 years (Figure 22 overleaf**). In other words, gold is now a bigger asset class than the world's main safe asset. The return of history is here.**

Figure 22: The value of above-ground gold exceeded the marketable value of US Treasuries last year for the first time in 40 years



Source : Deutsche Bank, St Louis Fed, Bloomberg Finance LP, US Geological Survey

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This paper builds on a suite of our thematic work for the Deutsche Bank Research Institute. The future of gold and the dollar in reserves will be fundamentally informed by the growing importance of the Global South, the competition and innovation in payments technology, and the evolution of some of the geopolitical foundations for dollar-based payments and savings behaviour like the petrodollar. Some key reference pieces from the Deutsche Bank Research Institute include:

The Global South: A strategic approach to the world's fourth bloc (May 2025)

What Iran means for the dollar: a perfect storm for the petrodollar (March 2026)

What do stablecoins mean for dollar dominance? (Sep 2025)

Geoeconomics 101: implications for the dollar, emerging markets and beyond (Jan 2026)

The shift in US leadership: building the euro for a new age (May 2025)

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OMFIF, Gold and the New World Disorder: Consequences of Central Banking Shifts, 2025-26 Perspectives

Appendix 1

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Deutsche Bank
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Q&A with Shreyas Gopal and Henry Allen

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Curious about the inspiration behind 'Rate Check'? We sat down with hosts Shreyas Gopal, FX Strategist and Henry Allen, Macro Strategist to discuss why now was the perfect time to launch Deutsche Bank Research's new podcast.

Q: What was the initial spark or idea that led to the creation of 'Rate Check,' and why did you feel that now was the perfect time to launch a new podcast focused on fixed income, FX, EM and commodity markets?
Shreyas: I've always found Deutsche Bank Research to be a very collaborative environment, and one where our best ideas are often first debated amongst, multiple different teams before they're presented to clients. The idea of Rate Check is to harness that collaboration and discuss our latest views in the format that's probably been the most frequently by clients – a podcast!

Q: Building on that, given the vast amount of financial news and analysis available today, what specific gap or need in the market do you believe 'Rate Check' will uniquely fill for its listeners?
Henry: The aim of this podcast is to be more markets-focused, contrasting with the more thematic approach in Research's other podcast Podzept. This focus allows us to provide targeted value, drawing on our DB Research experts who've covered these areas across several economic cycles. We're hoping to harness that experience, particularly where we believe widely-held market narratives might be missing the mark.

Q: Beyond simply explaining "what's moving markets and why," what do you hope listeners will do or take away after tuning into an episode of 'Rate Check'?
Shreyas: We're hoping that anyone with a markets focus will take something away every week, whether that's a short summary of what's been at the forefront of our recent client conversations, the key upcoming events, or something more detailed on a specific asset class from one of our experts. Oh, and we hope they subscribe, of course, so they never miss an episode!

Q&A with Shreyas Gopal and Henry Allen

Deutsche Bank
Research

M&A Spotlight

Megatrends supersede war fears

April 2026

Luke Templeman | Galina Pozdnyakova

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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Image: Microsoft Designer

Lingering effects from the war in Iran could impact the world economy

But it depends on how much energy infrastructure has been damaged

The IMF recently analysed the impact of the war in Iran on the global economy:

Current scenario
Global growth is projected to be 3.1% in 2026 and 3.2% in 2027, slower than its recent pace of about 3.4% in 2024–25, and to settle at about that rate in the medium term.

Adverse scenario
Larger and more persistent increases in energy prices. Global growth would slow to 2.5% in 2026, and inflation will reach 5.4%.

Severe scenario
More damage to energy infrastructure in the conflict region will cut global growth to only about 2% in 2026, while headline inflation would be just above 6% by 2027.

"The impact on emerging market and developing economies would be almost twice that on advanced economies."

Source: IMF, Deutsche Bank.

How 2026 could play out under our current assumptions about the world economy (DB forecasts)

S&P 500	↑	14.8%
Stoxx 600	↑	3.2%
10yr treasury yield	↑	4.45% (+20bps)
US GDP	=	2.4% Q4 yoy
Eurozone GDP	↓	0.4% Q4 yoy
UK GDP	↓	0.8% Q4 yoy
Crude oil (assuming the SoH reopens in May)	↓	$80 p/b

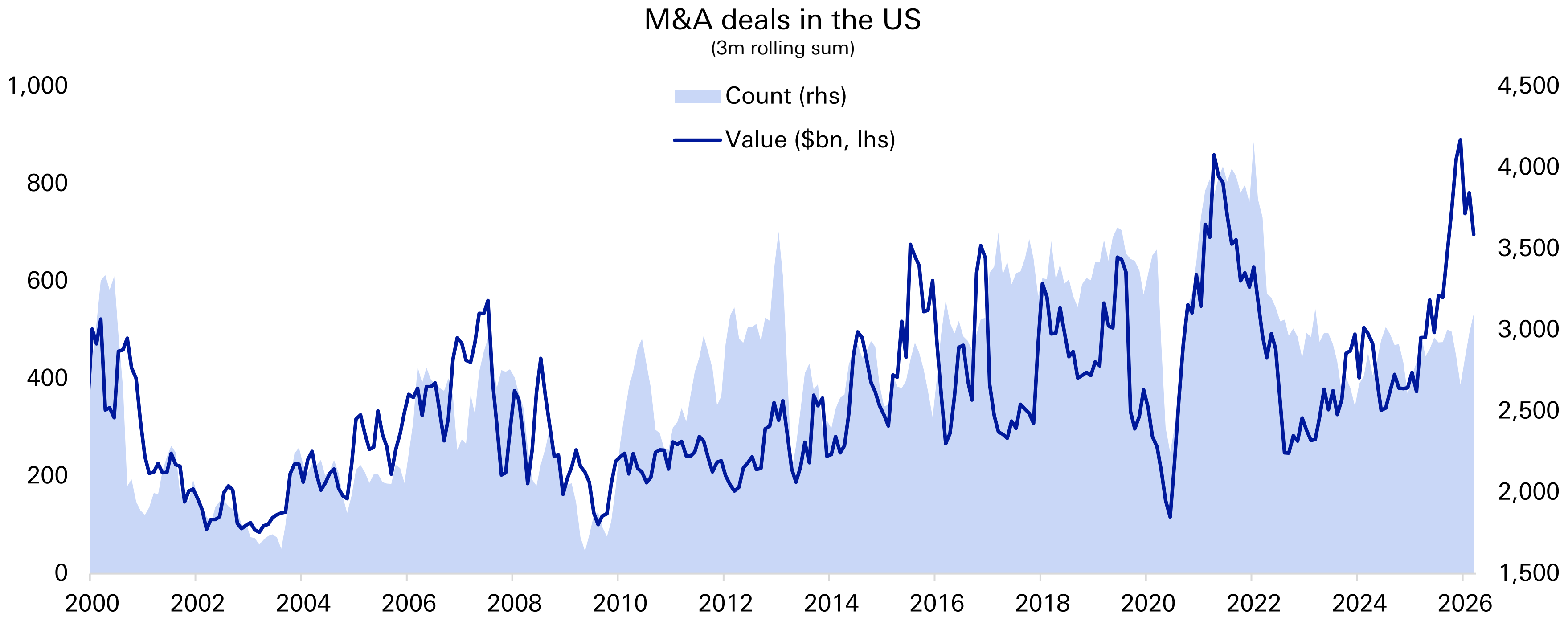
M&A themes in the shadow of the Iran war

1. Corporates are 'doing something big'	269 strategic deals in the US >$1bn in the trailing 12m as of Q1; exceeded 2021 peak
2. M&A as a growth driver	Organic revenue growth of ~5% in the US and ~4% in Europe still below prior averages – M&A being used to boost topline is a reversal of the trend in the 1st half of the 2020s
3. Asset efficiency	European divestments and carve-outs around their cyclical high, Strong trend up in US divestments
4. Defensive releveraging	US firms have delevered to 2015 levels. European firms around their multi-decade lows
5. Using M&A to hedge geopolitical risk	300% growth in US deals into Europe; 43% growth in European deals into the US
6. Operating through volatility	Focus on maintaining US profit margins at ~11%; Europe at ~8%
7. Executing without perfect visibility	20pp spread between CEO confidence and consumer sentiment in the US. Large deal execution continues unfazed despite consumers reporting their worst sentiment on record
8. Targets with autonomous supply chains	Q1 growth in M&A spread across sectors

Source: Bloomberg Finance LP, Dealogic, Deutsche Bank.

Q1 review: In the US, deal values fell after the exceptionally strong Q4 … but the number of deals is rising.

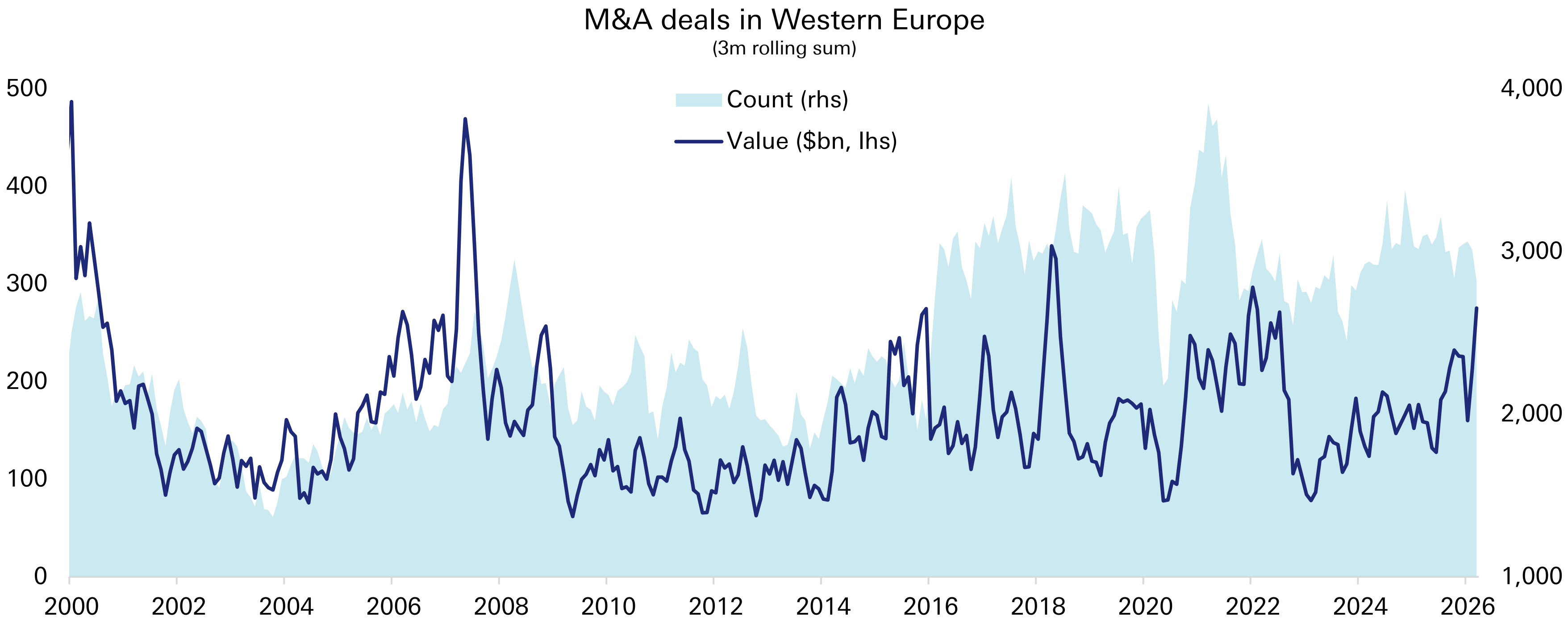
That points to a broadening M&A momentum beyond megadeals



Source: Dealogic, Deutsche Bank.

Q1 review: In Europe, large deals carried M&A despite some slowing in deal numbers

Corporates jumped into the market as peers announced deals and companies attempted to take advantage of rising dispersion and lower valuations in some quarters

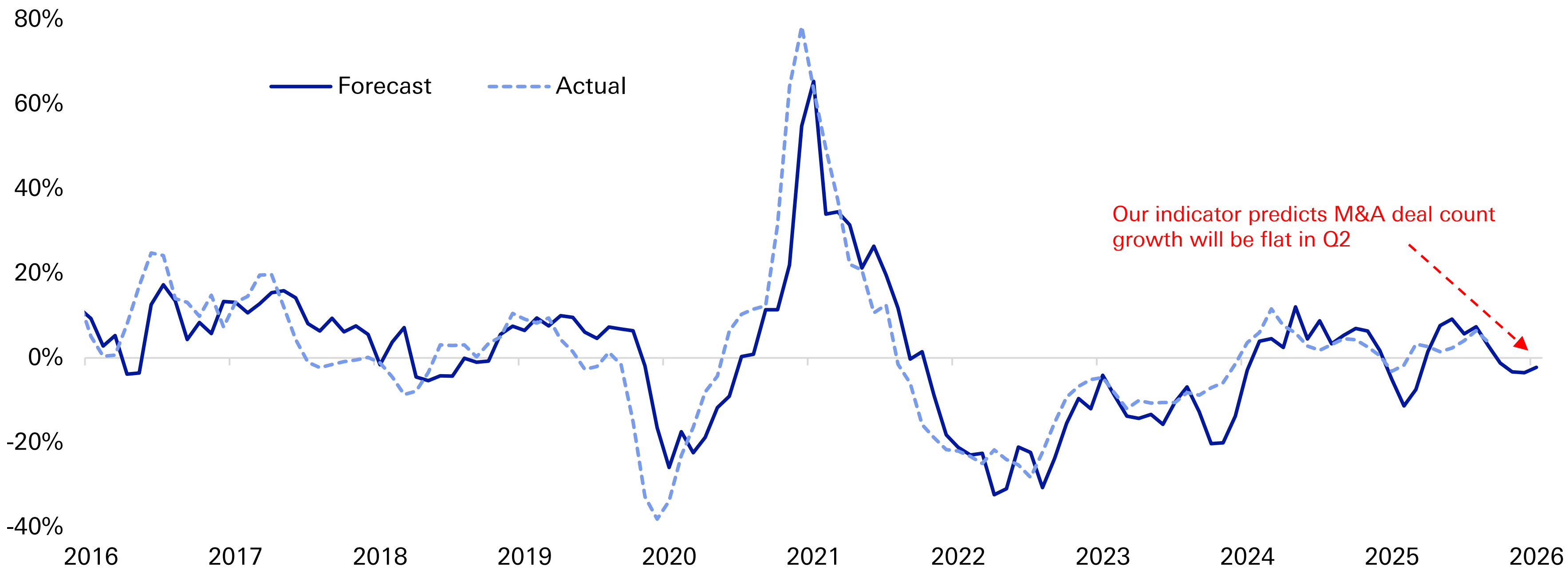


Source: Dealogic, Deutsche Bank.

Our M&A forecast – US

Our indicator for the US suggests that after a good Q1 for deal count growth, we could see a flat Q2 as economic data, especially sentiment and uncertainty indices, has dragged on our forecasts. H2 should then be a period of growth even if the mid-term elections provide some market choppiness.

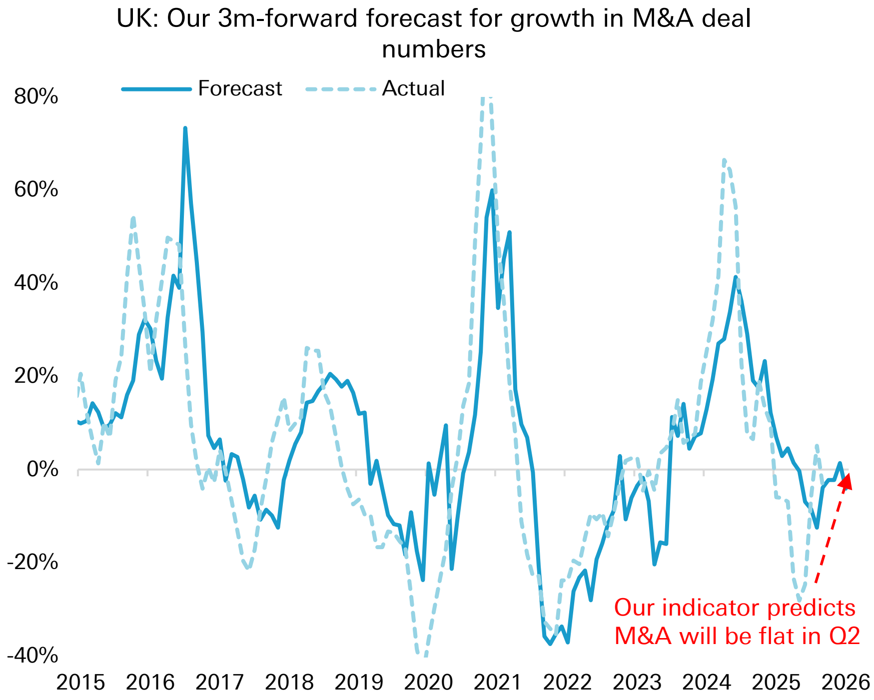
US: Our 3m-forward forecast for growth in M&A deal numbers

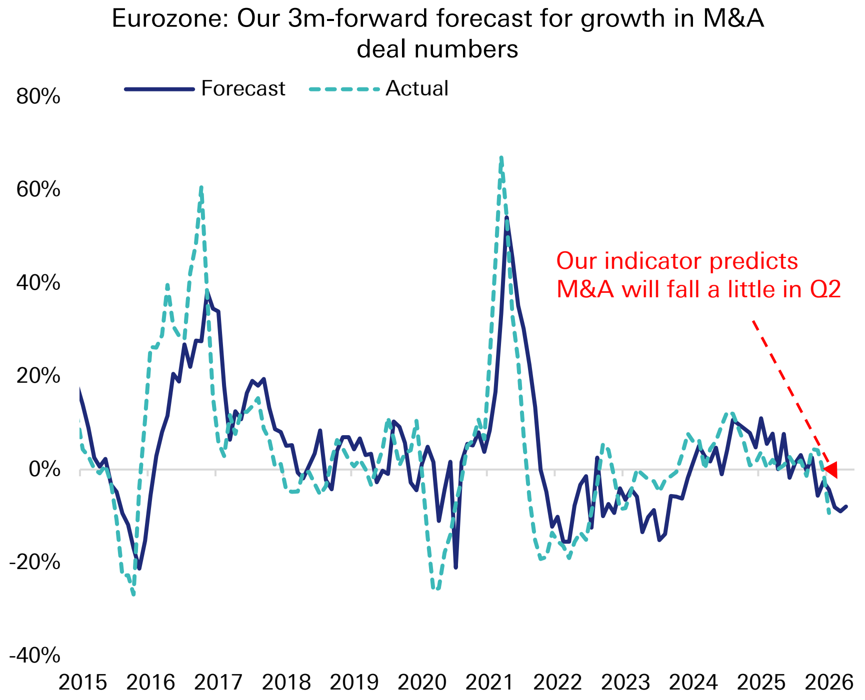


Source: Dealogic, Deutsche Bank.

Our M&A forecast – Europe & UK

We expect near-term UK dealmaking to be more resilient as Europe is weighed on by worsening market conditions. We then expect a recovery in H2





Source: Dealogic, Deutsche Bank.

Megatrends in global economies and markets are increasingly driving M&A

We recently published a brief outline of our quantitative model that analyses how key megatrends affect the S&P 500, GDP, inflation and interest rates here. We have applied this framework to quantitatively assess the effect on M&A markets.

Our quantitative megatrend model shows the impact of 6 trends on economies and markets

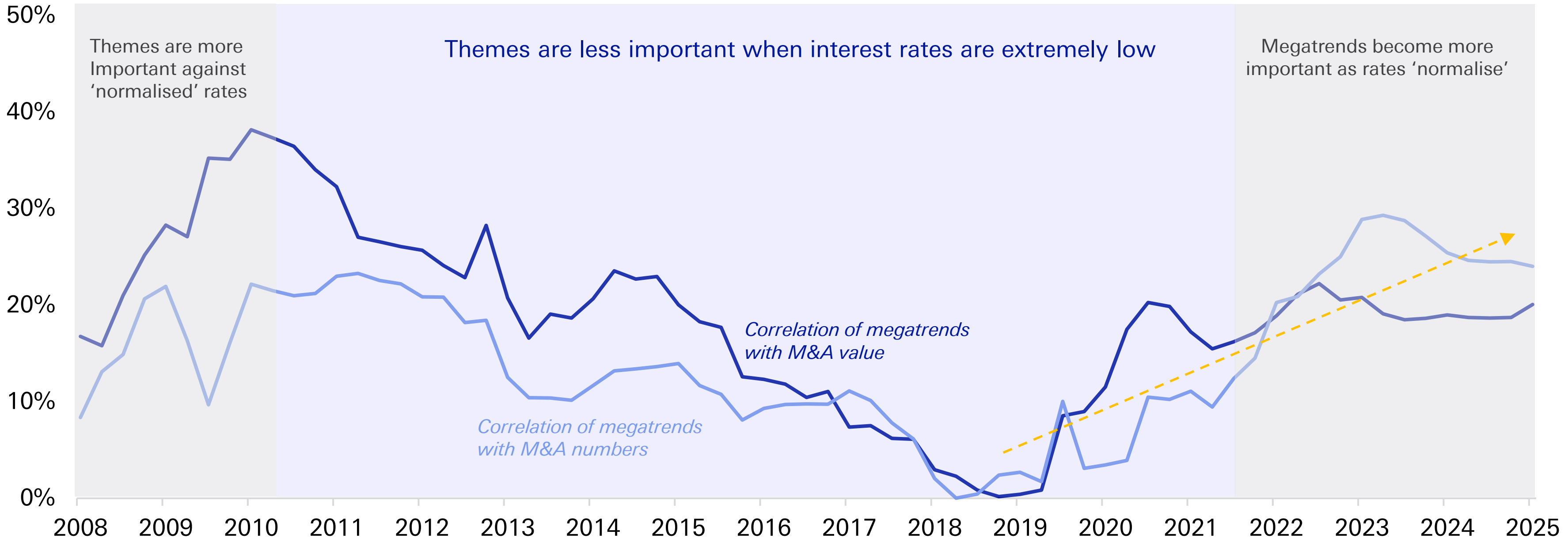
Source: Dealogic, Deutsche Bank.

Megatrends in global economies and markets are increasingly driving M&A cont.

The aggregate of the six megatrends has been increasingly important for US dealmaking after a slump in 2010s

M&A has become increasingly correlated with the growing impact of megatrends

Correlation between megatrend growth and M&A values and volumes (higher = more dealmaking growth attributed to thematic megatrends)

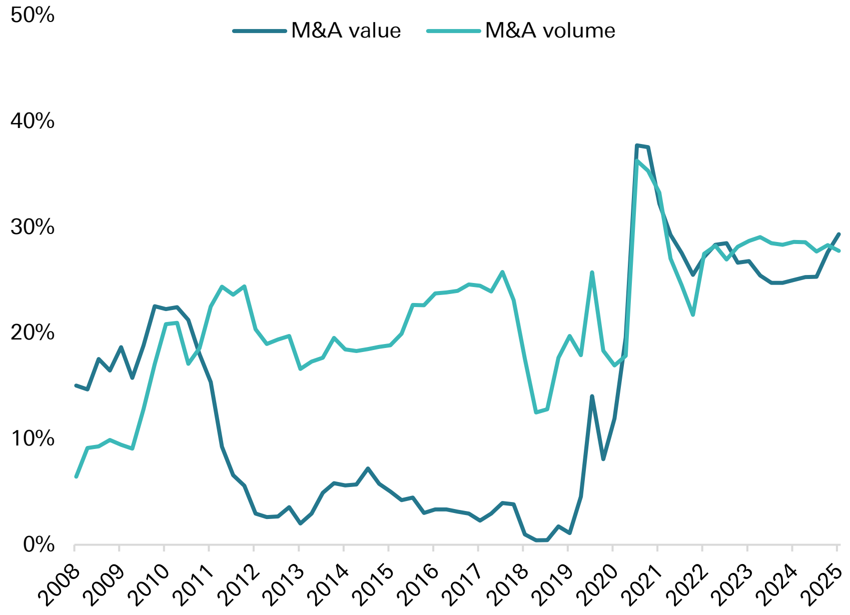


*Source: Dealogic, Bloomberg Finance LP, Haver Analytics, EIA, Deutsche Bank. * Note: rolling 10yr R^2*

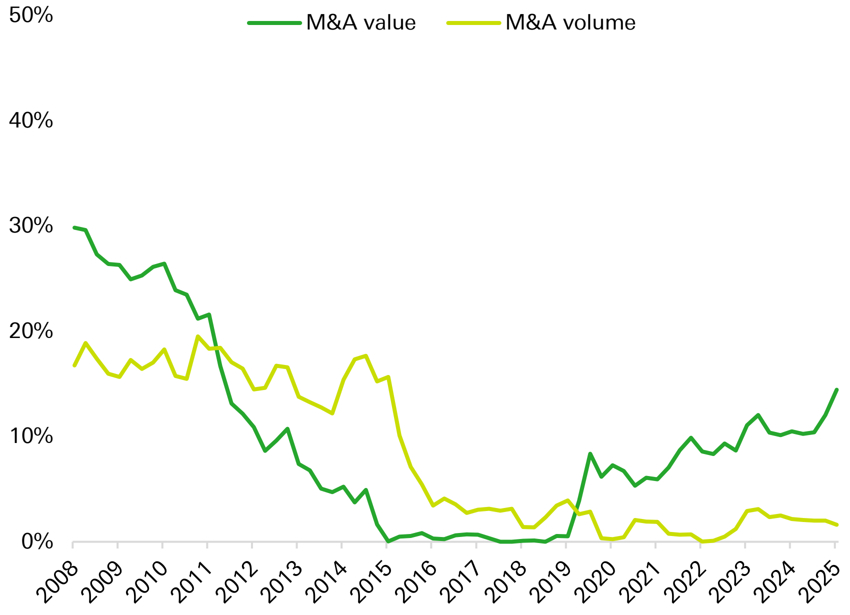
Two important megatrends for M&A are 'technology' and 'geopolitics & globalisation'. The link between these and M&A has been growing tighter.

M&A likes calm geopolitics, but it loves technological advancement. Large deals, in particular, have coincided with greater trade and globalisation in recent years.

Correlation between our **technology** megatrend indicator and M&A values and volumes *



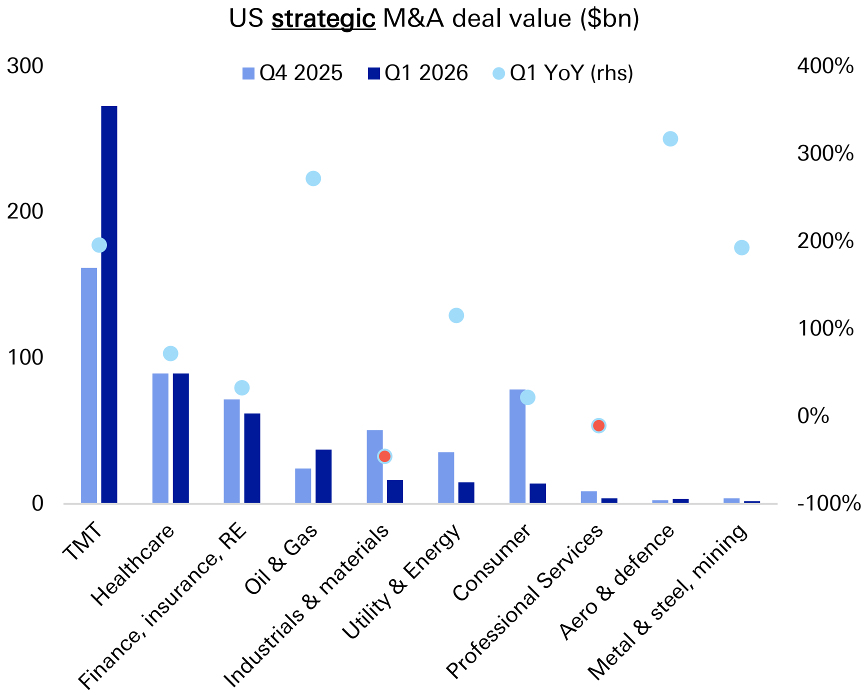
Correlation between our **geopolitics & globalisation** megatrend indicator and M&A values and volumes *

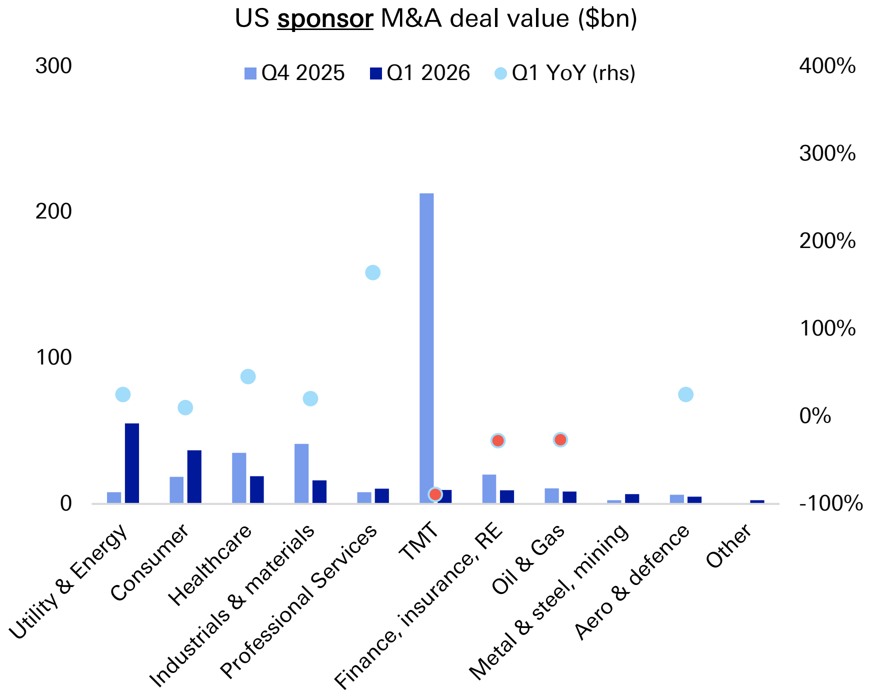


*Source: Dealogic, Bloomberg Finance LP, Haver Analytics, EIA, Deutsche Bank. * Note: rolling 10yr R^2*

Corporate buyers are ‘doing something big’ (our main theme of 2026 see here)

Demand for corporates to do transformational deals was strong even though it was a bit lower after the big Q4. The most notable drop was in sponsored TMT dealmaking amidst software jitters and the very strong tech dealmaking last year.

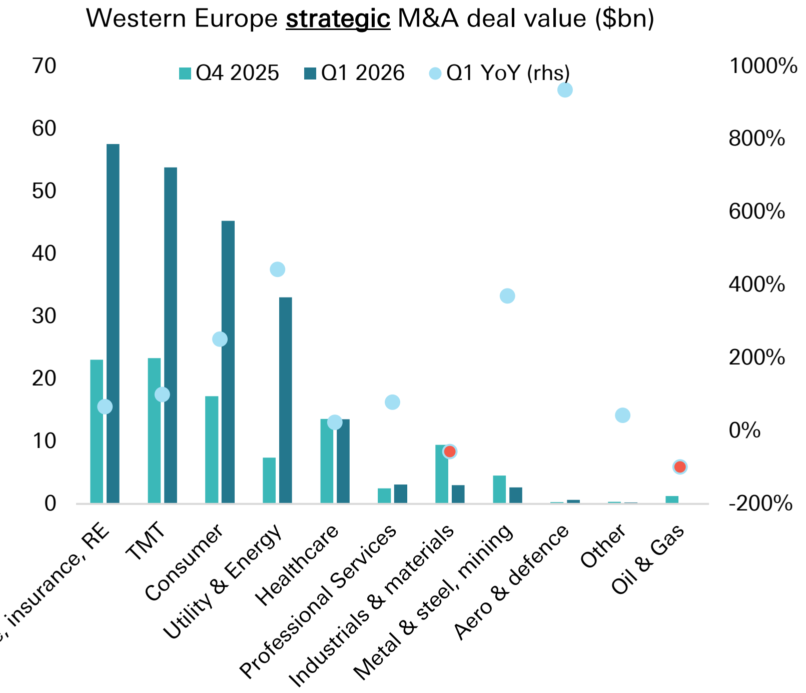


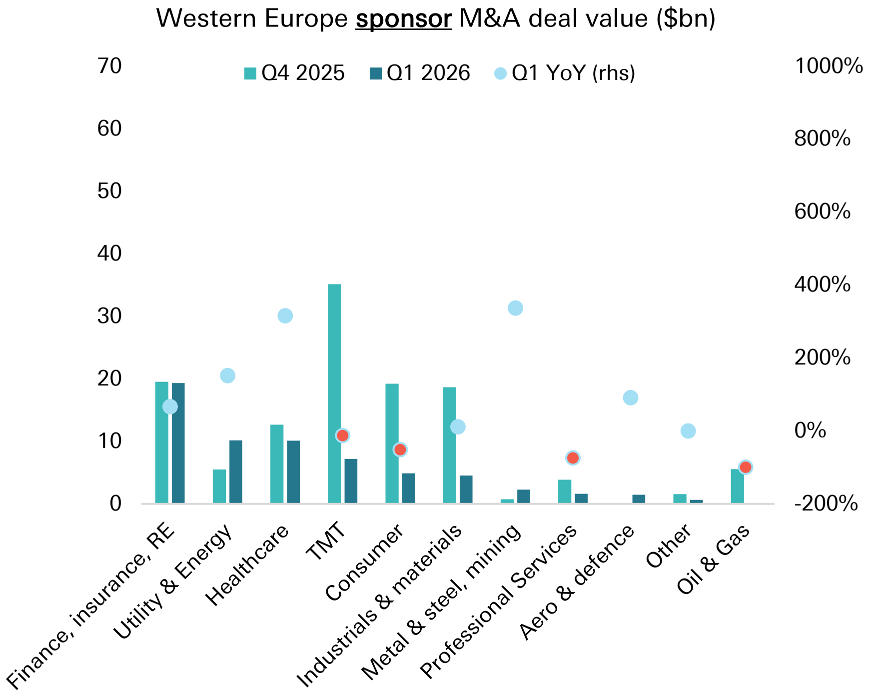


Source: Dealogic, Deutsche Bank.

European corporate buyers are also seeking acquisitions to beat out their peers

In Q1, Western Europe caught some of the residual momentum from the US – a few sectors in strategic M&A more than doubled last quarter. Sponsors, however, slowed down, especially in tech and economically-sensitive sectors.

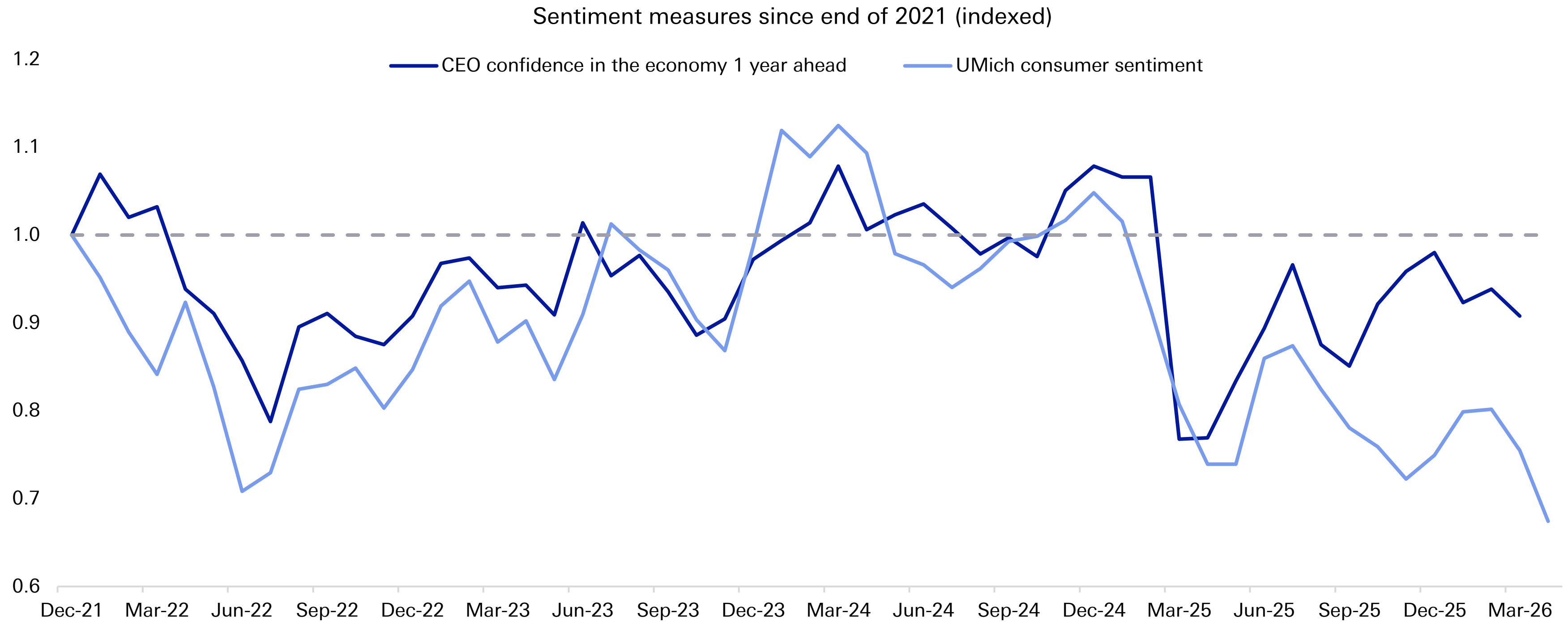




Source: Dealogic, Deutsche Bank.

CEO sentiment is higher than most people think …

There has been a divergence between consumer sentiment (most talked about) and business sentiment (not so talked about)

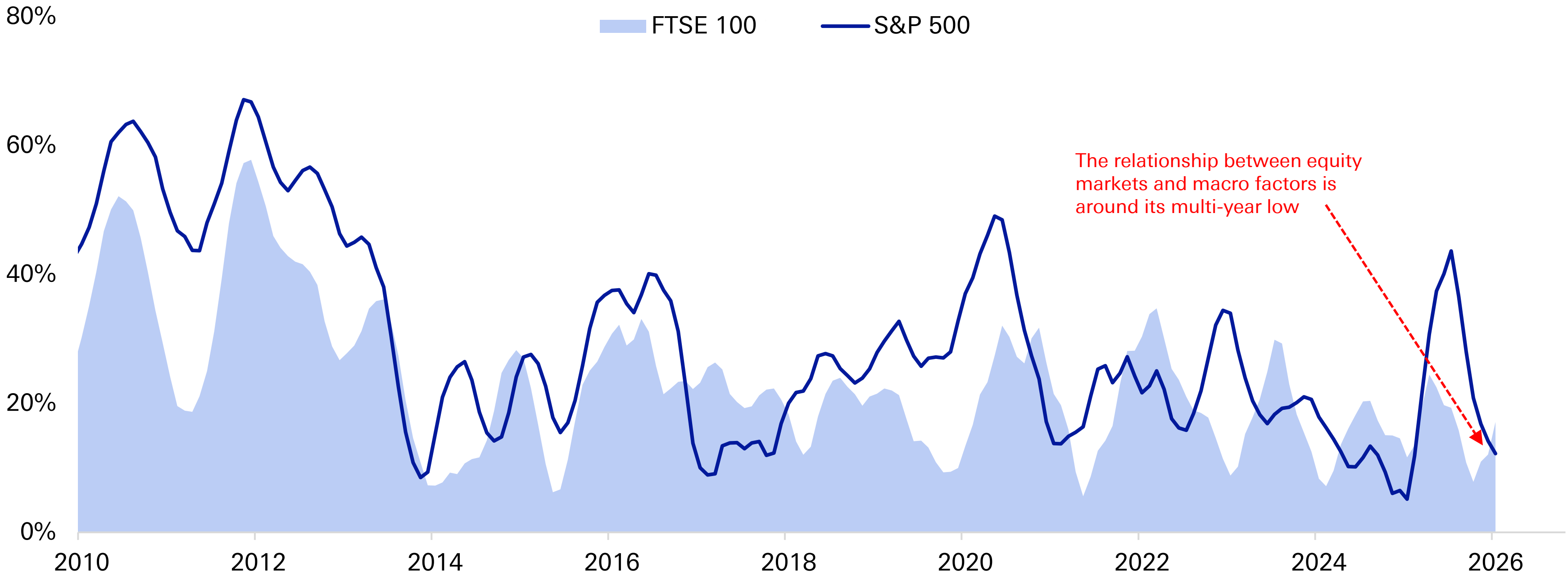


Source: Bloomberg Finance LP, Deutsche Bank.

… while the macro backdrop is less important than many expect

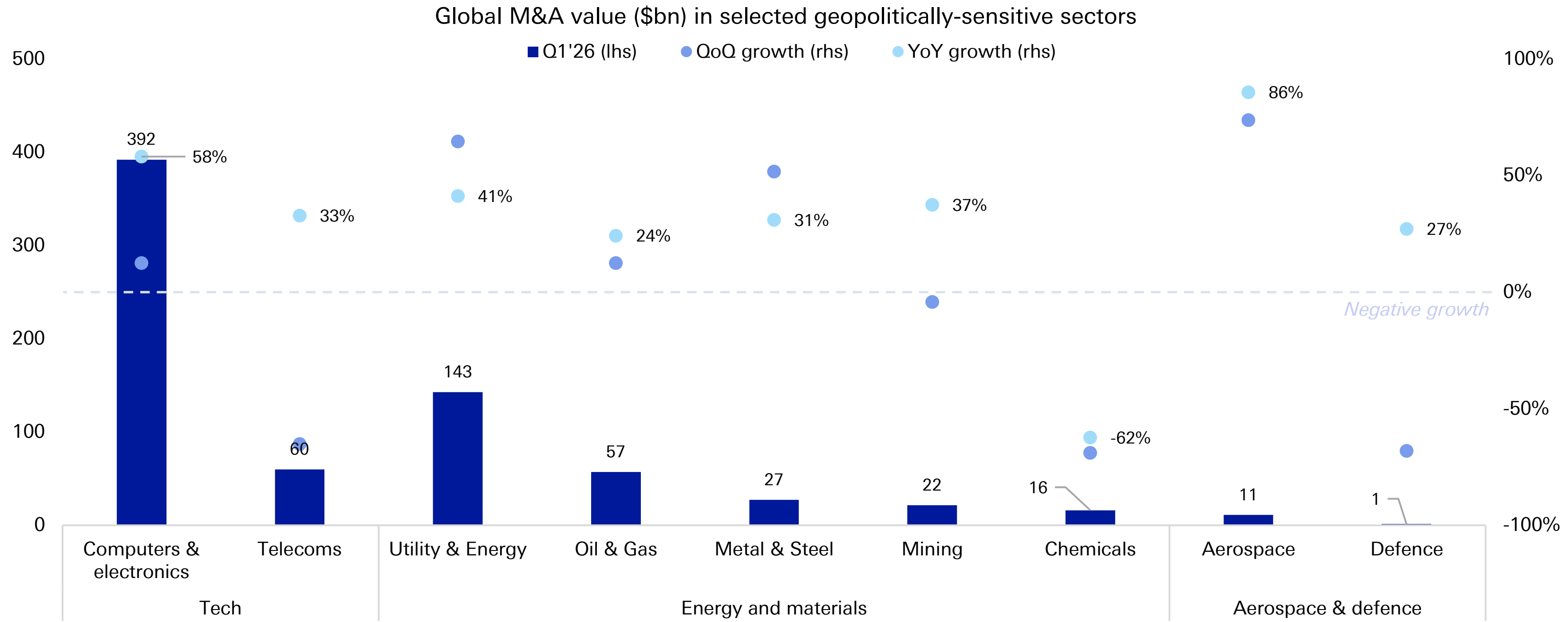
M&A markets closely follow broader stock markets which appear less concerned about the macro – we measure this by looking at the correlation of equity markets to several other markets that are highly-sensitive to macro movements.

Sensitivity of the S&P 500 and FTSE 100 to macro-driven assets (6m mov avg)



Source: Bloomberg Finance LP, Deutsche Bank.

Geopolitics and M&A: Globally, deals in sensitive sectors like energy and aerospace & defence did especially well in Q1

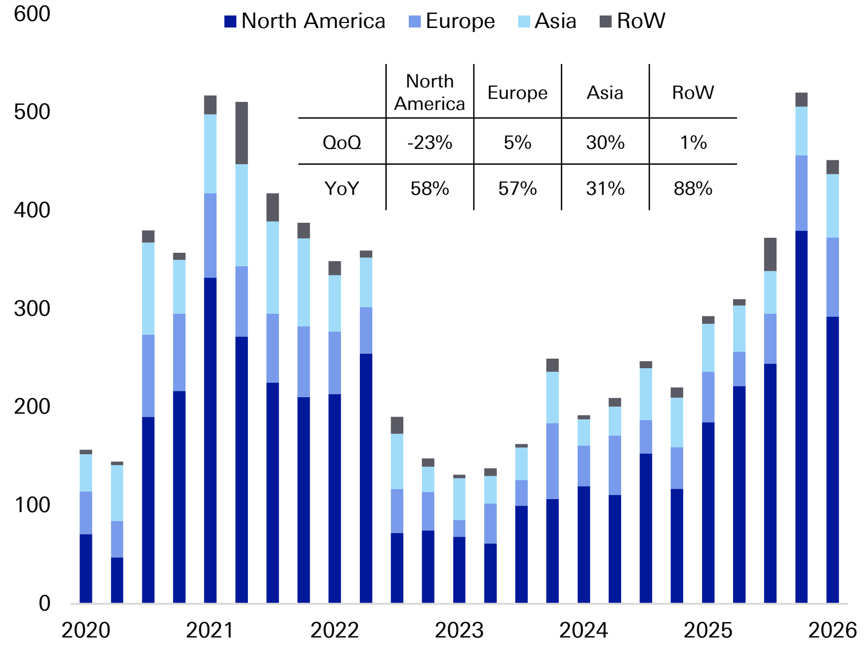


Source: Dealogic, Deutsche Bank.

Geopolitics and M&A: Growth in tech has been strong in Asia, but somewhat slow in North America and Europe after a very strong Q4.

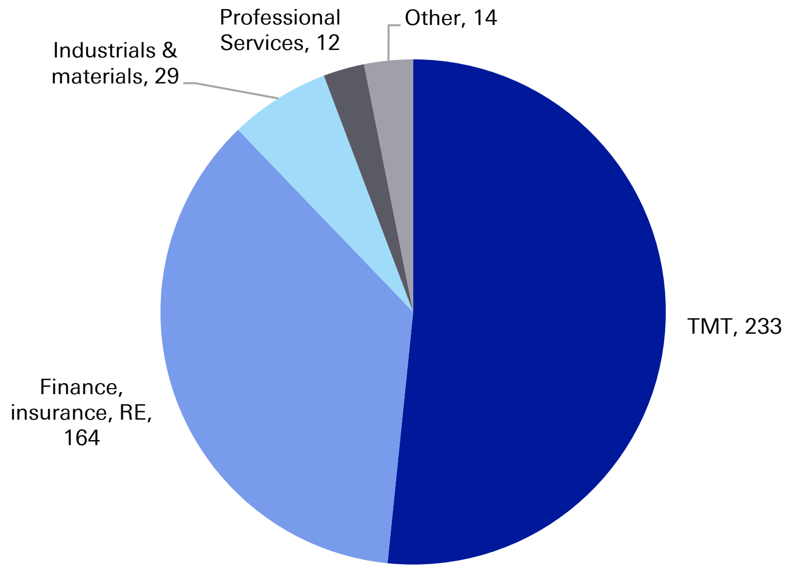
Deal values across all key regions were still up significantly YoY. Tech buyers and sponsors remain the key acquirors.

Global technology deals, by deal region ($bn)



	North America	Europe	Asia	RoW
QoQ	-23%	5%	30%	1%
YoY	58%	57%	31%	88%

Global TMT deals in Q1'26 ($bn) buyers' sector

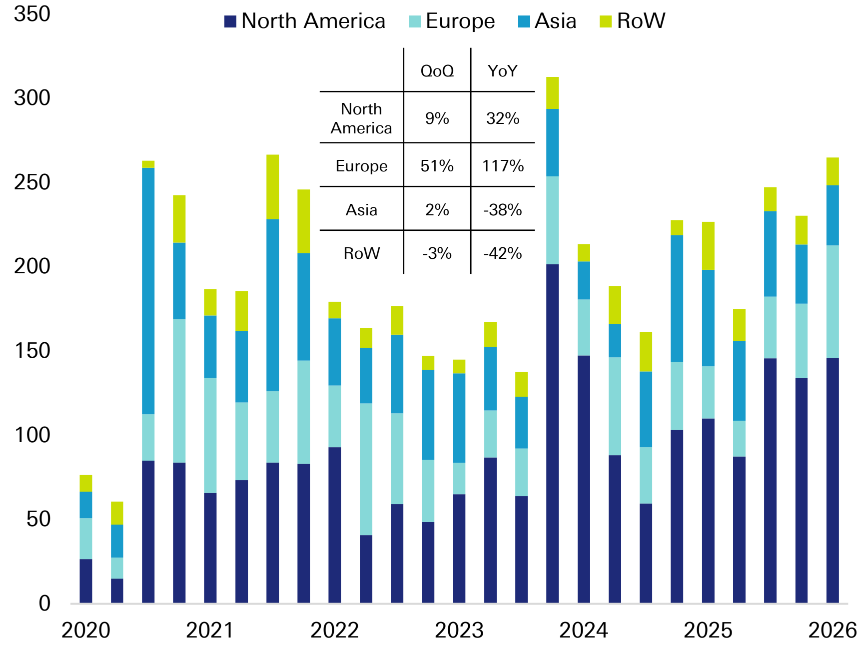


Source: Dealogic, Deutsche Bank.

Geopolitics and M&A: Energy and materials had a very strong Q1 in Europe, driven by sponsors

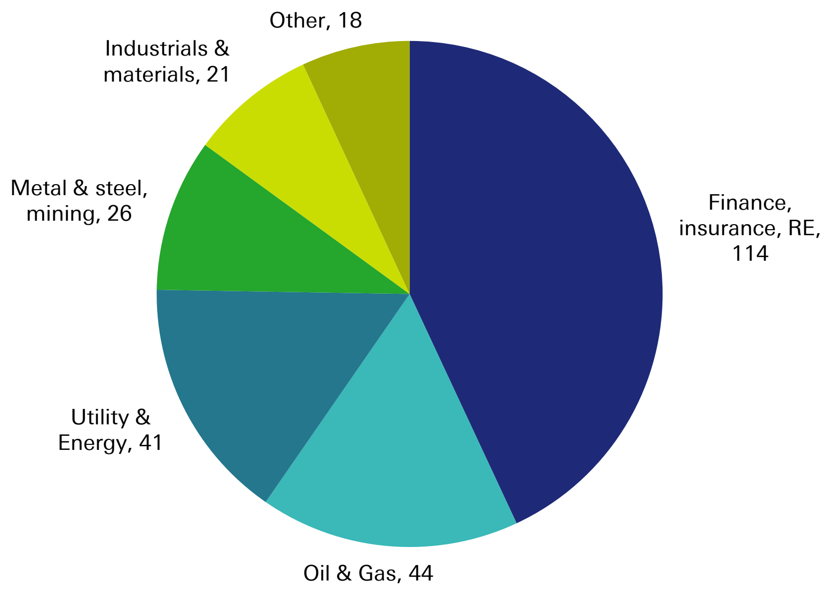
By contrast, growth in Asian dealmaking in these sectors has been lagging.

Global energy and materials deals, by deal region ($bn)



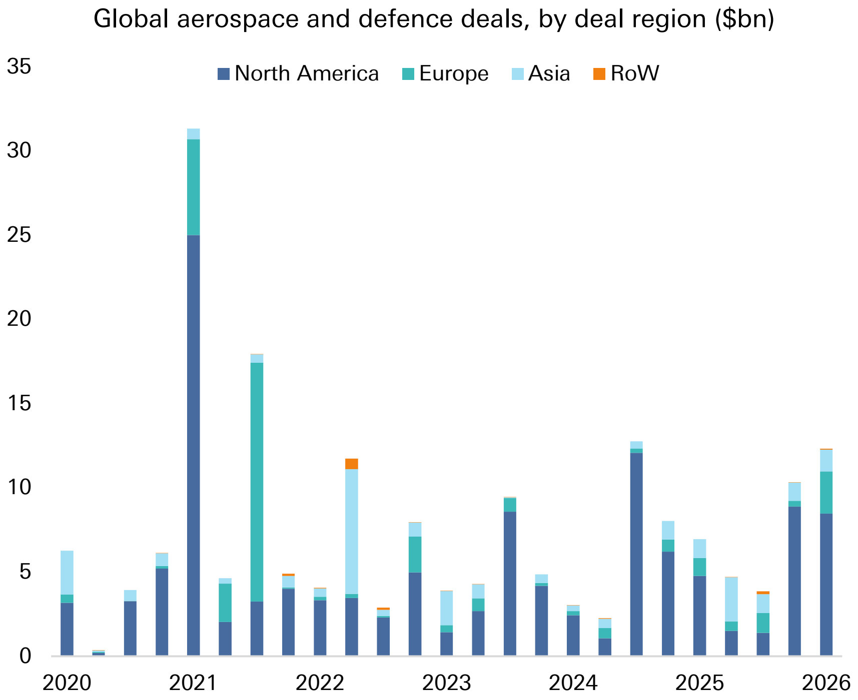
	QoQ	YoY
North America	9%	32%
Europe	51%	117%
Asia	2%	-38%
RoW	-3%	-42%

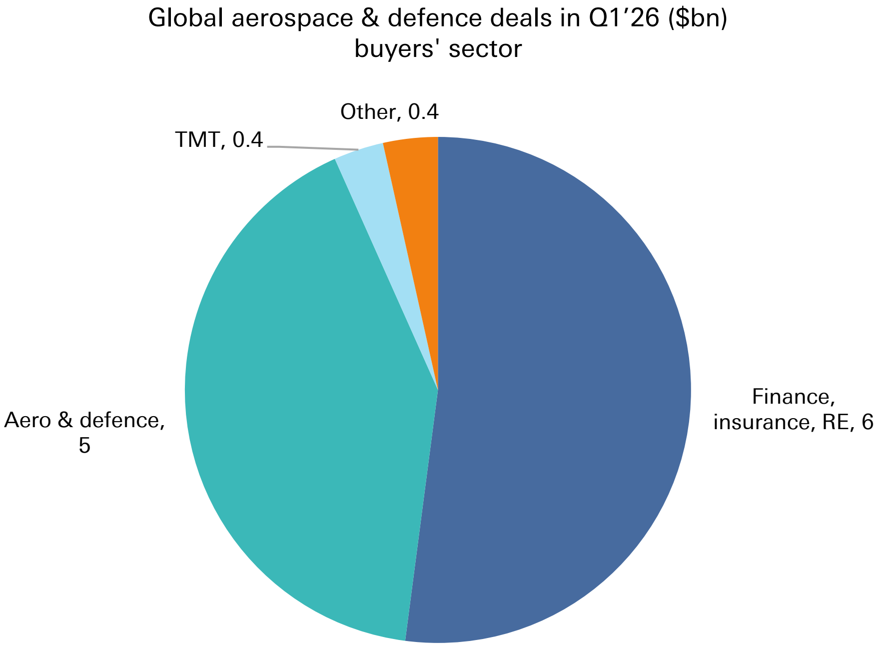
Global energy & materials deals in Q1'26 ($bn) buyers' sector



Source: Dealogic, Deutsche Bank.

Geopolitics and M&A: Aerospace & defence is still a nascent area of dealmaking, but growth was strong in Europe and especially from sponsors.



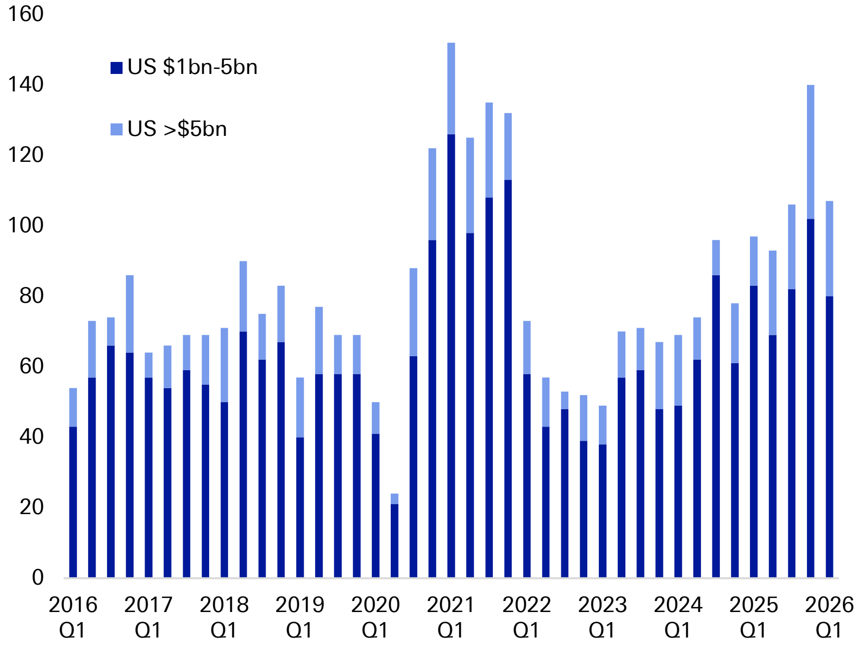


Source: Dealogic, Deutsche Bank.

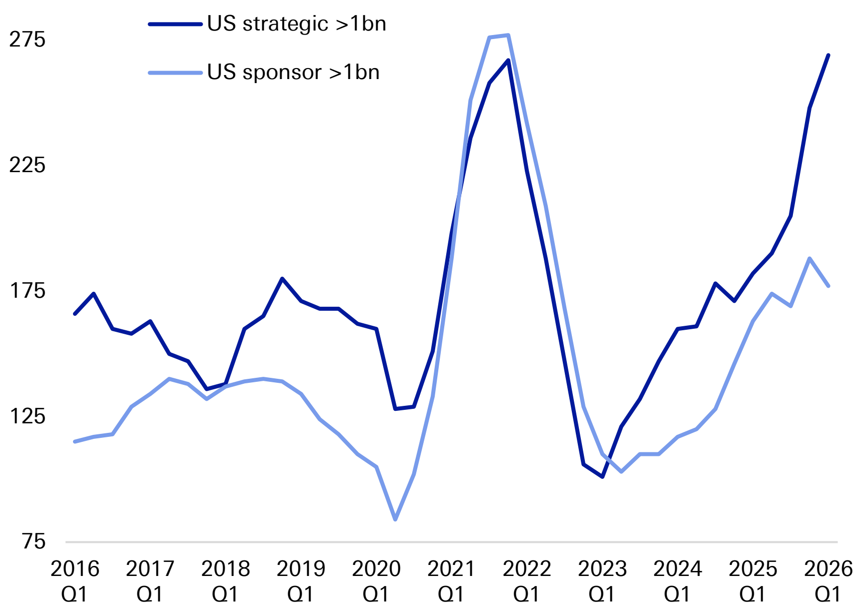
Megadeals – In the US, these have cooled from a very strong Q4 but remain in an upwards trend

Importantly, more and more megadeals are done by strategic buyers, with sponsors taking a break.

US, number of megadeals per quarter



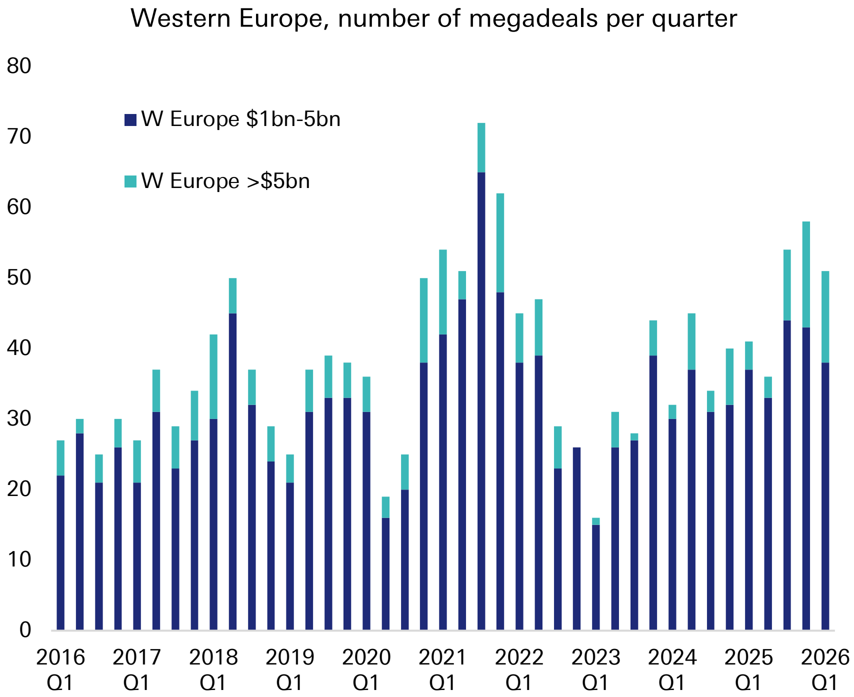
Deal count of large M&A by buyer (>$1bn, t12m)

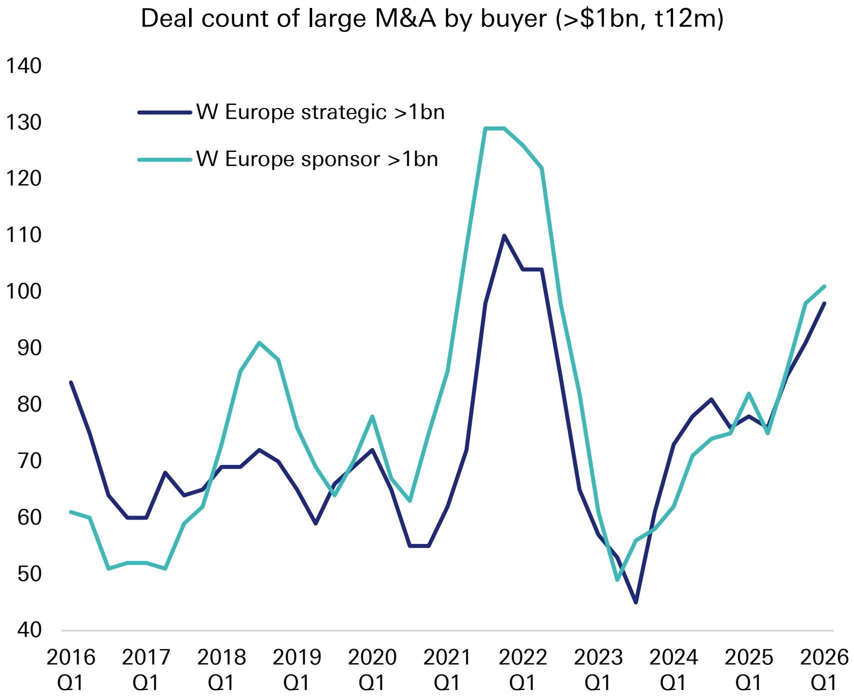


Source: Dealogic, Deutsche Bank.

It's a similar picture in Europe, with a third quarter in a row with a large number of +$5bn deals.

Both strategic and sponsor buyers are doing more +$1bn deals

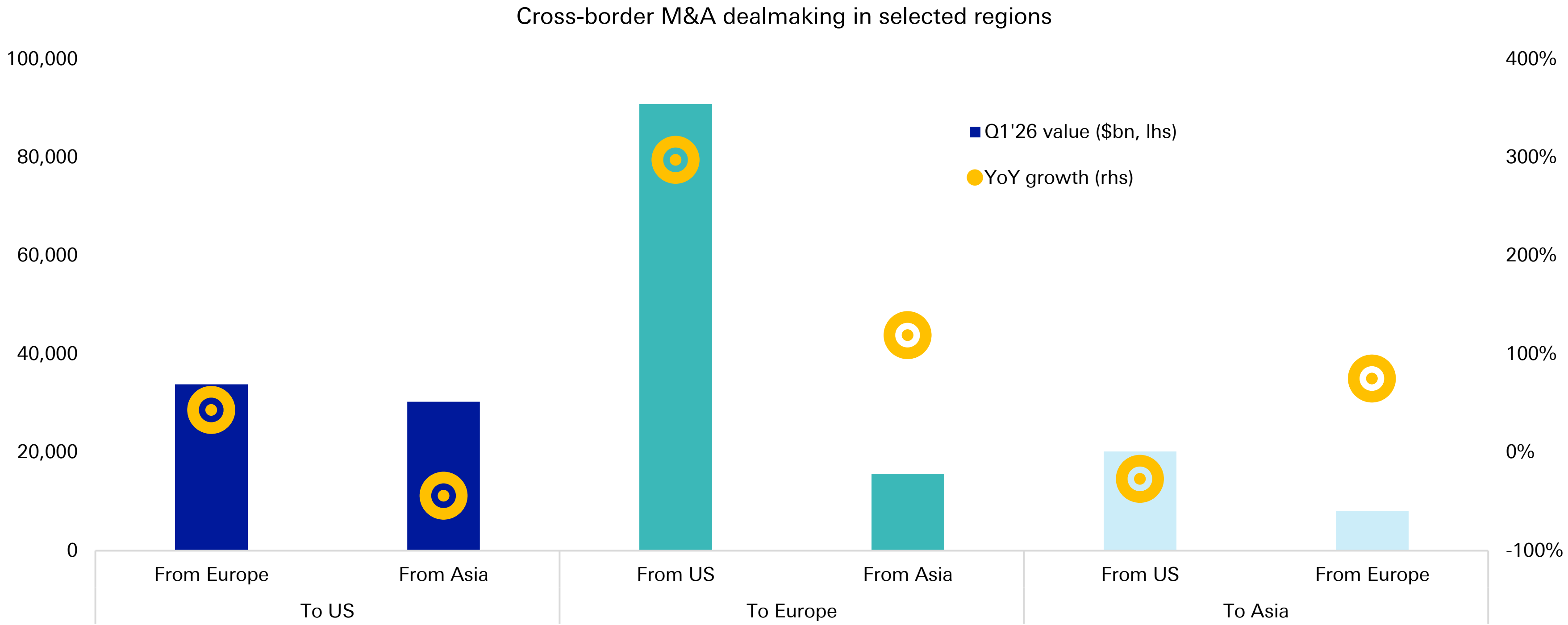




Source: Dealogic, Deutsche Bank.

Deal flow between continents – US and European companies are hedging geopolitical risk

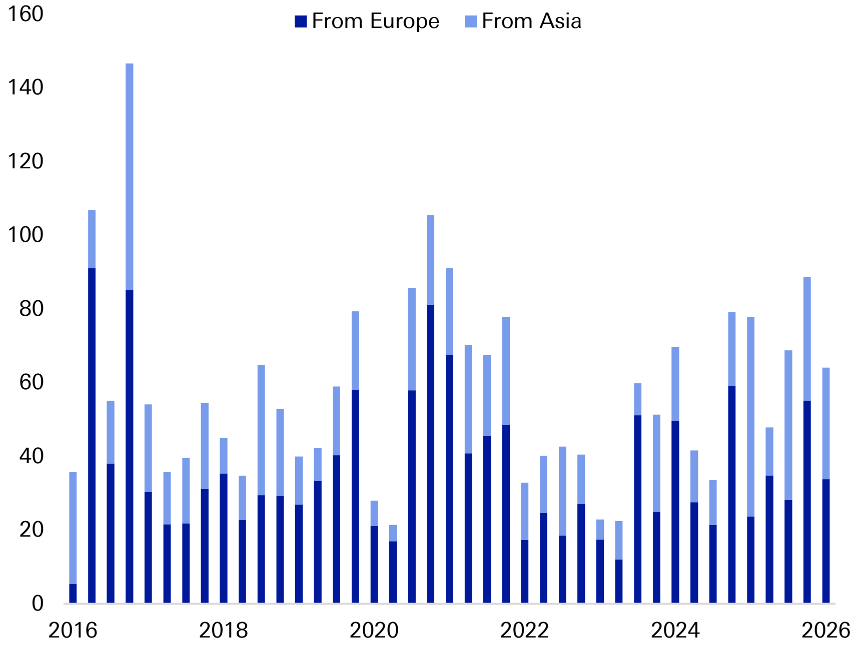
Europe has been the clear winner in terms of interest from abroad, especially the US. Dealmaking between the US and Asia has somewhat slowed.



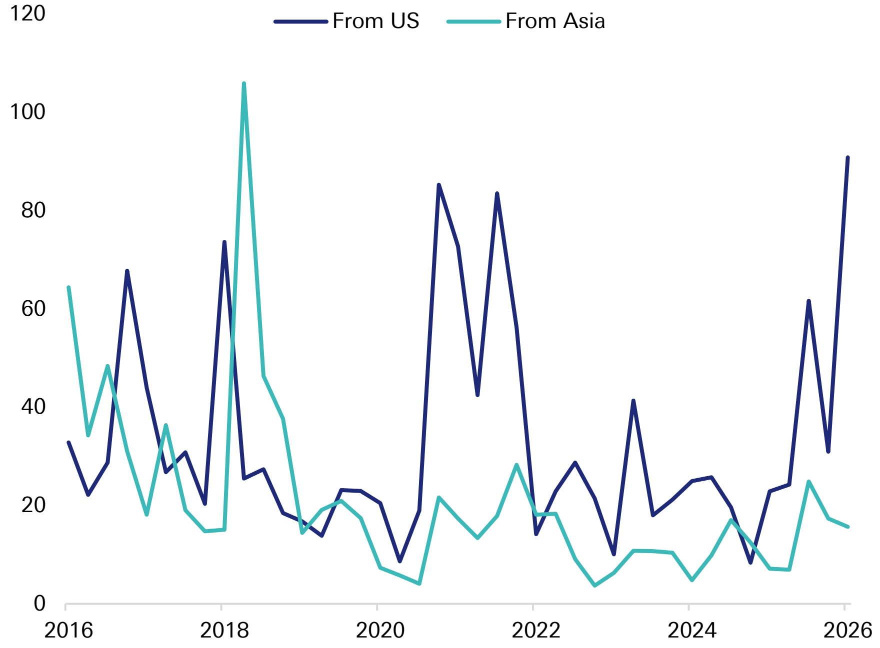
Source: Dealogic, Deutsche Bank.

Cross-border deals into Europe by US buyers have seen a stronger quarter than during the M&A boom of 2021. Dealmaking by Asian buyers has been steady in both the US and Europe.

Cross-border US M&A ($bn, quarterly)



Cross-border European M&A ($bn, quarterly)

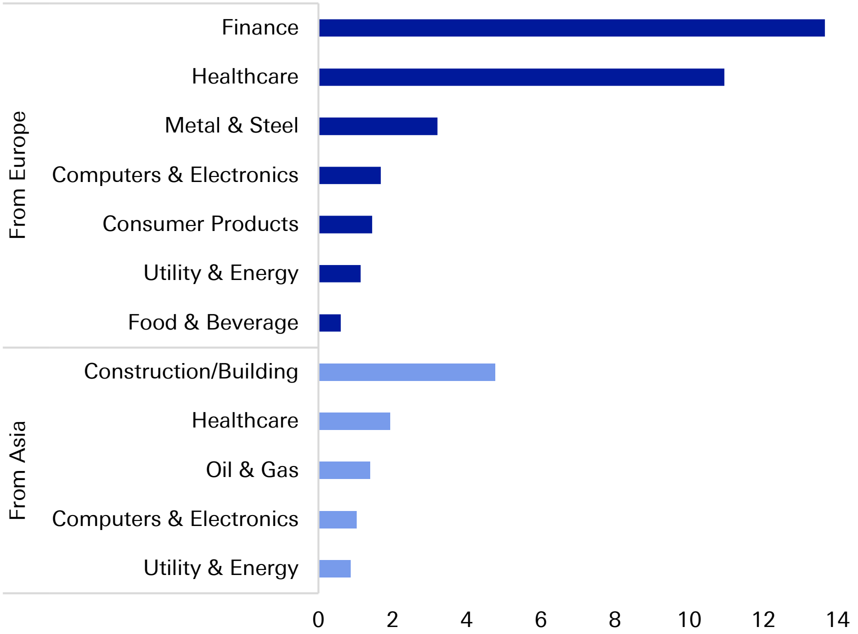


Source: Dealogic, Deutsche Bank.

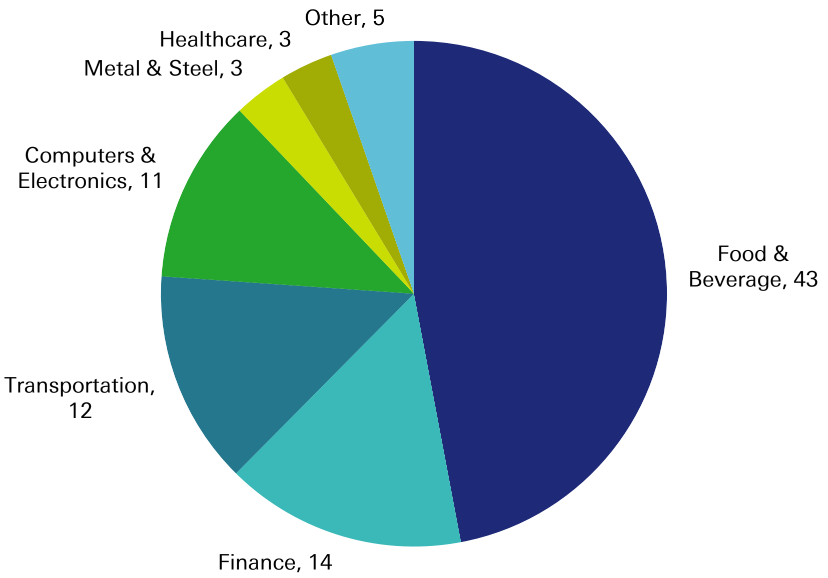
Cross-border US deals have cooled on tech (in line with the market)

Buyers from Europe were focused on defensive finance and healthcare. US dealmaking in Europe has been skewed towards staples, which along with finance and transport added up to c75% of Q1's value.

Cross-border US deals by sector ($bn value in Q1)



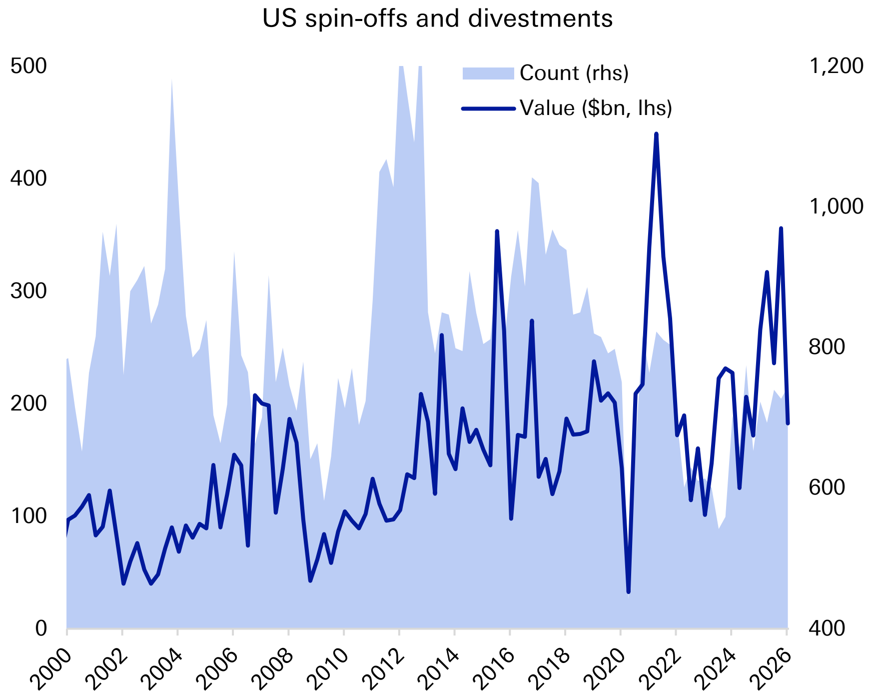
European cross-border deals with the US, top sectors in Q1 by value ($bn)

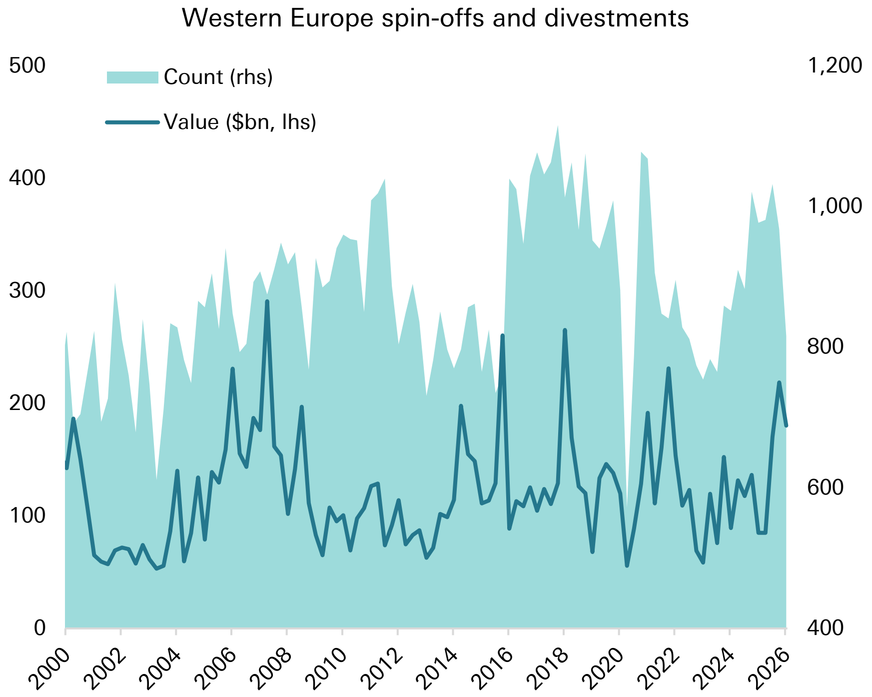


Source: Dealogic, Deutsche Bank.

Efficiency drive: Spin-offs, split-offs, divestments and asset carve outs

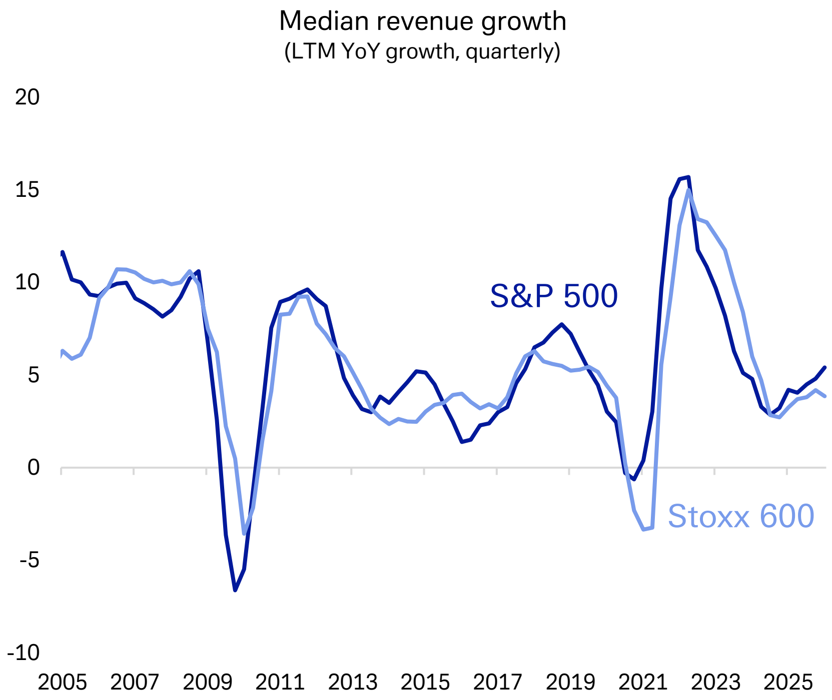
Momentum in Europe stronger than in the US but the push for more efficient operations and better 'asset turnover' is driving divestments as a broader trend. The plentiful dry powder at private equity firms means there are many willing buyers.

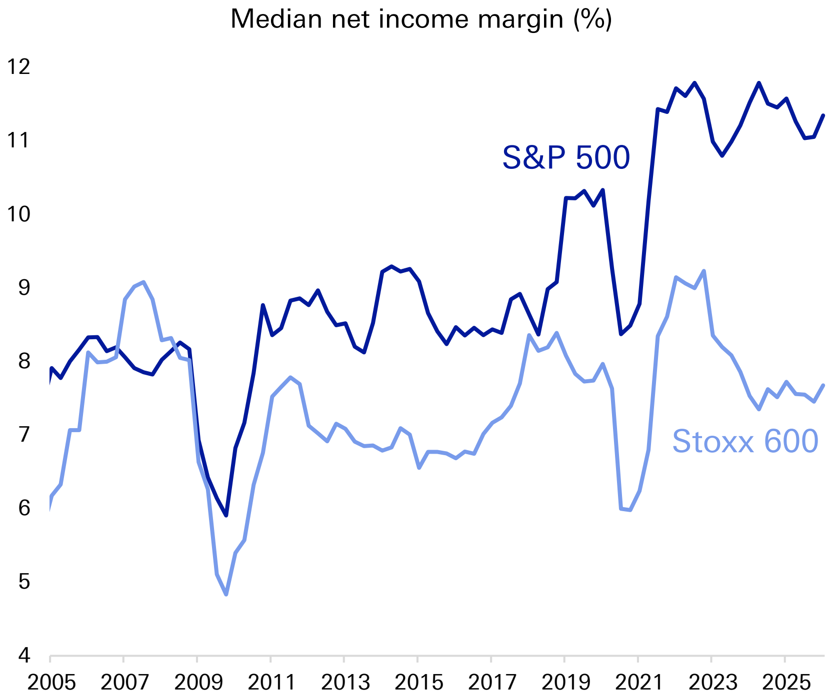




Source: Dealogic, Deutsche Bank.

Growth- and efficiency-centred M&A is becoming a priority as corporates recover from the downtrend in revenue growth and the plateau of profit margins





*Source: Bloomberg Finance LP, Deutsche Bank. * Note: excluding financials, real estate and utilities.*

Appendix 1

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Deutsche Bank
Research Institute

This energy shock isn't like 2022, let alone the 1970s

April 20, 2026

Authors

Henry Allen
Macro Strategist

The strong market rebound of recent weeks has led to comparisons with the first month or so after Russia's invasion of Ukraine in 2022. Back then, there was a similar recovery, but it was followed by even bigger declines as the scale of the stagflationary shock became clear.

That's understandably led to questions about a repeat. But those comparisons are unfair on several grounds:

— First, to state the obvious, oil and gas prices are lower today than back then. That applies to both front-end prices and expectations. For instance, even if you compare a similar point one month into the Ukraine conflict, Brent futures were pointing to an extended oil shock, with 6-month futures above $100/bbl, whereas today they're currently beneath $80/bbl. In other words, investors aren't pricing in an extended stagflationary shock.

— Second, the oil price comparison is even more favourable today if you adjust for inflation and lower energy intensity. That's because we've had cumulative inflation of around 10% since 2022, and energy intensity has continued to decline. So, even if oil did get to $100/bbl again, the impact is smaller today than it was then.

— Third, inflation is starting from a lower base today, meaning that central banks aren't being forced into the aggressively hawkish responses we saw after the 2022 shock.

— Fourth, the macro data has been resilient. That differs from the more serious oil shocks of the past, when there were immediate data downturns as soon as the shock occurred, e.g. the first oil crisis of 1973, or the Gulf War in 1990. By contrast today, the inflation impact is clear, but the data has remained expansionary on both sides of the Atlantic.

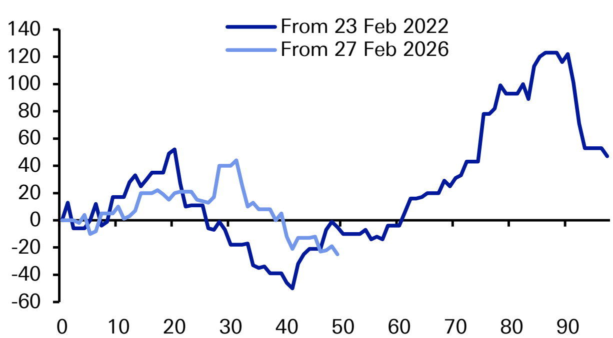
For markets, the key point is today's shock isn't hitting the severity thresholds of past oil shocks, on a whole range of metrics. The reason 2022 saw even bigger market downturns after the first month,was because it was accompanied by a much more severe energy shock (particularly on the gas side), along with high inflation that led to the most aggressive rate hiking cycle in a generation, and genuine fears of a recession. Today, we are not meeting those severity thresholds. Therefore, the resilience we've seen across multiple asset classes makes more sense than it might first appear, and isn't just a sign of complacency.

Why today's energy shock is a milder version of past episodes

The rapid recovery in risk assets has caught many off guard. In fact, it's just the third time since WWII that the S&P 500 has seen 3 consecutive weekly gains above 3%. That's extended to other asset classes, with HY credit spreads in the US and Europe now tighter than when the conflict began.

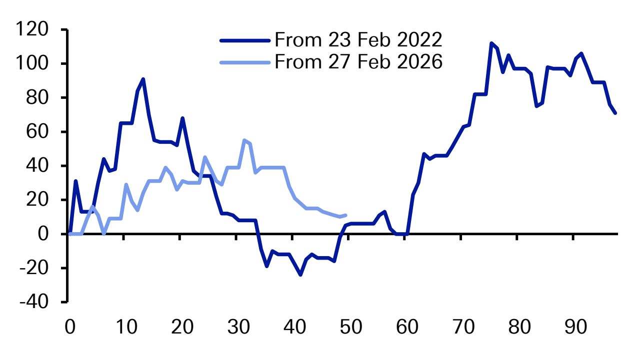
In response, some have made comparisons with 2022, when there was also an initial recovery in risk assets, only to be followed by even deeper declines. So bearish voices today have suggested that we're not out of the woods yet and, if a more sustained stagflation shock is realised, then the playbook is for deeper losses ahead.

Figure 1: Change in US HY spreads in 2022 and 2026 (bps) - an initial recovery in 2022 was followed by even more widening



Source : Haver Analytics, Deutsche Bank

Figure 2: The move in EUR HY spreads followed a similar path as well



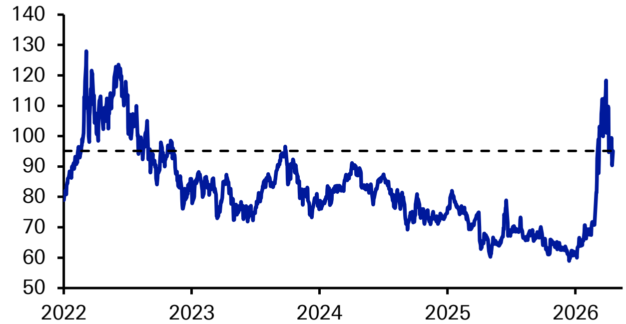
Source : Haver Analytics, Deutsche Bank

However, we think those comparisons are overplayed. In fact, from an economic standpoint, there are several points for optimism today relative to 2022, let alone some of the bigger oil shocks of the past that led to outright recessions.

1. Oil and gas prices are at a lower level now than in 2022, both at the front end and in expectation.

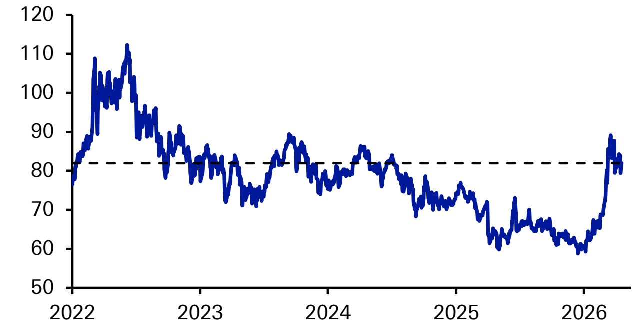
- It almost seems too obvious to state but, with energy prices at a lower level than 2022, it makes sense that markets aren't going to have as severe a reaction.
- Critically, expectations on the futures curve are also sharply lower. Even if you go back to the same point in the Ukraine conflict, just over a month in, investors got to a point where they expected Brent crude would be above $100/bbl in six months' time. That hasn't happened today.
- Of course, there's pushback on the basis that expectations are simply expectations. But in markets, expectations affect reality. For example, a key driver of the 2022 bear market was the belief that central bank rate hikes formed part of a classic boom-bust cycle that would end in recession. Similarly in 2025, expectations of a global trade war, and even a potential US recession, led to huge financial turmoil after 'Liberation Day', even before the tariffs had taken effect. We cannot isolate expectations from market performance.

Figure 3: Brent crude oil prices ($/bbl) are clearly lower today than they were in 2022



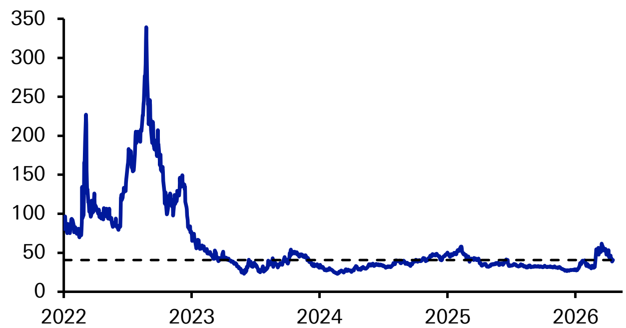
Source : Bloomberg Finance LP, Deutsche Bank

Figure 4: The 6-month Brent future ($/bbl) are lower as well, showing that investors aren't pricing a sustained oil shock like they were back then



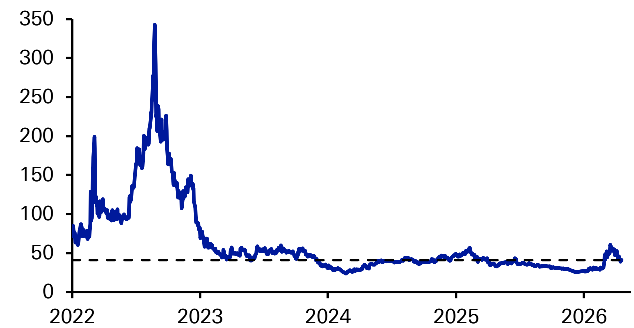
Source : Bloomberg Finance LP, Deutsche Bank

Figure 5: European natural gas front-end futures (EUR/MWh) are a fraction of their 2022 levels



Source : Bloomberg Finance LP, Deutsche Bank

Figure 6: 6-month European natural gas futures (EUR/MWh) are also substantially lower

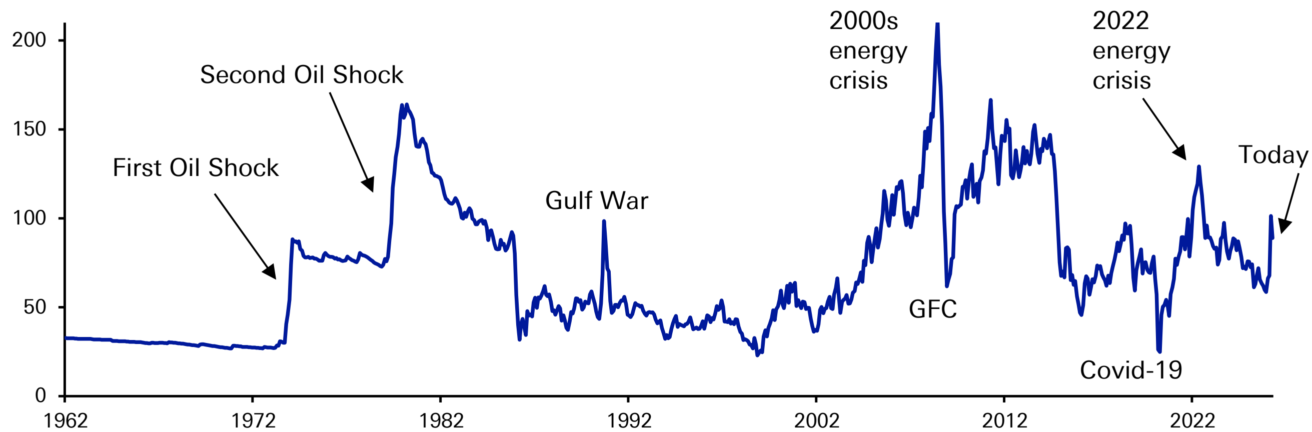


Source : Bloomberg Finance LP, Deutsche Bank

2.The energy price comparison becomes even more favourable today if you adjust for inflation and lower energy intensity.

- When we think about the impact of an energy price shock, we clearly need to account for inflation. Back in 2022, Brent crude oil spent around five months when it was at least $100/bbl. But since then, both the US and the Eurozone have seen cumulative inflation of at least 10%. So a given level of oil prices won't hit quite as hard as it once did, even a few years ago. Indeed, if we look at a long-term series of oil prices that adjust for inflation, this shock looks far less pronounced than shocks of the past.

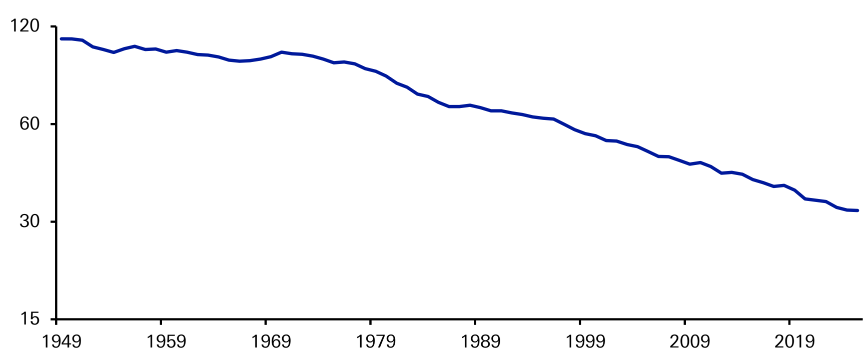
Figure 7: WTI Oil Prices ($/bbl) in real terms - the 2026 shock doesn't particularly standout relative to history, particularly with the pullback since March



Source : Finaeon, Bloomberg Finance LP, Deutsche Bank

- Another factor to adjust for is energy intensity. That's the amount of energy required to generate each unit of GDP, and it's been falling progressively in recent decades, particularly given the shift from manufacturing to services and progress on energy efficiency. But again, it means that energy shocks hit with less severity each passing year when they do happen. Even in 2025, US energy intensity was down 6% from its 2022 levels.

Figure 8: US Energy Intensity (1970 = 100), Log Scale



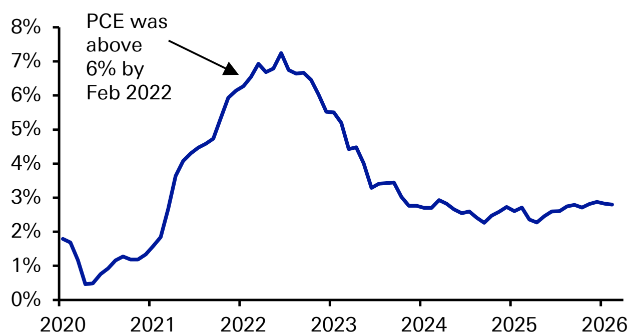
Source : EIA, Deutsche Bank
Note: Energy intensity calculated from the ratio of total primary energy consumption to real GDP

3. Inflation is starting from a lower base today, so central banks aren't being forced into aggressively hawkish responses.

- A key difference with 2022 (and the 1970s oil shocks) is that inflation is broadly around target levels going into this shock. In Feb 2026 at least, US PCE inflation was at 2.8% and Euro Area CPI was at 1.9%.
- That's a very different scenario to early 2022, inflation was around 6% in the US and the Euro Area, meaning the energy shock pushed it even further above target.
- Indeed, it's worth remembering that in early 2022, central banks had already begun a hawkish pivot, even before Russia's invasion of Ukraine

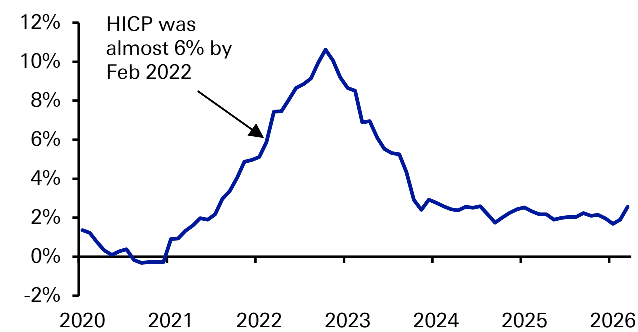
that February. For instance, the Fed was already signalling three rate hikes for that year in its December 2021 dot plot. That's a big difference with today, as the Fed was still signalling cuts last month in their dot plot, even after the Iran conflict had been underway for a few weeks.

Figure 9: US PCE inflation (year-on-year % change)



Source : Haver Analytics, Deutsche Bank

Figure 10: Euro-§Area HICP inflation (year-on-year % change)

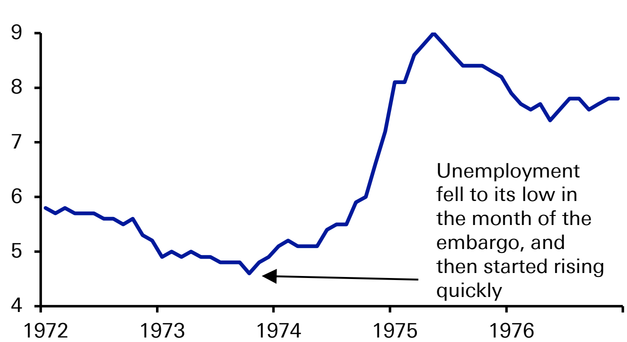


Source : Haver Analytics, Deutsche Bank

4. The macro data has been resilient thus far. That differs with the more serious oil shocks of the past, after which there was an immediate data downturn.

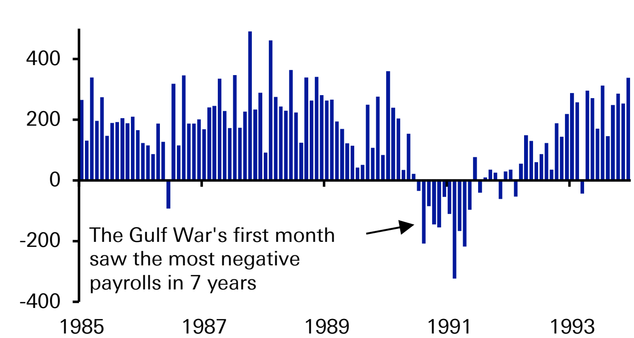
- Although we don't have a huge amount of data, the limited amount we have isn't pointing towards contraction, so again that fits into the resiliency of risk assets. Indeed, the US jobs report for March had the biggest payrolls increase in 15 months, alongside a lower unemployment rate. Meanwhile in Europe, the March PMIs slowed a bit, but were still in expansionary territory.
- Some have argued that it's still early days for this conflict, but in plenty of previous oil shocks, the impact was felt immediately. For example, in August 1990 when the Gulf War began, US payrolls saw their biggest contraction (-208k) in 7 years. Similarly, when the first oil shock hit in October 1973, the unemployment rate immediately started a clear rise from the following month.

Figure 11: US unemployment picked up straight after the first oil shock in 1973



Source : Haver Analytics, Deutsche Bank

Figure 12: When the Gulf War began in August 1990, US payrolls saw their biggest contraction in 7 years



Source : Haver Analytics, Deutsche Bank

Conclusion

Whilst the 2022 comparisons are understandable given it's the most recent big energy shock, we think those comparisons are overplayed. After all, the scale of the energy shock is smaller today, particularly if you adjust for inflation and lower energy intensity. Moreover, inflation is at lower levels to start with, avoiding the need for a significantly hawkish response. Some have suggested that the data will eventually turn negative but, in several of the more serious oil shocks of the past (e.g. 1973, 1990), that happened immediately, and didn't take several months, which again backs up the idea that today's shock isn't as severe.

Clearly, there's a lot of volatility and caution is warranted. But the reason 2022 saw even bigger market downturns after the first month was because it was accompanied by a much more severe energy shock (particularly on the gas side), along with high inflation that led to the most aggressive rate hiking cycle in a generation, and genuine fears of a recession. Today, we are not meeting those severity thresholds.

Appendix 1

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Deutsche Bank
Research

Crypto adoption: The comeback ?

April 2026

Marion Laboure | Camilla Siazon

Deutsche Bank Research Institute

Deutsche Bank
Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Crypto adoption: The comeback ?

**** In this chartbook, we will be focusing only on the US, UK, and EU. Please reach out if you would like to receive more data on the other countries.****

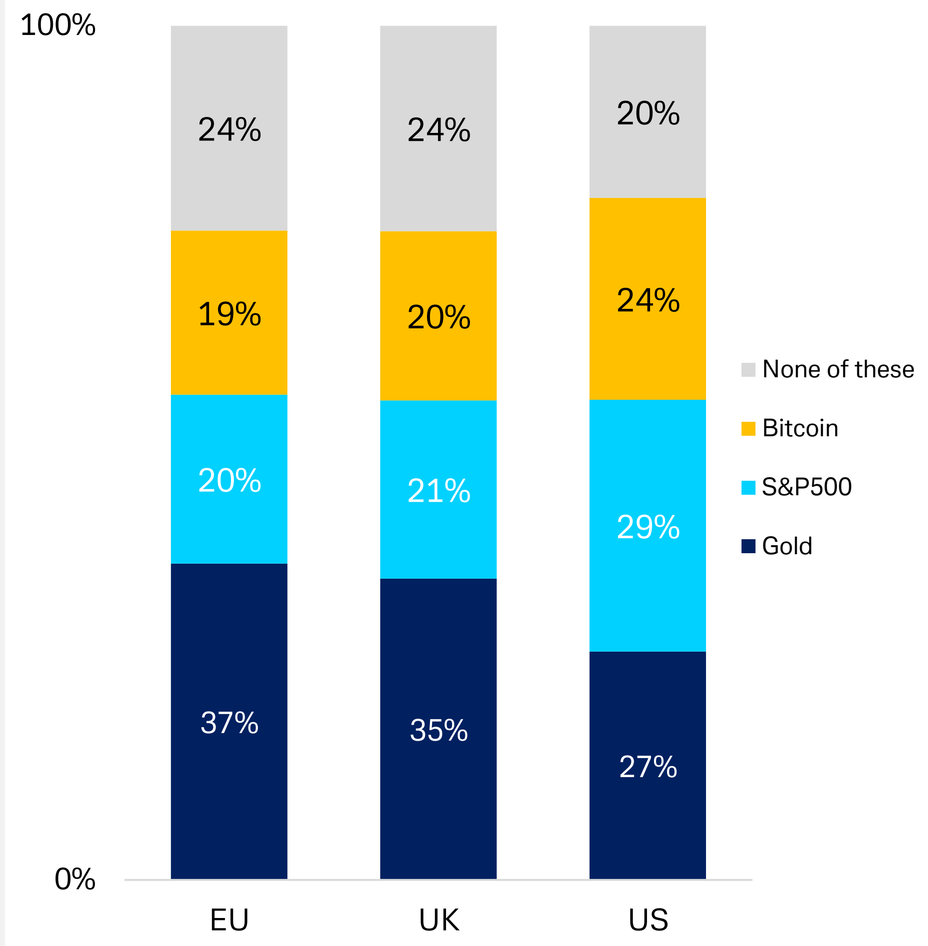
Q1 2026 saw crypto continue to struggle amidst macro and geopolitical shocks, with Bitcoin down -22%. However, **March saw some recovery, supported by strong US ETF inflows and limited retail re-engagement**. Our latest retail crypto survey chartbook reflects this, showing a rebound in US consumer adoption.

This edition leverages our proprietary dbDataInsights analysis across the US, EU and UK, surveying 3,400 consumers to explore new trends: holder investment appetites, consumer prediction of future Bitcoin prices, and crypto investments types.

Main takeaways:

1. After a steady slide the last few months, **US crypto adoption has recovered to July 2025 levels**. In the UK, adoption fell slightly but is still higher over the long term, while Europe remains steady.
2. **The majority expect Bitcoin to be lower than today**, and very little anticipate a return to the $120K record again.
3. **Consumers continue to favour the S&P 500 and Gold over Bitcoin**, though the gap is narrower in the US.

If you were to make a new investment, which, if any of the following asset classes would you select? (1-3 years)



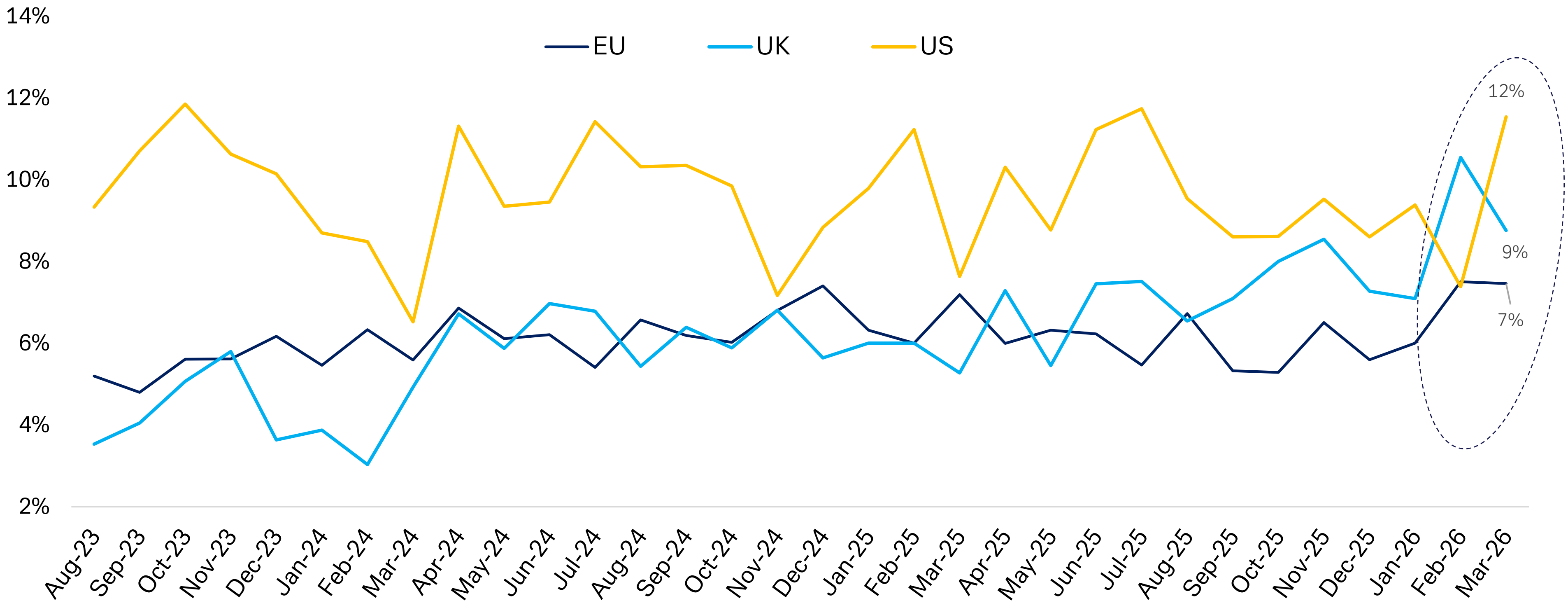
Source: Deutsche Bank dbDataInsights.

REGION

After steadily declining since July 2025, US crypto adoption rates recovered in March

- ❑ US adoption rates recovered to 12% in March from its trough in Feb (7%). This pattern fits with Bitcoin's wider price rebound (+2.20%) and return of inflows into institutional Bitcoin ETFs (~$1.3bn) in March.
- ❑ In the UK, the rate of adoption decreased slightly to 9% in March but it is still higher over the longer-term, suggesting a more sustained interest in crypto amongst consumers.
- ❑ Europe remains steady at 7%.

Consumers who have personally invested in crypto (e.g. Bitcoin, Ether)



Source: Deutsche Bank dbDataInsights. Question: In the last 12 months, which if any of these financial activities have you personally done? Please select all that apply: personally invested in crypto (e.g. Bitcoin, Ether).

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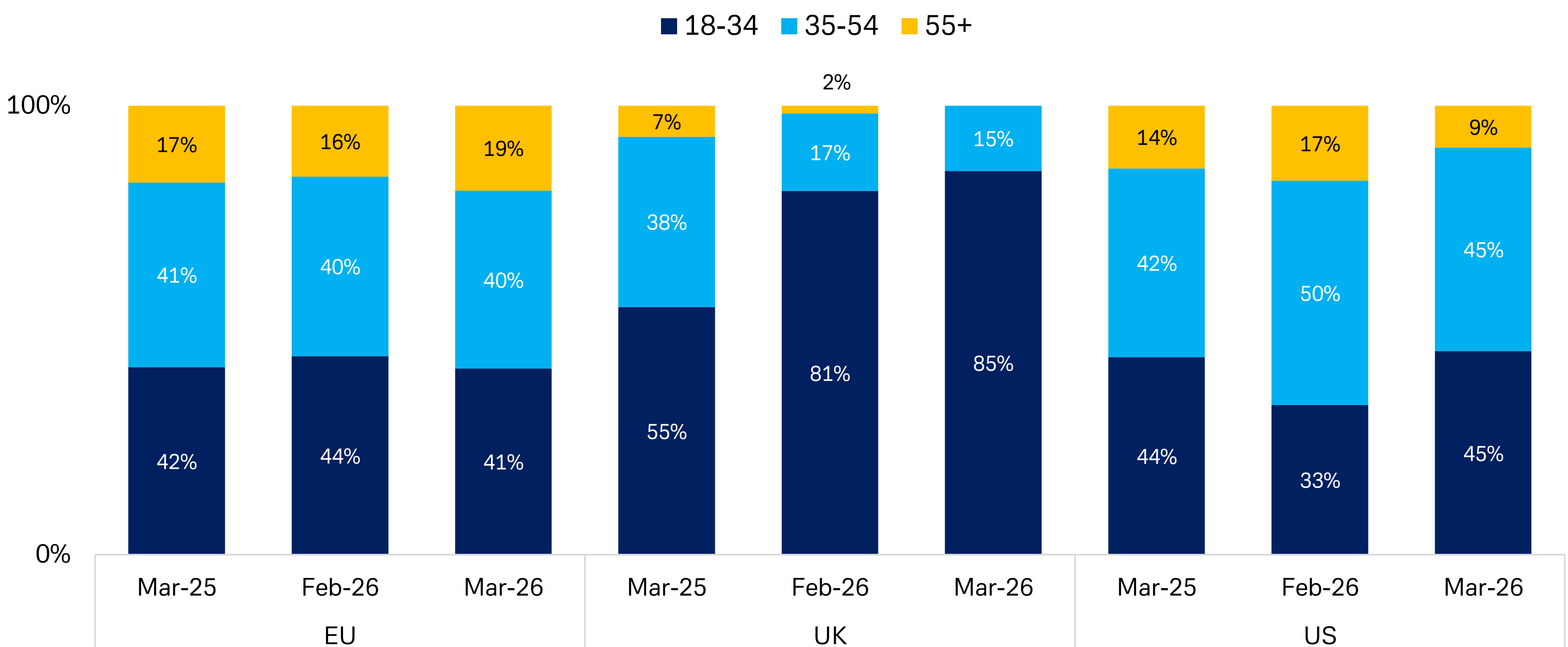
AGE

Of those who have said they personally invest in crypto, the demographic was relatively split between those aged 18-34 and 35-54 years old in the US and EU

- ❑ In the US and EU, the crypto adoption rate between 18-34 and 35-54 age groups remained evenly distributed in March.
- ❑ The biggest rise has been amongst young consumers (aged 18-34) in the UK. Crypto usage in this age group jumped in March (85%) vs. one year ago (55%).
- ❑ Those 55 years or older continued to show the lowest crypto adoption rate, underscoring the lack of crypto understanding amongst the older respondents. In the UK, no one above 55 years old said they invest in crypto.

Retail crypto adoption by age



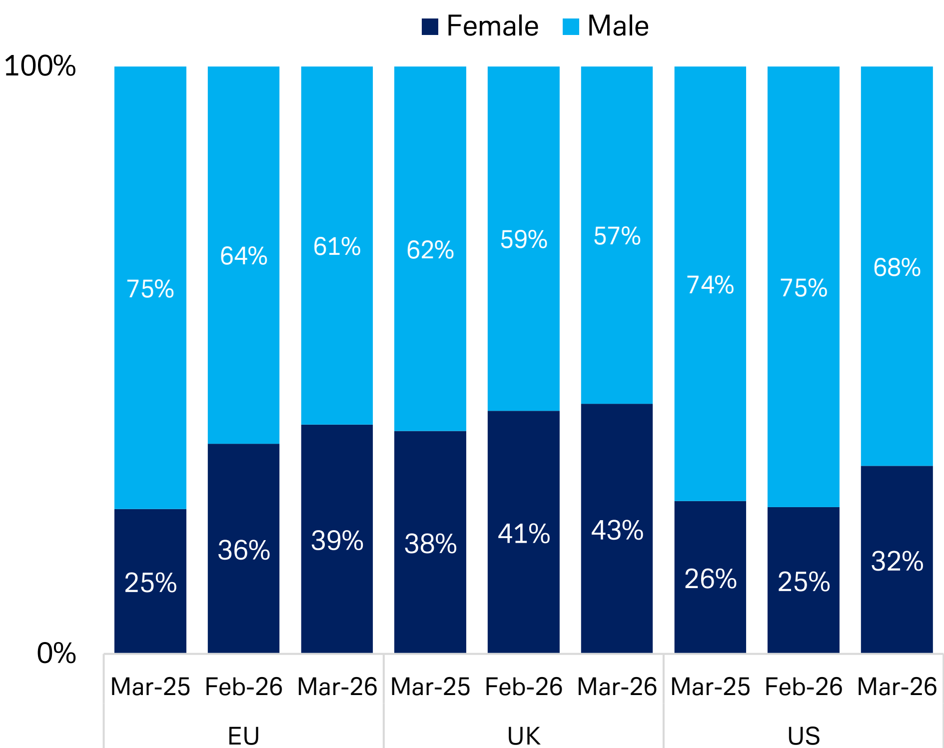
Source: Deutsche Bank dbDataInsights.

GENDER

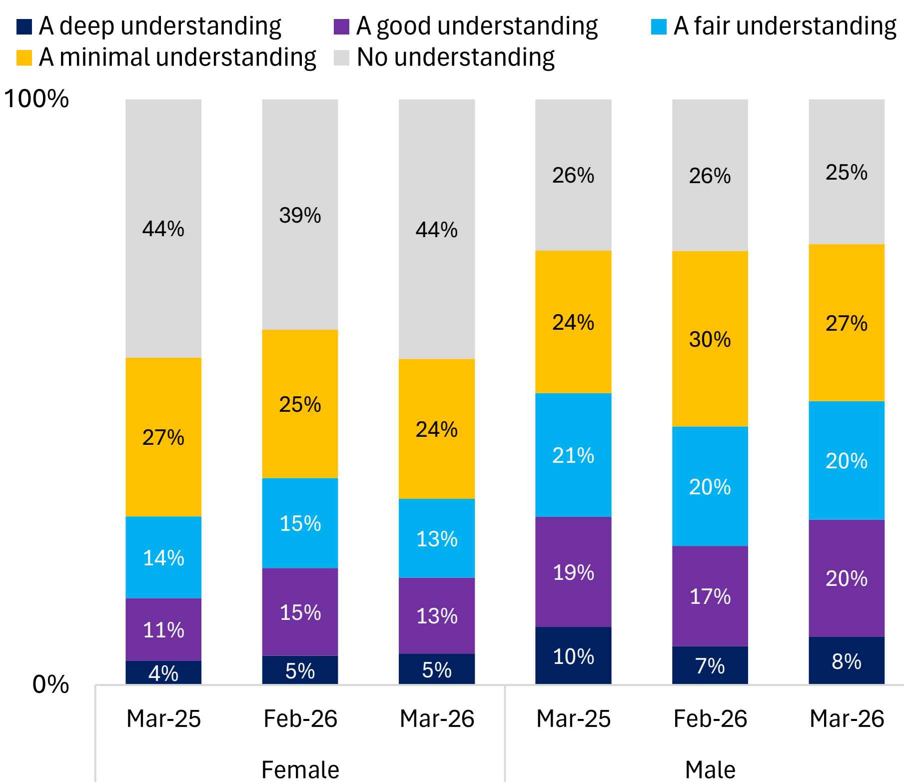
Male consumers continue to lead crypto adoption, but the rate amongst females is growing

- Of those who said they personally invest in crypto, the majority in the EU (61%), UK (57%), and US (68%) were men.
- Still, female crypto adoption also rose in March, compared to one month and one year ago. The UK (43% female vs 57% male) had the closest distribution.
- Male respondents continue to exhibit greater levels of crypto knowledge than women. March 2026 figures show that 28% of average male respondents believe they have a good/deep understanding of crypto vs. 18% of average female respondents.

Crypto adoption rates by gender and country



Levels of crypto knowledge by gender



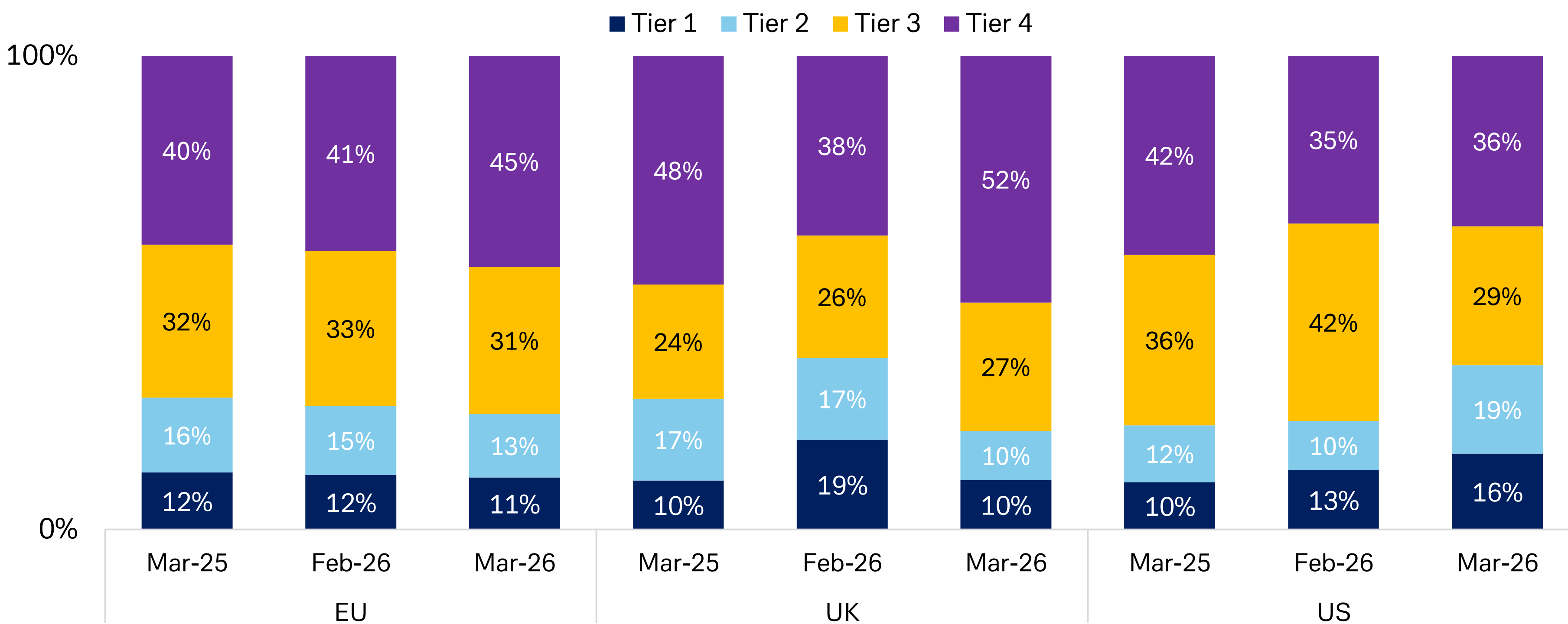
Source: Deutsche Bank dbDataInsights.

INCOME LEVEL

Of those personally investing in crypto, consumers continue to be relatively well-off, with 65% earning above $50K in the US

- In the US, 36% of respondents who earn above $100K said they have personally invested in crypto.
- Interestingly, the level of lower-income earners (up to $25K for the US) investing in crypto also rose to 16% vs 10% from March 2025.
- The UK saw the biggest rise. Crypto adoption amongst Tier 4 (£50K) income earners rose to 52% versus 38% the month prior.

Retail crypto adoption according to income*



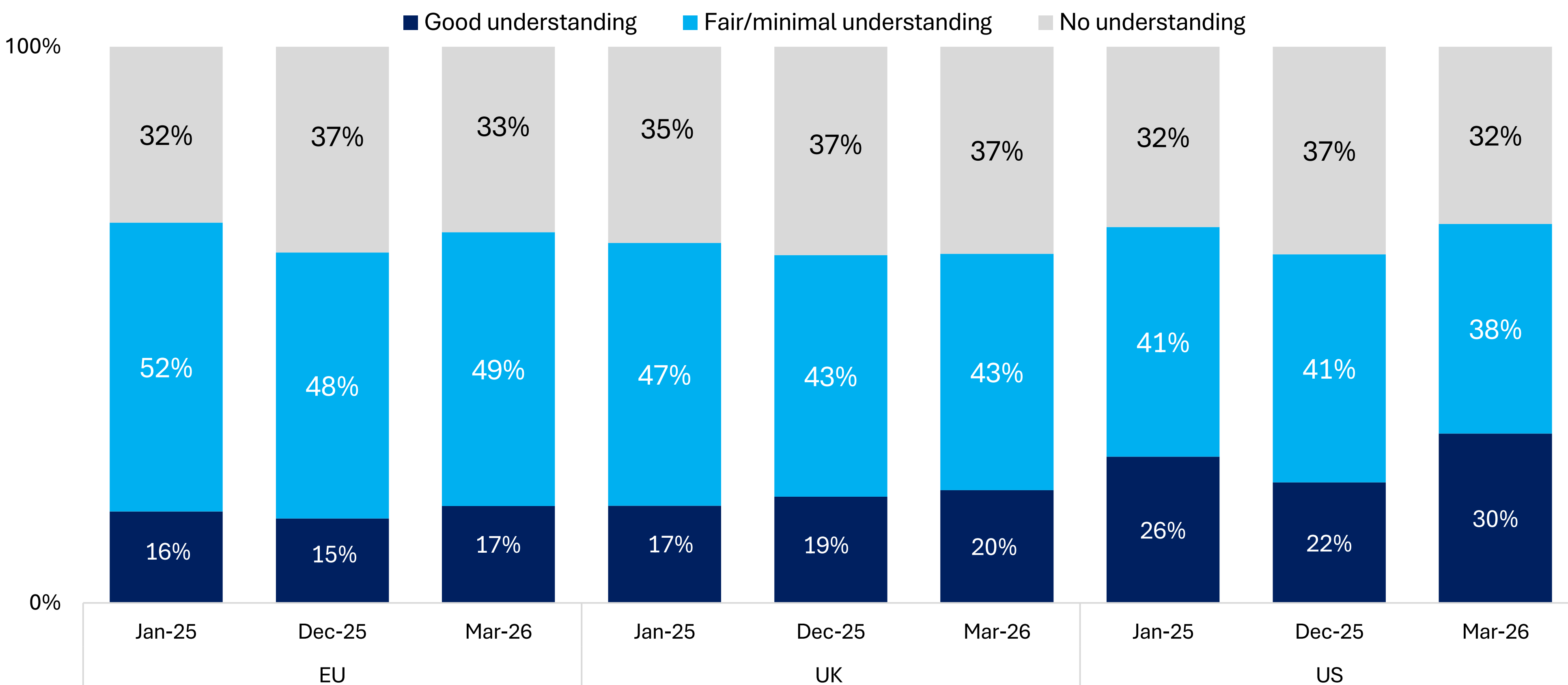
*Tier 1: up to $25K for US; up to £20K for UK; up to €20K for EU.
Tier 2: $25K - $50K for US; £20K - £30K for UK; €20K - €30K for EU.
Tier 3: $50K - $100K for US; £20K - £30K for UK; €20K - €30K for EU.
Tier 4: $100K+ for US; £50K+ for UK; '€50K+ for EU.
Source: Deutsche Bank dbDataInsights. Survey for March.

UNDERSTANDING

Levels of consumer crypto knowledge were fairly neutral, with no significant spikes or drops observed

- Between Dec 2025 and Mar, consumers' levels of understanding showed mixed trends.
- The US rose 8 percentage points to 30% saying they have a good understanding of crypto, in line with increased adoption behaviours. The UK (19% to 20%) and EU (15% to 17%) registered a more marginal increase in levels of understanding between Dec and Mar.

Q. What level of understanding, if any, do you currently have of crypto?

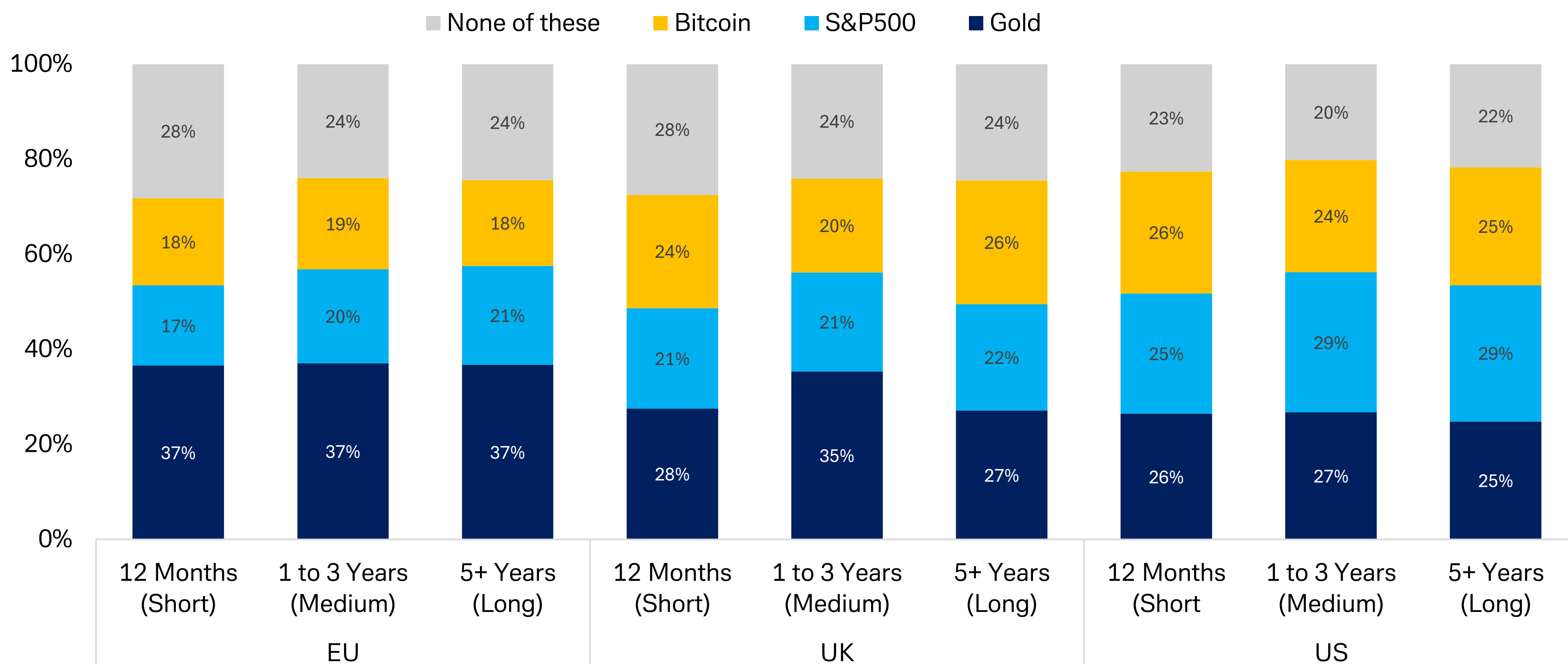


Source: Deutsche Bank dbDataInsights.

INVESTMENTS
Consumers continue to favour gold and the S&P 500 over Bitcoin, although the degree varies

- While gold is a popular investment across investment horizons in Europe (37%), the US is more split between the S&P 500 (25%), gold (26%) and Bitcoin (26%) especially in the short term (12 months).
- In the UK, the split is also evenly distributed, although a majority favours gold (35%) vs. the S&P 500 (21%) and Bitcoin (20%) as the medium-term investment (1-3 years).

Q. If you were to make a new investment for the following periods which, if any, of the following asset classes would you most likely select.



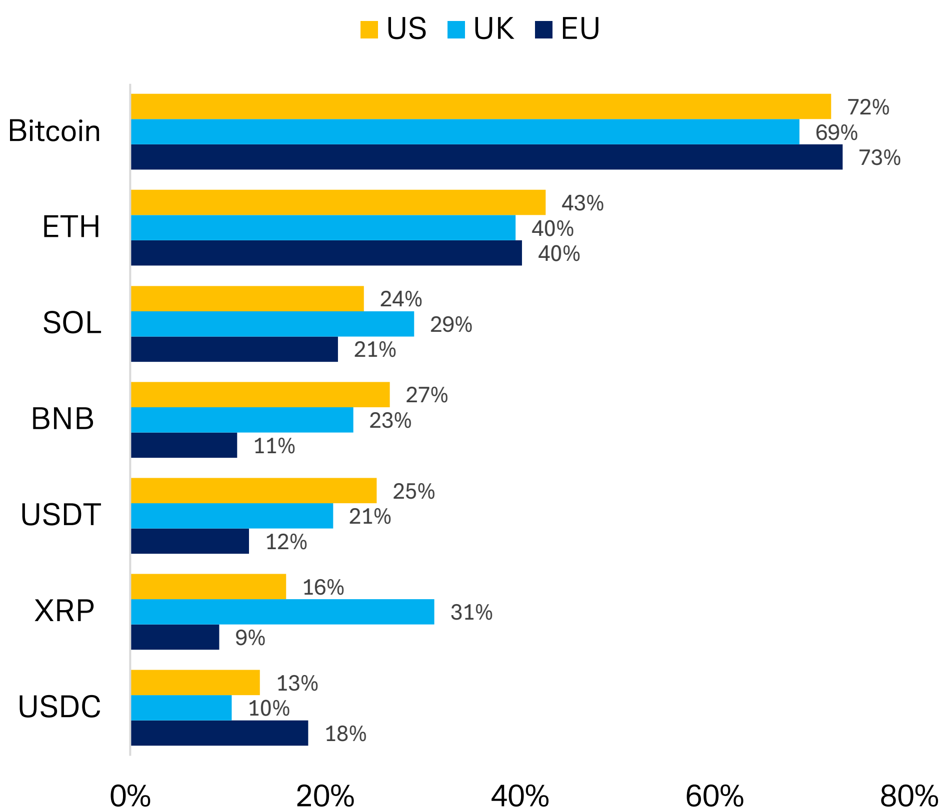
Source: Deutsche Bank dbDataInsights.

BITCOIN REMAINS KING

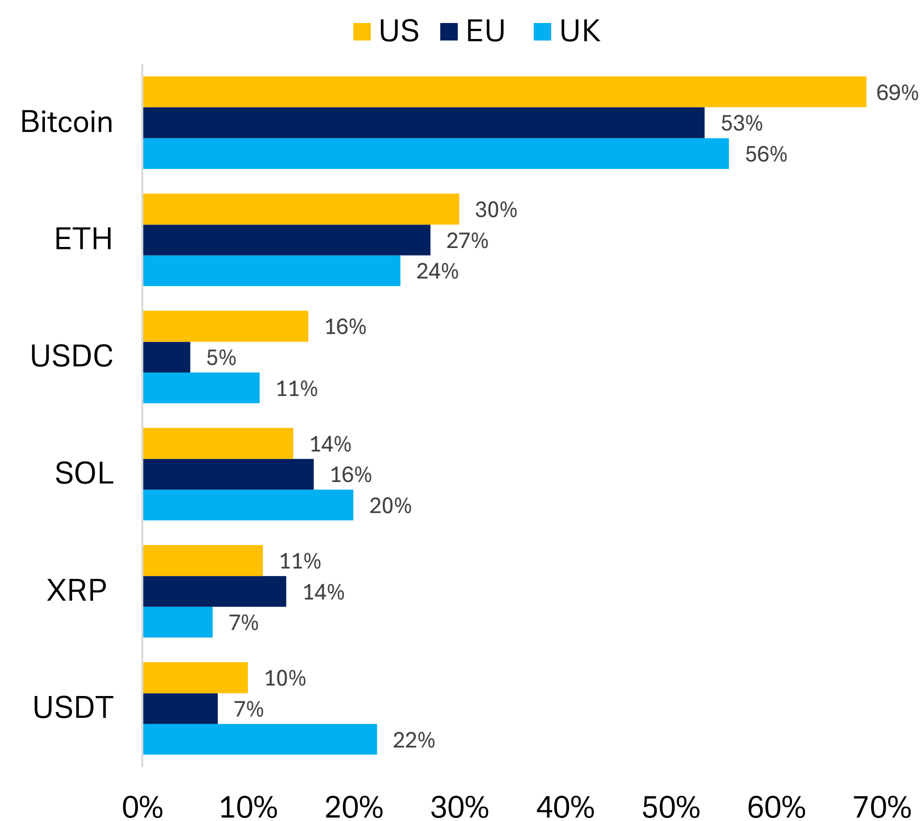
Consumers still choose Bitcoin as their crypto of choice

- ❑ Bitcoin consistently stands out as the most held crypto currency, with around 70% ownership across all regions, and far surpassing stablecoin ownership like Tether or USDC.
- ❑ It also stands as the top choice for future investment across the US (69%), UK (56%), and EU (53%).

Q. Which, if any, of the following cryptocurrencies do you currently hold/invest in? Please select all that apply.



Q. Which two of the following cryptocurrencies would you most want to invest in over the next 6-12 months, regardless of your current holdings?



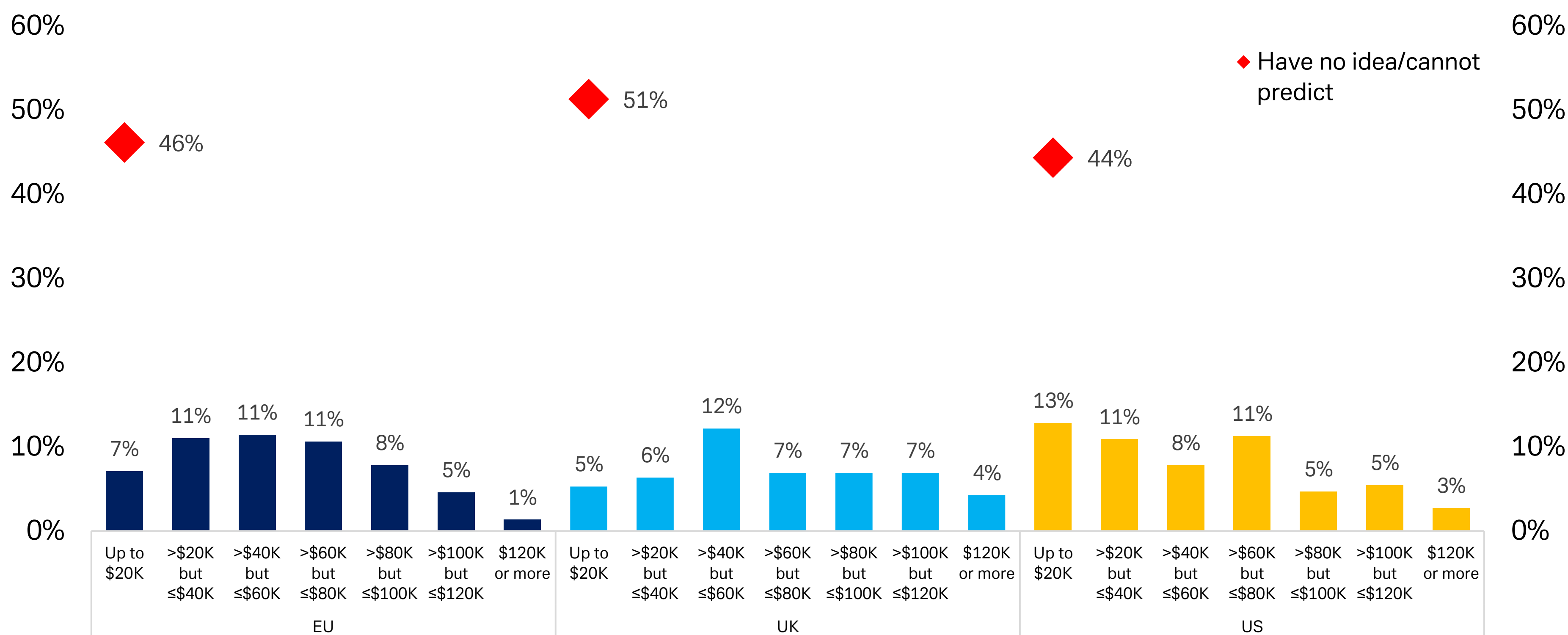
Source: Deutsche Bank dbDataInsights.

WHERE WILL BITCOIN BE BY THE END OF 2026?

Consumers seem bearish on Bitcoin's prices, believing it will be lower than what is now (around $75k). The majority say they don't know / can't predict.

- In the US, 19% believe Bitcoin's price will be between $20k – $60k by end-2026. This falls in line with the EU (22%) and UK (18%).
- Notably in the US, 13% believe that Bitcoin's price will land below $20k in 2026. The last time Bitcoin's price was that low was in January 2023 – a steep slide considering it hovered around $125k 6 months ago.
- And very little believe Bitcoin will reach its high again by end-2026 in the EU (1%), UK (4%) and US (3%).

Q. What do you predict the traded value for Bitcoin will be at the end of 2026?



Source: Deutsche Bank dbDataInsights.

Appendix 1

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Beyond the barrel: Europe's hidden energy vulnerability

13 April 2026

Authors

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Recent supply chain stress, highlighted by European PMI data, has renewed focus on the continent's external vulnerabilities. In this note, we argue that Europe's exposure is twofold:

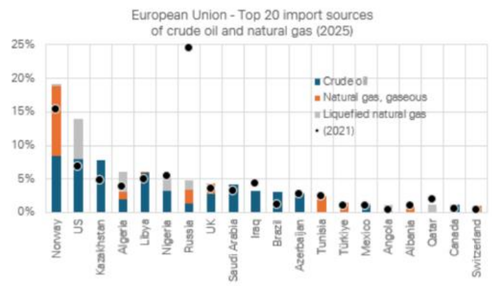
- A direct, tangible dependency on the Gulf for specific commodities like oil products (e.g., jet fuel, diesel), oil derivatives, chemicals and metals, representing a measurable cost.
- A more elusive and structural indirect risk from "hidden" energy embedded within industrial imports, particularly from Asia.

This indirect exposure is qualitatively different, comprised of refined products and electricity (not just crude oil) and threatens shortages of specific, non-fungible industrial inputs like plastics, chemicals and metals. This upstream vulnerability is far harder to substitute than direct commodity imports and, as we've argued previously, is a prime example of a non-linear risk that can amplify stagflation and complicate monetary policy.

Note: 'Direct' exposure hereafter refers to Europe's reliance on imported goods from a specific region, for instance energy from the Middle-East. 'Indirect' exposure refers only to the "embodied" content of an import – the energy consumed along the value chain to produce that good before export to Europe.

A ceasefire's reprieve, a system's vulnerability. The fragility of the agreed two-week ceasefire in the Middle East was underscored this weekend by the failure of US-Iran talks. While any pause in hostilities offers a welcome reprieve for global supply chains, this development highlights that underlying risks remain in place.

Figure 1: Supply chain pressure is increasing rapidly. EU direct energy exposure to GCC States not as pronounced as to Russia in 2022



Source: Deutsche Bank Research, NY FED, S&P Global, Eurostat (datasets nrg_ti_gas and nrg_ti_oil) for the right-hand chart. GCC countries refers to a sum of Bahrain, Iran, Kuwait, Qatar, Saudi Arabia and the UAE. 2025 trade share covers January to September.

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Recent data has already provided a clear warning signal of this vulnerability, with the March PMI releases for Europe showing a notable lengthening of supplier delivery times and a fresh acceleration in input costs (Figure 1, left).

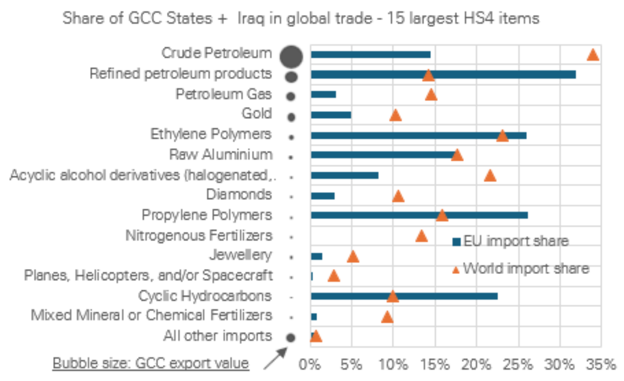
While this offers a glimpse of what could happen if the conflict escalates again, the transmission mechanism to Europe is more complex than a simple repeat of 2022. There are two distinct channels of vulnerability when it comes to this energy shock: a direct, measurable impact on the cost of specific imports – largely energy –, and a more subtle, structural and indirect risk embedded within global value chains, potentially impacting the vulnerability of various inputs.

Channel 1: A tangible, product-level exposure. The direct risk channel has two layers. The first is the overall cost of energy. As we quantified in a recent note, the euro area's energy import bill should increase by around 0.8% of GDP, based on current market pricing (Brent at 98 USD/bbl and gas at 47 EUR/MWh).

The second risk is a more specific dependency on the region central to the crisis in terms of physical supply. This is most visible not at the continent-wide level – where suppliers like Norway, the US, and Kazakhstan now dominate Europe's energy supply (Figure 1, right) – but at the product-specific level, where Gulf countries can provide upwards of 15% of EU imports. A key example is refined products, where Europe sources a notable share from the region (Figure 2). While Europe's significant domestic refining capacity (supplying 81% of the domestic market in refined petroleum in 2022) provides a crucial buffer, the import dependency remains.

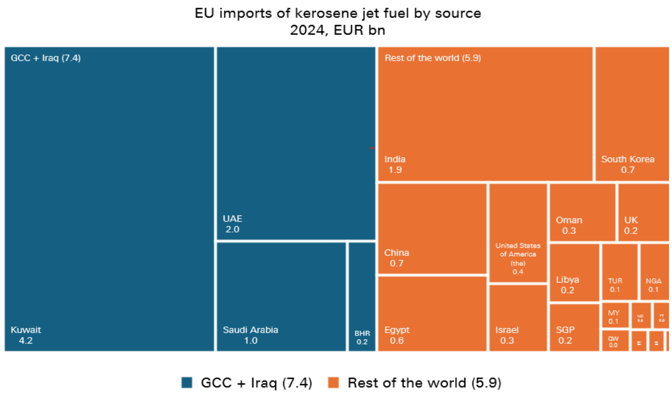
For certain types of refined petroleum products, such as kerosene, the share of GCC countries as a source for EU imports reaches 56% (Figure 3). On this point specifically, our in-house specialists think that the air transport sector has 45-55 days of effective cover, higher than the normal 30-35 days thanks to the release of reserves by the International Energy Agency. Uncertainty prevails, however, as the ceasefire has done little to ease the Strait of Hormuz situation.

Figure 2: The EU's reliance on the Gulf is meaningful in specific categories, although generally lower than the rest of the world



Source: Deutsche Bank Research, BACI

Figure 3: An example of high exposure is kerosene, with an import share of 56%

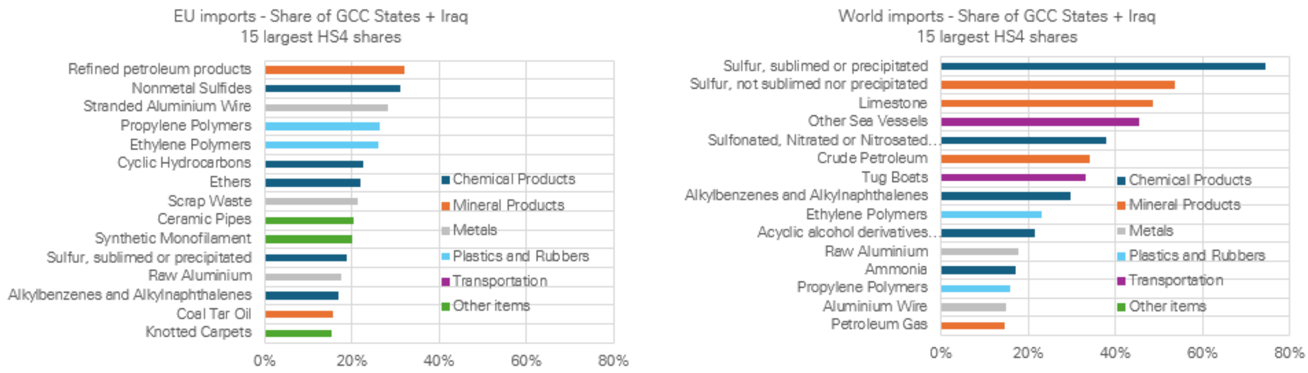


Source: Deutsche Bank Research, Eurostat

Furthermore, granular data reveals pockets of high dependency for specific petrochemicals, plastics, and metals (Figure 4, left). These concentrated exposures represent a measurable cost to Europe's import bill and a vulnerability for the industries that rely on them. In short, while Europe's product-level exposure may be lower than in other parts of the world, it is real and cannot be dismissed. The second channel of risk, however, is qualitatively different and arguably more complex.

Channel 2: A hidden, structural risk in value chains. While Europe's visible exposure is specific, the GCC and Iraq region's role in supply chains more globally creates a broader bottleneck risk. The region's global importance extends beyond crude oil, and it holds a commanding position in key industrial inputs, particularly in mineral and chemical products (Figure 4, right).

Figure 4: The granular set of products where the Gulf makes a large fraction of EU and world imports



Source: Deutsche Bank Research, BACI

For instance, the region accounts for over 70% of global exports of certain forms of sulfur and 48% for limestone. This is complemented by significant global shares in chemical feedstocks like sulfonated/nitrated derivatives (38%) and plastics like ethylene polymers (23%).

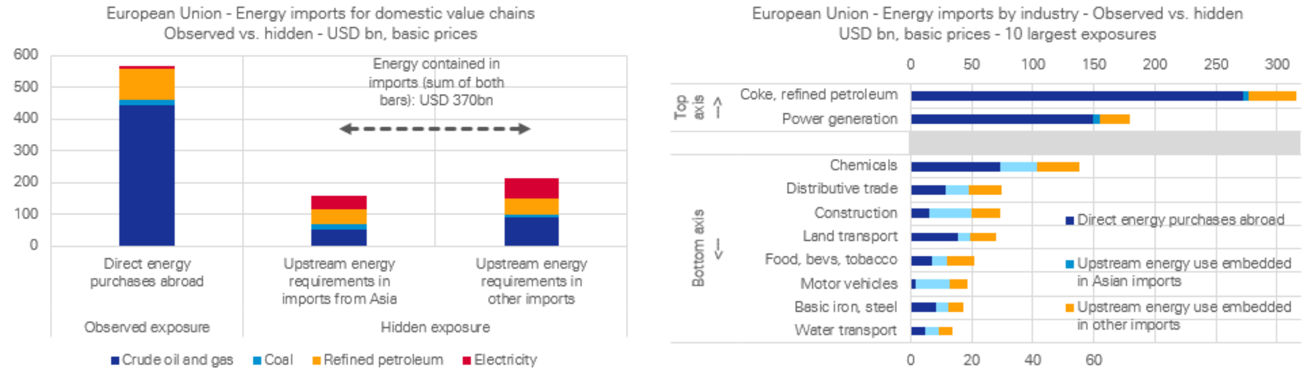
This global bottleneck risk is one indirect threat to Europe. Another transmission channel is through second-round effects via Asia, whose industries are more reliant on these Gulf supplies. Recent headlines (see here, here and here) already flag this, with reports of Asian producers cutting output due to soaring feedstock costs.

We quantify this indirect channel by calculating the energy "embodied" in Europe's imports – that is, the energy consumed all along the global value chain to produce imported goods. Based on OECD Input-Output tables for 2022 – the last year available –, we find that in addition to Europe's USD 567bn (3.8% of EU GDP) in direct energy imports, the goods imported into Europe for use in supply chains required an additional USD 370bn (2.5% of GDP) of indirect energy to be consumed in their production (Figure 5, left). A significant share of this – USD 170bn (1.1% of GDP) – was embedded in goods and services from Asia alone. These measures capture total energy use across production stages, including energy used within the energy sector itself, and is therefore not 100% comparable to observed import flows.

Crucially, this hidden exposure has a different risk profile. While direct imports are dominated by crude oil, the indirect energy footprint has a larger share of refined products and electricity. This shifts the focus of risk away from just oil wells and towards the stability of foreign refining capacity and power grids – critical, yet less monitored, factors for European industrial resilience.

Pinpointing upstream vulnerabilities. What inputs could create disruptions? The impact of this indirect exposure is not uniform across the European economy. Beyond direct energy consumers (refined petroleum and power generation) sectors such as chemicals, manufacturing, and construction carry a significant indirect energy footprint tied to their Asian supply chains (Figure 5, right).

Figure 5: Energy dependency – What is observed and what hides in imports. The exposure of many manufacturing sub-segments is largely indirect



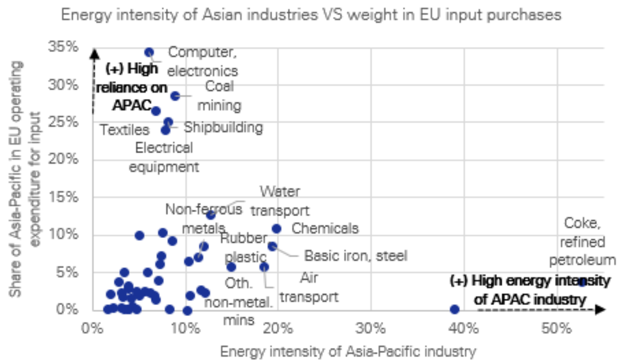
Source: Deutsche Bank Research, OECD Inter-Country Input-Output Table (2022). Analysis covers imported energy only. The OECD's $567bn figure differs from Eurostat's ~$700bn primarily due to valuation (OECD uses 'basic prices', Eurostat uses 'CIF') and scope (OECD tracks by exporting industry, not purely by product). The $567bn measure also only accounts for energy as an input (intermediate demand).

The transmission risk is most acute for inputs that combine two characteristics: high sourcing from Asia and high energy intensity. This makes basic chemicals and industrial materials – like plastics and steel –primary vectors through which cost and availability shocks can propagate into European industries (Figure 6).

Having measured total energy requirements along value chains, we now isolate the pure energy cost absorbed by each economy. This strips out non-energy components of the energy sector's output (intermediate inputs as well as margins, taxes and labour costs), providing a cleaner measure of underlying energy exposure. This shifts the focus from production requirements to income absorption/value added contributions from energy as an input.

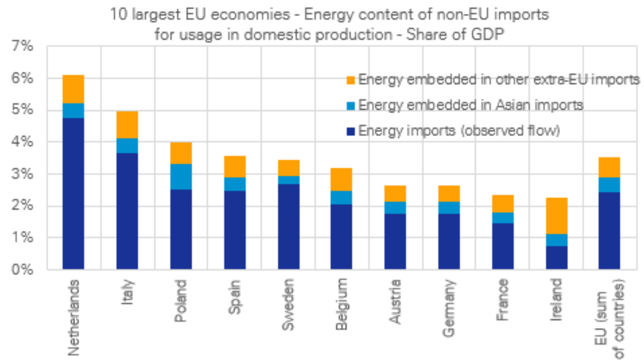
This country analysis reveals a remarkably widespread "hidden" burden. Factoring in this absorbed energy cost adds an amount equivalent to between 0.8% and 1.5% of GDP to the import bill for most major European economies. This effectively increases their total energy dependency by a third to a half, compounding already high exposure in countries like Italy, Spain, and Poland (Figure 7).

Figure 6: Which inputs from Asia could be at risk in terms of supply?



Source: Deutsche Bank Research, OECD ICIO

Figure 7: Energy absorption shifting higher when hidden energy is captured



Source: Deutsche Bank Research, OECD ICIO

Conclusion: A structural vulnerability. The key takeaway is that Europe’s exposure in terms of supply is twofold. The direct part (trade exposure to Gulf countries on crude oil, gas, oil-derivatives, chemicals and metals is tangible and should not be overlooked. The indirect part is more elusive, structural and difficult to manage: energy that arrives embedded within industrial inputs – with a precise fit in those chains – is much harder to substitute than fungible commodities like oil or natural gas, for which Europe's sourcing has already shifted to other suppliers.

This creates the potential for shortages of specific, critical components, a prime example of the non-linear risks we highlighted in a recent piece. The nature of such a structural vulnerability means it is often less responsive to broad macroeconomic tools and requires a more targeted level of monitoring. The ultimate message is that even countries with limited direct dependence to the Gulf face a meaningful – and arguably less manageable – upstream vulnerability through their reliance on global industrial supply chains.

The lingering risk of shortages amplifies the stagflation risks already prevalent from the energy shock (Figure 6), and complicates the conduct of monetary policy. Our view is that the risk of persistent inflation will likely drive the ECB to hike rates to 2.50%, the upper end of the range of neutral rates. This would signal the ECB’s commitment to price stability without overburdening growth when there are downside risks.

Appendix 1

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Deutsche Bank
Research

What Q1's emerging themes mean for 2026 and beyond

April 2026

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Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Contents

Thought experiment

If, at the beginning of 2026, someone asked how far the S&P 500 would fall if the Strait of Hormuz was closed, what would you have answered?

Probably more than the 6-7% peak-to-trough drop in Q1

The result reflects not just that corporate earnings forecasts remain resilient and investors are fearful of being out of the market if there is a sudden ceasefire in Iran and markets rebound. It also reflects how some of the broader themes and megatrends in markets are influencing the outcome.

Our thanks to Krystle Coe for her assistance with this presentation

1. What themes are driving markets?

The themes driving markets in Q1 – based on AI analysis of our daily market commentaries

1 Escalating geopolitical tensions in the Middle East

2 Stagflationary fears and hawkish central bank response

3 "Picks and shovels" AI investment theme

4 Equity market correction and increased volatility

5 Heightened oil price volatility and supply chain adaptation

10 Escalating geopolitical tensions in the Middle East

9 Landmark legal rulings on social media and AI

8 The weakening euro

7 Slowing economic momentum in Europe

6 Disruption in the semis and technology sectors

How much did each key theme in Q1 impact the S&P 500 (as a proxy for risk assets more broadly)?

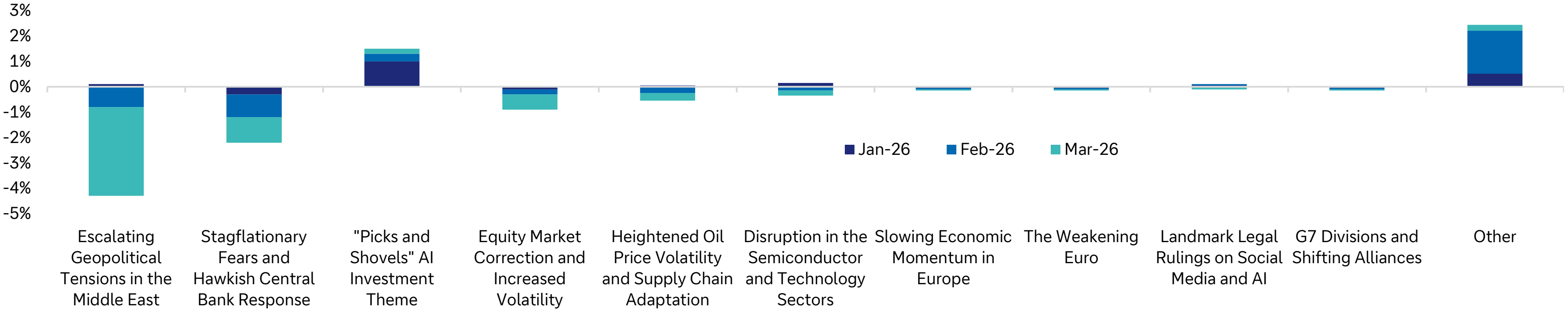
Iran conflict
It mattered … a lot … but the reality is that other factors support markets

AI disruption
The sell-off is just one of many over the last 3 years. Good tech earnings for Q1 & 2 may assuage fears

Fed policy
Central banks typically don't react to supply-side shocks, but the Fed may delay cuts

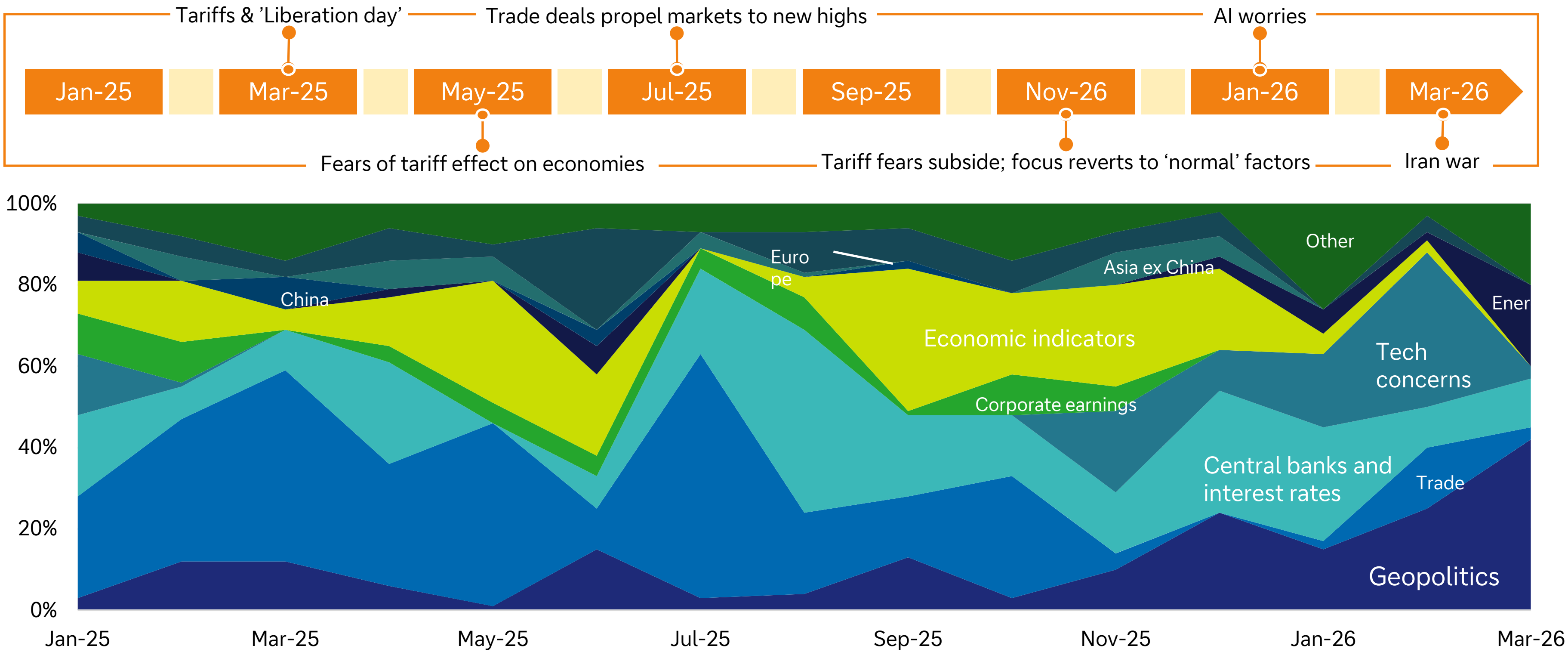
Energy markets
Investors are not pricing in a recession, but if oil remains above $90, they may suddenly do so

Impact of Q1's key themes on the S&P 500 (AI assisted by dbLumina)



Source: dbLumina, Deutsche Bank.

Our AI analysis shows the importance of each 'theme category' to markets each month



Source: dbLumina, Deutsche Bank.

The impact of today's themes 1 year ahead

We asked our AI model, dbLumina, to analyse the impact of the themes for the quarter. The 'positive' and 'negative' indicators are relative to expectations for the year at the beginning of the quarter. For example, a net negative score for the S&P 500 does not mean we expect the index to fall. Rather it means we see the rising probability that the result will be lower than previously expected.

Theme		GDP Growth	S&P 500	Interest Rates[1]	Inflation[1]
1	Middle East Conflict	Significantly Negative	Significantly Negative	Moderately Positive	Significantly Positive
2	Stagflation & Hawkish central banks	Significantly Negative	Significantly Negative	Significantly Positive	Significantly Positive
3	"Picks & Shovels" AI	Significantly Positive	Significantly Positive	Slightly Positive	Slightly Positive
4	Market Correction	Slightly Negative	Moderately Negative	Neutral	Neutral
5	Oil Price Volatility	Moderately Negative	Moderately Negative	Moderately Positive	Significantly Positive
6	Tech/Semiconductor Disruption	Slightly Negative	Moderately Negative	Neutral	Neutral
7	Slowing European Economy	Slightly Negative	Slightly Negative	Neutral	Neutral
8	Weakening Euro	Slightly Negative	Slightly Negative	Neutral	Slightly Negative
9	Landmark Legal Rulings	Slightly Negative	Moderately Negative	Neutral	Neutral
10	G7 Divisions	Slightly Negative	Slightly Negative	Neutral	Neutral

	GDP Growth	**S&P 500**	**Interest Rates[2]**	**Inflation[2]**
Sum of positive/negative indicators (positive is +1, negative is -1)	-9	-9	-1	-6

1) For interest rates and inflation, "positive" implies upward pressure on rates/prices
2) A score >0 for interest rates and inflation is 'good' ie. they are expected to go lower.

The impact of today's themes 3 years ahead

We are far more optimistic into the medium term. That appears to reflect the fact that the background economic and corporate fundamentals are relatively solid. Still, there are numerous trends working against global economic growth, although these factors will be specific to individual countries.

Theme	GDP Growth	S&P 500	Interest Rates[1]	Inflation[1]
1 Middle East Conflict	Slightly Negative	Slightly Negative	Slightly Positive	Slightly Positive
2 Stagflation & Hawkish Banks	Neutral	Slightly Positive	Moderately Negative	Moderately Negative
3 "Picks & Shovels" AI	Significantly Positive	Significantly Positive	Moderately Negative	Moderately Negative
4 Market Correction	Neutral	Slightly Positive	Neutral	Neutral
5 Oil Price Volatility	Slightly Negative	Neutral	Slightly Positive	Slightly Positive
6 Tech/Semiconductor Disruption	Moderately Positive	Moderately Positive	Slightly Negative	Slightly Negative
7 Slowing European Economy	Slightly Negative	Slightly Negative	Neutral	Neutral
8 Weakening Euro	Neutral	Neutral	Neutral	Neutral
9 Landmark Legal Rulings	Neutral	Slightly Positive	Neutral	Neutral
10 G7 Divisions	Slightly Negative	Slightly Negative	Neutral	Neutral

	GDP Growth	S&P 500	Interest Rates[2]	Inflation[2]
Sum of positive/negative indicators (positive is +1, negative is -1)	-1	+3	+1	+1

1) For interest rates and inflation, "positive" implies upward pressure on rates/prices
2) A score >0 for interest rates and inflation is 'good' ie. they are expected to go lower.

Emerging themes – part 1

We asked dbLumina to help us identify the key emerging themes that have the potential to most impact markets and economies in the short term

1 The 'AI hardware' squeeze and supply chain bifurcation

In Q1, there was a clear "bifurcation" between AI beneficiaries (hardware) and laggards (software), as investors grew more discerning.

12-Month Outlook: The focus will shift from the conceptual "AI disruption" to the tangible constraints and opportunities in the hardware manufacturing process.

We expect increased volatility in semiconductor stocks as investors scrutinise margin pressures, capex guidance, and the ability to maintain technological leadership.

US policy on the export of advanced chips will likely impact earnings for a concentrated but critical group of companies.

2 Sovereign fiscal fragility and the rising cost of debt

Mounting government deficits and rising debt service costs may become more prominent, particularly given the energy price shock.

The dollar's drop and then rally in Q1 partly reflected growing unease about the US fiscal and policy path. In Europe, fiscal concerns in France widened spreads in a sign of how rapid the issue can impact markets.

12-Month Outlook: We expect fiscal discipline to emerge as a continuous market narrative rather than a sporadic one.

Yield spreads may spike rapidly as markets price in a 'great divergence' in fiscal health. This may lead to near-term headwinds for equity markets in more-indebted nations and could lead to renewed domestic political tensions, particularly in Europe.

Emerging themes – part 2

A key learning from the ‘emerging theme’ analysis is the focus on some longer-term themes that have been pushed aside by the conflict in Iran

3 Reshoring and the proliferation of “friend-shoring"

This builds on the theme of ‘middle powers’ building alliances between themselves, restructuring global supply chains. In Q1, Canada was particularly active with Prime Minister Carney visiting numerous middle power countries to seek closer economic ties. Meanwhile, the EU and India progressed a trade deal and, within Europe, there was a greater push to build out domestic military capabilities.

12-Month Outlook: ‘Middle power’ countries are likely to increase their capital investment and M&A activity in sectors deemed strategically important, such as semiconductors, critical minerals, and defence.

This will benefit industrial, materials, and energy sectors in politically aligned countries. However, it may also lead to higher costs and inefficiencies for multinational corporations as they are forced to reconfigure their supply chains.

This may drag on margins and add low-level, persistent inflation.

4 Strain in US-consumer and labour market dynamics

There is a worrying paradox of weak consumer sentiment but a resilient (potentially fragile) economy – a trend confirmed by economic reports and surveys in Q1. In particular, the "two-speed" economy became more apparent, with reports noting job cuts in sectors like government, finance, and warehousing (healthcare remained strong). This suggests a lack of broad-based strength.

12-Month Outlook: The health of the US consumer is our most closely watched variable for the US economy and the S&P 500. Any slowdown will put the Fed in a difficult position, forcing it to choose between fighting inflation and supporting a weakening economy. A key indicator is the performance of consumer discretionary and staples stocks.

Current evolving risks

The conflict in the Middle East has acted as a catalyst, amplifying pre-existing inflationary pressures and forcing central banks into a hawkish corner, thereby creating a significant risk of a global stagflationary shock

1 The stagflationary shock: A vicious cycle

The global economy is facing a stagflationary threat

The conflict in Iran has caused a dramatic energy price shock which has occurred in an environment where inflation was already proving persistent.

Key central banks have responded with cautious or hawkish rhetoric, signalling that inflation remains a priority even at the cost of economic growth.

2 Geopolitical escalation and fragmentation

The conflict in Iran is the epicentre of global risk and has quickly metastasised with the involvement of non-state actors (Hezbollah, Houthis etc), shipping disruption, and attacks on regional neighbours.

The divisions within the G7 have become clear with European allies showing frustration at the lack of consultation. That further risks a fracturing of traditional Western alliances

3 Equity market de-risking and fragile sentiment

Global equity markets are in a correction phase with the S&P 500's peak-to-trough drop being more severe than the average response to geopolitical shocks.

The positive narrative around the "Picks and Shovels" of AI, while still present, has been completely overshadowed

4 European economic vulnerability

Europe is particularly exposed as it faces an energy shock and a sharp economic slowdown.

The ECB is in an exceptionally difficult position, forced to maintain a hawkish stance to fight imported inflation despite a weakening domestic economy

Emerging risks underestimated by the market – part 1

While investors are currently grappling with the immediate impact of the Iran conflict, there are several second- and third-order consequences that appear to be underestimated

1 European sovereign debt concerns

The underestimated risk

The combination of a hawkish ECB, slowing growth, and the need for increased fiscal spending on defence and energy security is sowing the seeds of a renewed sovereign debt crisis in the Eurozone.

Historical lesson & analysis

The crisis of 2011-2012 showed how quickly concerns over the fiscal position of one member state can create systemic contagion. The yields on peripheral debt are a potential canary in the coal mine. A hawkish ECB, may have a difficult choice to make, particularly if one of the options involves removing the safety net that has protected the Eurozone for a decade.

The divisions in the G7 suggest that a coordinated political solution, which was crucial to resolving the last crisis, will be much harder to achieve this time.

2 The “picks and shovels" AI bubble

The underestimated risk

While the "picks and shovels" narrative appears robust, it is not immune to a bubble. The massive capital expenditure into the AI supply chain is predicated on future returns that are still largely theoretical.

Historical lesson & analysis

History is replete with infrastructure-led bubbles, from the railway booms of the 19th century to the dot-com fibre-optic rollout. In each case, the building of the infrastructure was a real economic activity, but the eventual returns on that capital were wildly overestimated, leading to a collapse.

The risk is that the market has capitalised various firms based on a gold-rush mentality without fully accounting for the risk of overcapacity or technological obsolescence.

Emerging risks underestimated by the market – part 2

Just two of the big things that markets take for granted is the leadership of Big Tech firms and the ability of companies to adapt their supply chains. But what if developing events erect permanent obstacles?

3 The unravelling of the "Big Tech" social contract

The underestimated risk

The recent court verdicts against Google and Meta are likely not isolated events but the start of a fundamental, and costly, regulatory and legal realignment for Big Tech.

Historical lesson & analysis

The recent rulings are comparable to the landmark rulings against Big Tobacco. There, it took decades for the full financial and social impact of the tobacco litigation to be fully realised. Markets appear to be viewing today's verdicts as one-off legal setbacks. However, they may mark a pivotal shift in the "social contract" for technology firms.

The legal precedent could open the floodgates to class-action lawsuits and force a complete overhaul of business models built on user engagement and data monetisation. In turn, this will lead to structurally lower growth and profitability for some of the market's largest constituents.

4 The illusion of supply chain adaptation

The underestimated risk

While markets have been calmed by signs of supply chain adaptation, such as Saudi Aramco re-routing oil, these are suboptimal and less efficient solutions that will create a drag on global productivity.

Historical lesson & analysis

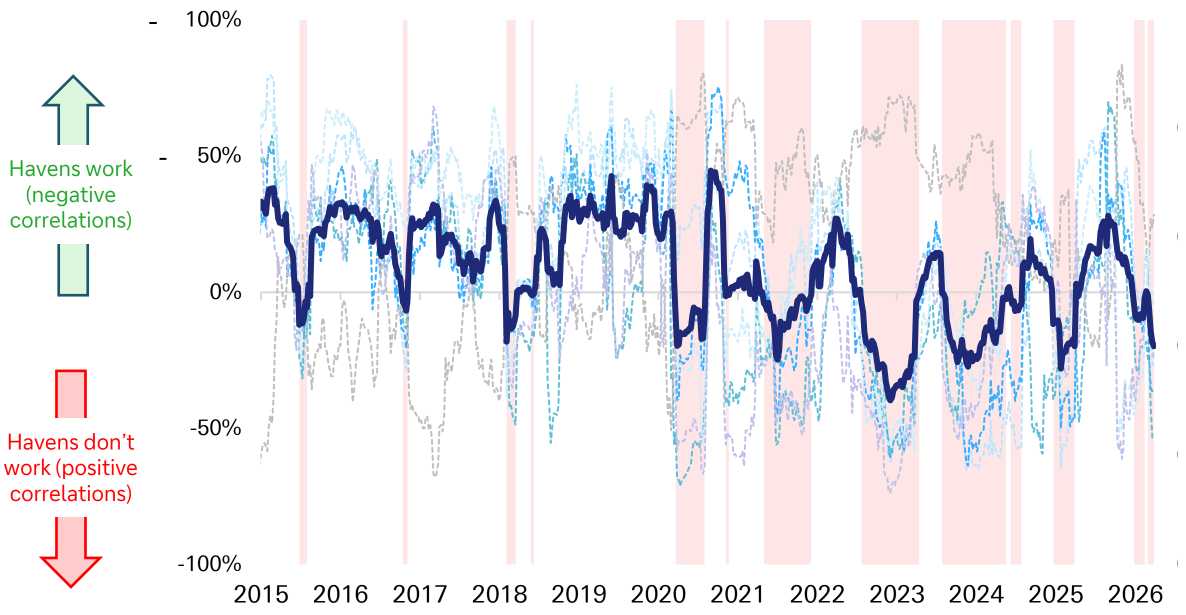
The response to covid and the Ukraine war showed that supply chains can adapt. However, that adaptation came at the cost of higher inflation and lower efficiency. The 'friend-shoring' and re-routing currently being celebrated are, in essence, a tax on the global economy.

Moving from a just-in-time, efficiency-focused model to a just-in-case, resilience-focused one has a cost that may be underestimated in long-term growth and inflation forecasts. This structural inefficiency will linger long after the immediate crisis has passed.

The mysterious case of the disappearing haven assets

Our 'haven indicator' shows that traditional havens have been largely absent in the 2020s

Our haven indicator (20-wk rolling, avg across pairwise correlations with S&P 500)



Source: Bloomberg Finance LP, Deutsche Bank.

How our haven indicator works

We take six traditional haven assets:

1. Gold
2. US dollar
3. Japanese yen
4. Swiss franc
5. 10yr treasury
6. 10yr bund

We then analyse their individual correlation with the S&P 500 (our proxy for 'risk') over time. Finally, we take an average of the six correlations.

The red shading indicates periods where the average haven **did not work** (ie. positively correlated with the S&P 500). This includes:

1. Iran war in 2026
2. US tariff crisis in 2025
3. Peak fear about interest rate rises in 2022
4. Initial covid turmoil in 2020

2. The market impact of specific themes

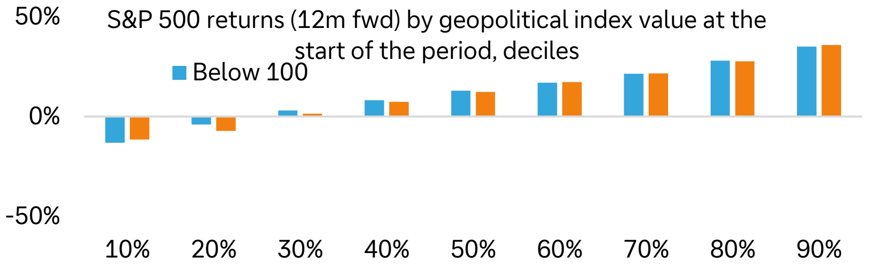
Oil shocks and markets

See our in depth reports here and here

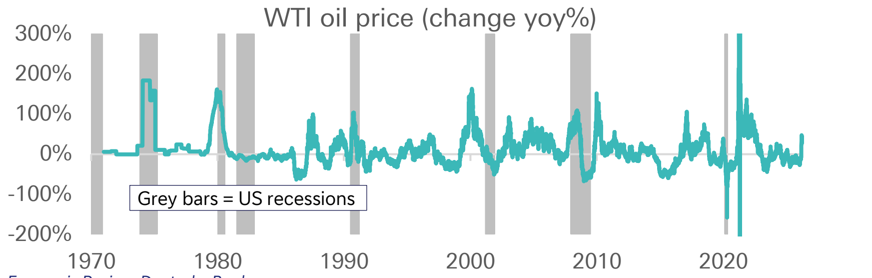
The moves in 10yr treasury yields are highly dependent on whether oil markets and thus US inflation are affected

- Linear models suggest that even very large oil price shocks may have muted effects on the US economy. But the assumption of linearity may be at odds with the historical observation that oil price shocks tend to coincide with recessions.
- Historical episodes of geopolitical tensions show little evidence of prolonged US equity drawdowns, with the median return of 9.8% in the 12 months after event starts. This holds regardless whether an event was related to the Middle East.
- The indifference to geopolitical uncertainty in the medium term is present not only in the US stocks. Among different economies since 1900, it's mostly European and Australian/Canadian stocks that experienced notably lower returns when the geopolitical risk index has been above 100 (a level it has been at for a few months now).
- Among commodities, oil is a better proxy for geopolitical risk in markets (outside covid) than gold even with the US less reliant on oil than in the 1970s. The impact on gold has actually been marginally lower when the geopolitical risk index is above 100 in periods since 2010
- Gold correlations also point to a lack of broader geopolitics as the key driver, especially recently – gold has been negatively correlated with crude and US defense stocks in the last couple of months
- What about defence stocks? – those respond to fiscal actions due to geopolitics and have been more responsive to the slides in geopolitics risk more than the upsides recently. This is to be watched given the run up in valuations

Whether geopolitical risk is high or low, S&P 500 returns after 12m tend to be similar



Recessions often coincide with sharp increases in oil prices.



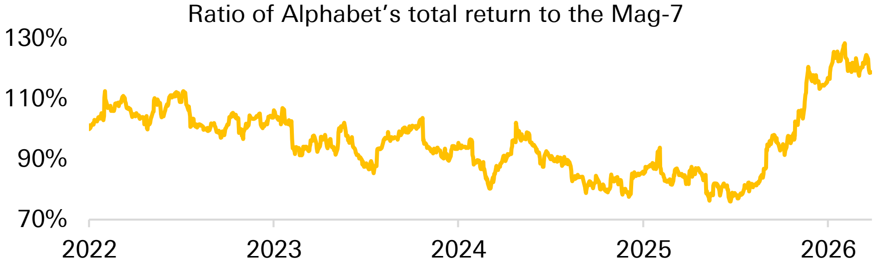
Source: Bloomberg Finance LP, Finaeon, Caldara, Dario and Matteo Iacoviello (2022), "Measuring Geopolitical Risk," American Economic Review, Deutsche Bank.

Software and technological disruption

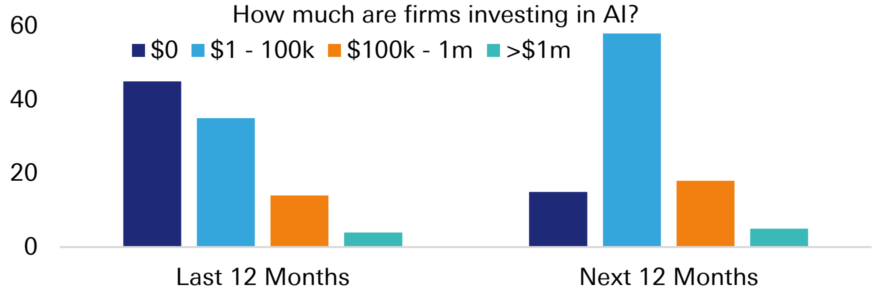
This sell-off is not the first … but the frequency of new AI tools has accelerated in recent months

- The software scare saw the largest sell-off in US software and services stocks vs the broader market since the ChatGPT release in late 2022. Notably, the divergence between "hard" and "soft" IT industry groups was unusually wide.
- Semis and tech hardware were the relative winners in tech. Notable losses occurred in a small number of management and development real estate firms
- Other parts of the AI ecosystem have been notably resilient. Meanwhile, the drop in software & services means there is a technical reinforcement to the selloff
- Amid all the AI investment themes, there is more-or-less a consensus that power demand will surge. This helps shows that concerns about other parts – demand and supply, are becoming more frequent.
- On top of the Google example, following the DeepSeek announcement, Nvidia's stock fell around 20%. Revenue estimates have doubled since then.
- There is also a lot of uncertainty about the impact on the more tangible firm outcomes, such as labour costs and customer satisfaction
- We aren't ignoring the real issues. Several factors support the "AI disruption" narrative. The issue is that the sell-off has been so indiscriminate. Thus, deleveraging and momentum selling of broader industries was a key driver.
- Lastly, public credit markets are the key to a real risk-off event. These show a relatively small exposure to technology (8% in $HY and 10% in IG) and recent spread widening is still modest. Private credit is more exposed but several factors including dry powder limit the risk of a proper turmoil as discussed here.

The Google example: Alphabet lagged after the release of ChatGPT but revenues continue to be strong which assuaged market's concerns about its LLM progress



Will software companies' clients substitute with their own AI tools? Many firms want to increase AI spend but the investment may not fully replace solutions from specialised firms.



Source: Bloomberg Finance LP, Duke University, FRB Richmond and FRB Atlanta, The CFO Survey - Q4 2025 (November 11 - December 1, 2025), Deutsche Bank.

Central bank policy amidst an oil shock

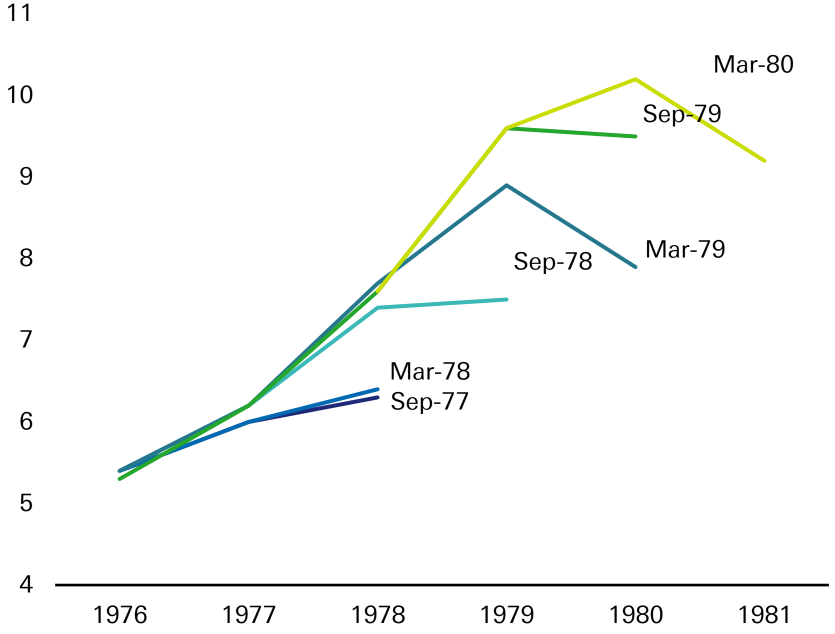
See our in-depth reports here and here

US interest rates are likely to remain on hold through to September

- Central banks face another energy shock, and were criticised last time for acting too late. This time, we can already see how they're trying to avoid a repeat, and are sounding less dovish (and sometimes hawkish) for a given level of inflation.
- Unlike in 2022, there's now an explicit acknowledgement that inflation has been above target for an extensive period. Indeed, Fed Chair Powell said in his March press conference that looking through energy shocks has "always been dependent on inflation expectations remaining well anchored", and he also mentioned the "broader context of five years now of inflation above target."
- ECB President Lagarde said the central bank will "do all that is necessary to ensure inflation is under control and French and Europeans don't suffer the same inflation increases like those we saw in 2022 and 2023."
- There is already discussion about the prospect of rate hikes pre-emptively, even before inflation has moved higher, rather than the more reactive conversation that took place in 2021-22.
- For the Fed, the latest trends suggest that a wait-and-see approach could gain broader support from more officials in the near term, absent a sudden deterioration in the labour market or a spillover of energy prices into core inflation. Our baseline expectation remains that the FOMC has likely delivered its final rate cut under Powell's leadership, and rates will remain on hold until September.

It is easy to underestimate inflation during an energy shock

The Fed's staff inflation forecasts during the 2nd oil shock in the 1970s (%)



Source: Bloomberg Finance LP, Deutsche Bank.

Private markets

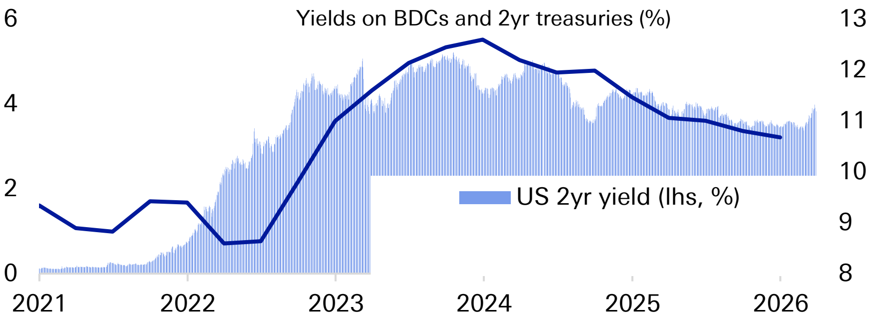
See our in-depth reports here and here

More transparency and communication are likely as private capital targets a more risk-averse and liquidity-demanding investor base to grow.

- In spite of the negative headlines and market performance, the largest managers continued with their exceptional fundraising and AUM growth. The focus is now on keeping the fundraising momentum, including among retail investors, in 2026.
- Where will the AI disruption fallout hit? There will be losers in private markets but price action in private capital-related assets that can be traded is likely misinterpreting the impact
- Any serious fallout from the actual impact of AI and credit fears on private capital is likely to be concentrated in the smaller, lower-quality corners of the industry.
- Many negative headlines have come from liquidity rather than credit risks amid withdrawal requests
- Dry powder continues to rise and is a sizable buffer to sector-specific problems.
- Outside software, exits are growing quickly – this should assuage investor concerns about private equity and free up more capital to do deals.
- There is thus some scope for a reversion in the recent selloff.
- We expect corporate buyers to step back into M&A. In fact, this may become private managers' preferred exit route given that public markets remain volatile amid geopolitical tensions.
- The biggest risk is a confidence crisis that grips wider financial markets even if it is not based on strong fundamental problems

Potential losses on investments is low down the list of concerns for private capital managers – the economy and public markets are far more impactful

Reported dividend cuts can relate to falling interest rates rather than solely outright credit distress given that BDC yields are driven by short-term rates



Source: Bloomberg Finance LP, Deutsche Bank.

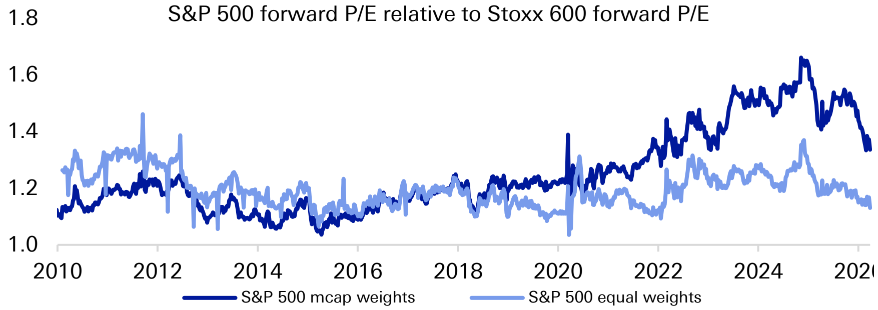
Corporate earnings – what the latest set of results tells us

See our in-depth report here

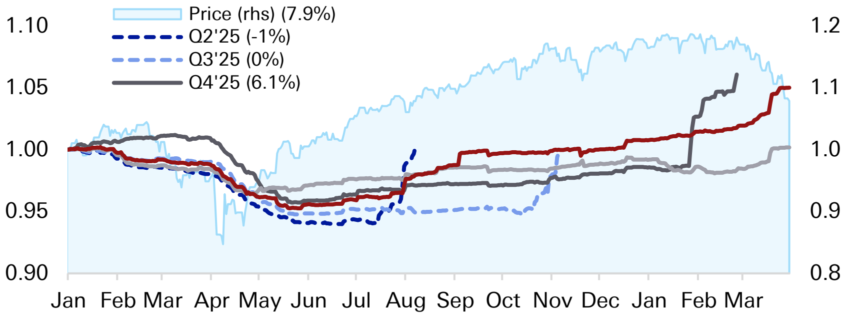
US stocks had a good earnings season with Q4 consensus EPS meaningfully moving up It was more of a mixed picture in Europe

- It's been another strong season in the US, with more dispersion in Europe. Notably, since the start of the earnings season, sectors that posted larger earnings beats have performed worse. This sets up for more gains once investor sentiment around the rotation trade and AI stabilises.
- The rotation into a wider set of stocks (despite strong earnings from tech) is a sign of persistent risk appetite. It also shows that once the dust settles on software fears, a rerating is likely. The IT sector now trades effectively on par with the market, lower than during the 2022 tech wash out in relative terms.
- There has also been a strong yield preference among investors, with dividend stocks in favour before the Iran conflict as the Nasdaq has been slipping. High-growth names including those outside TMT have also been performing strongly until March, showing the rotation has been rather exclusively out of tech as opposed to a full risk-off move.
- Zooming out, investor aversion to software means the US now significantly trails rest of the world. But perhaps more importantly, this shunning of US large cap tech pushed other DM markets to unusually high valuations. Only 4 economies out of 23 are now relatively cheaper vs own history over the last 5 years than the US.
- The equal-weighted version of the S&P 500 is now lower relative to European markets than the average since the data begins in 2010.
- Earnings will be another catalyst, in addition to valuations, that may rerate US markets v RoW higher (despite momentum headwinds).

The P/E premium of the S&P 500 over the Stoxx 600 is now also the lowest since 2022, wiping out the gains made since investors' AI enthusiasm proliferated.



S&P 500 earnings evolution since 2025



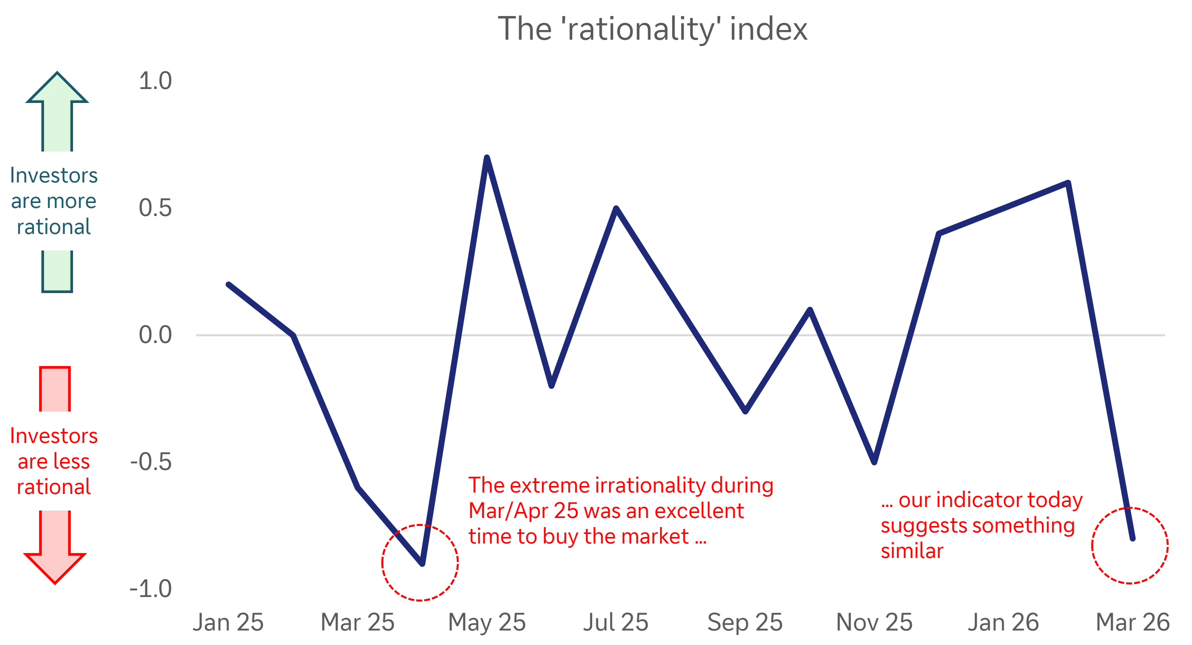
Source: Bloomberg Finance LP, Deutsche Bank.

3. Investor psychology over the last quarter

Are investors rational? Right now, our indicator says “No”

Investors appear to be good at analysing markets during normal times and terrible during periods of uncertainty

Our dbLumina gauge indicates investors have been irrational during market drops and periods of geopolitical tensions



Source: dbLumina, Deutsche Bank.

What does it mean to be ‘rational’

March 2026 Extreme Irrationality (Panic/Fear):
The escalation of the Iran conflict led to a spike in oil prices and a broad market sell-off, characterized as a "stagflationary shock". This overwhelming fear, with investors largely selling off risk assets, indicates a highly emotional and irrational response, despite brief moments of optimism.

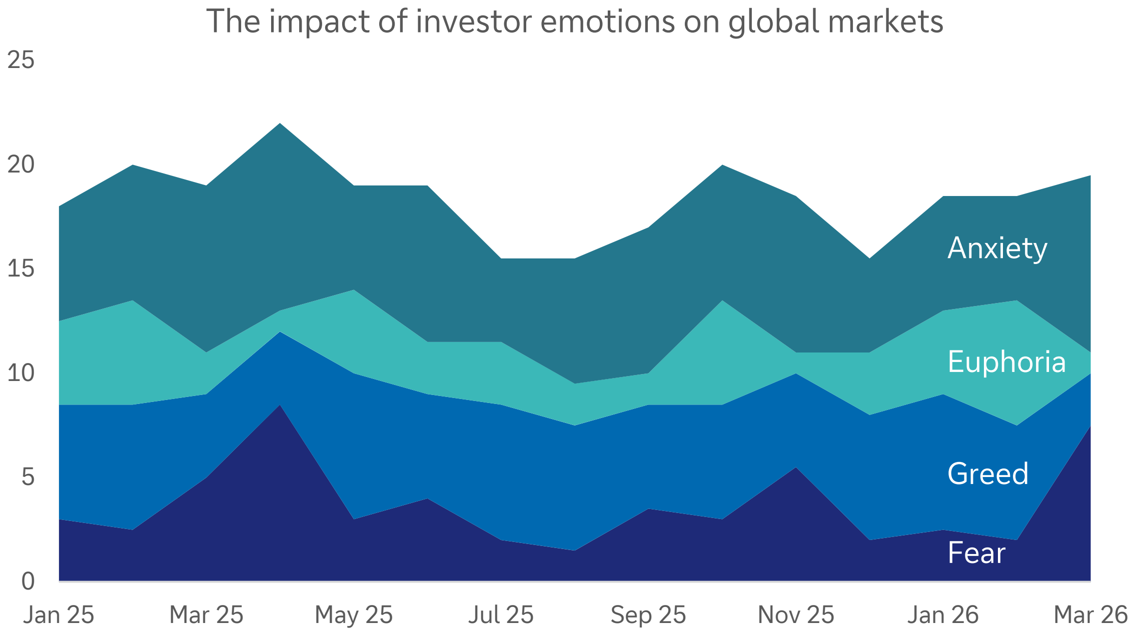
Fear Indicators:

Sharp, disproportionate reactions to threats, widespread sell-offs on announcements, panic, flight to safety beyond what might be fundamentally warranted, strong emotional language in descriptions (e.g., "roiled," "plunge," "blistering sell-off," "panic selling,“).

The impact of four market-moving emotions: fear, greed, euphoria and anxiety

dbLumina tells us that anxiety and greed can go hand-in-hand even as fear and euphoria wax and wane.

Anxiety and fear ruled Q1



Source: dbLumina, Deutsche Bank.

Status-quo bias is strong. That is creating more market volatility

Investors have been reluctant this decade to consider the upside and downside before the event. So, when something 'bad' happens, market drops can be more aggressive than expected.

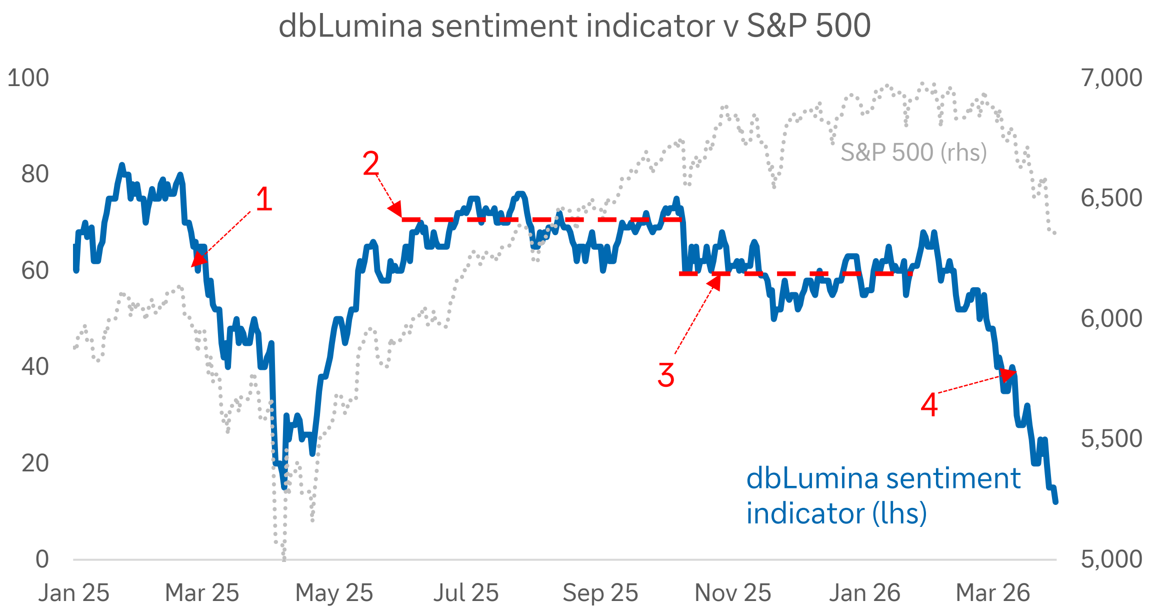
Some examples:

1. Demography and fiscal largesse is slow moving and predictable
2. Rush to price in the downsides of tech disruption
3. Fear of considering sovereign risk
4. Fear of underperforming peers is greater than the desire to beat them
5. Unwillingness to price in the upside of iterative technology improvements
6. Narrow vision of what globalisation is and the fear that it has peaked

We created our own gauge of sentiment. It uses AI to interpret macro, economic, and market signals.

This is our response to the fact that many confidence indices today fail to reflect reality. That is because they can fail to take into account the technological and social megatrends that now affect markets.

It is too early to say our gauge has predictive qualities but …



Source: Bloomberg Finance LP, dbLumina, Deutsche Bank.

1. Indicator drops 20 points in Feb-25 before the most dramatic stock market declines relating to tariffs.

2. Consistent strong sentiment as markets recovered from the tariff-induced sell-off.

3. Lower sentiment as markets top-out on valuations and software/PC nerves grow. Yet, no catalyst for a large drop.

4. Our sentiment indicator commenced a steep drop on 3-Feb. It fell 20 points before the war in Iran kicked off.

4. How Q1s themes fit in with our megatrend framework

How Q1’s themes fit inside our megatrend model

The most important megatrends signals in Q1

Technology
- The drop in the Nasdaq and low productivity

Sovereign debt & the gov. role
- Rising interest costs and fiscal policy uncertainty hurt, partially offset by robust treasuries demand

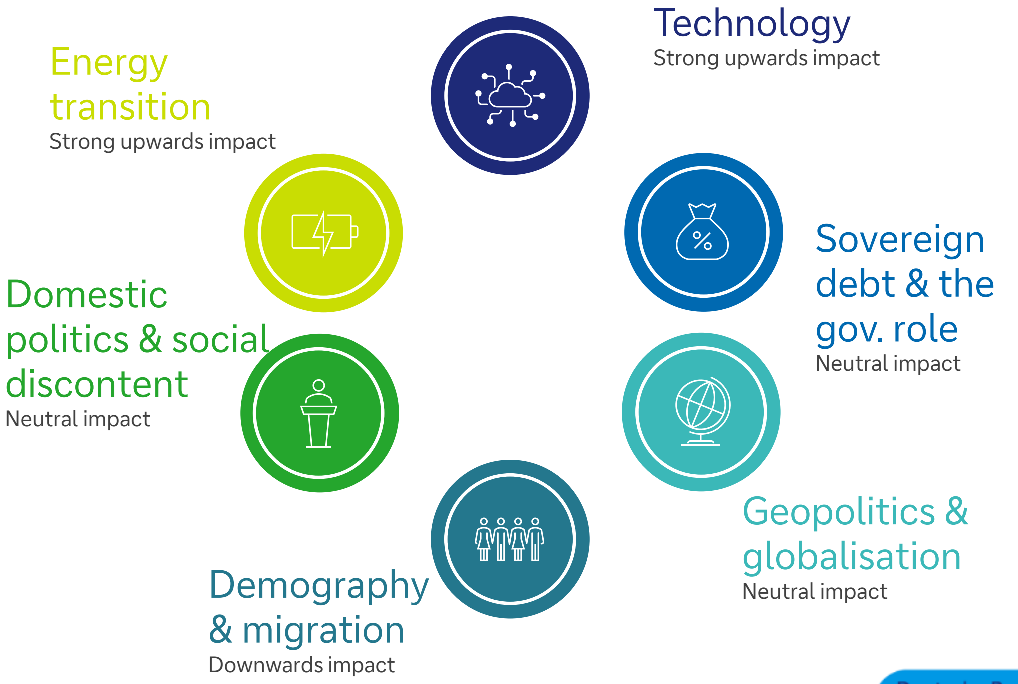
Geopolitics & globalisation
- The deteriorating geopolitical backdrop is balanced somewhat by robust global activity

Energy transition
- Rising prices of key metals

Domestic politics & social discontent
- Affordability and consumer sentiment remain a drag

Demography & migration
- A slow-burn in the working-age population

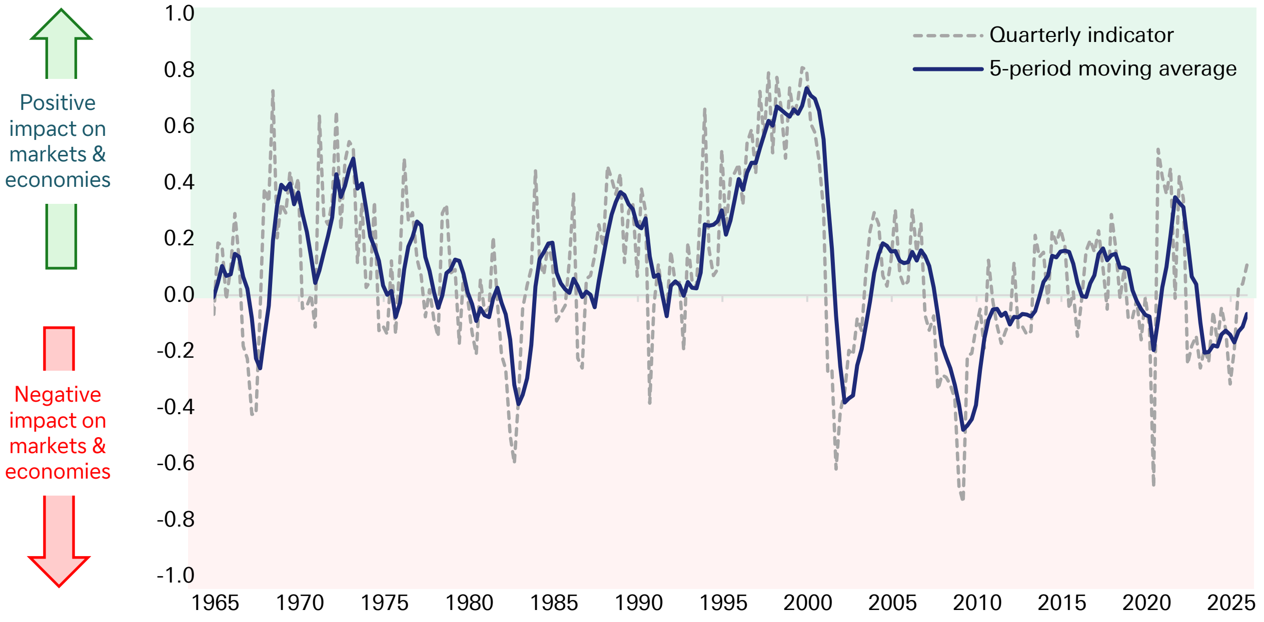
The overall impact of the ~60 signals on our megatrend model is positive for markets and economies. Our indicator has been improving in recent quarters:



Our megatrend tracker incorporates ~60 indicators

This has risen over the last 12 months, indicating that the broader megatrends are improving and so should have a positive impact on markets and economies

An aggregate of 6 megatrends



Source: Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

How the indicator works

We start with the following six megatrends:

- Technology
- Demography and migration
- Sovereign debt and deficits
- Geopolitics and globalisation
- Domestic politics and social discontent
- Energy transition

We then analyse ~60 economic, market, and social indicators that relate to each of the megatrends and analyse their impact on four key variables:

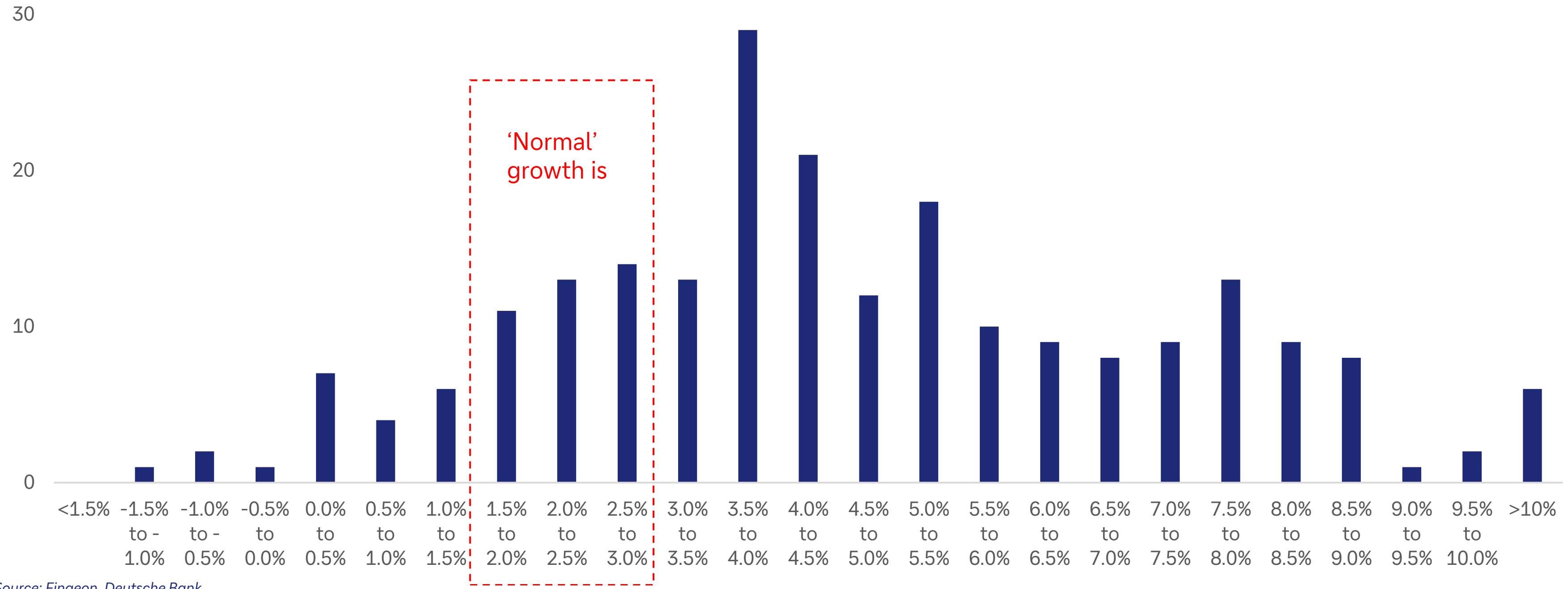
- GDP
- Equity prices
- Treasuries
- Inflation

When we aggregate the results, it gives us an indication of how the six megatrends have affected economies and markets over time.

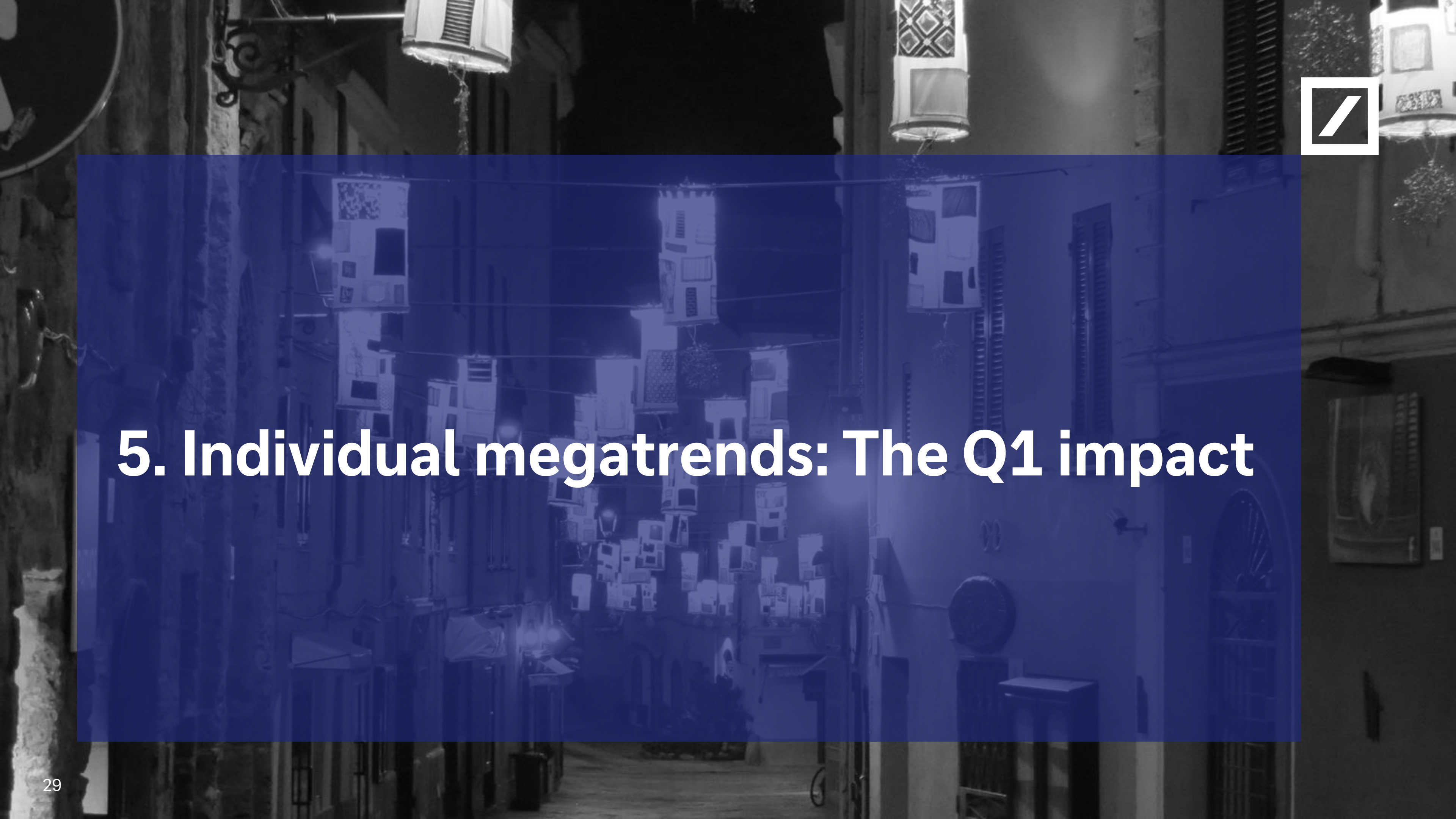
History shows that assuming ‘normal’ economic growth of around 2.5% into the medium term is very risky – in fact 74% of years have seen growth over 3% pa (on a rolling 10yr basis).

This is particularly the case when a world-changing technology is involved

Frequency of US GDP growth 1790-present (rolling 10yr, annualised)



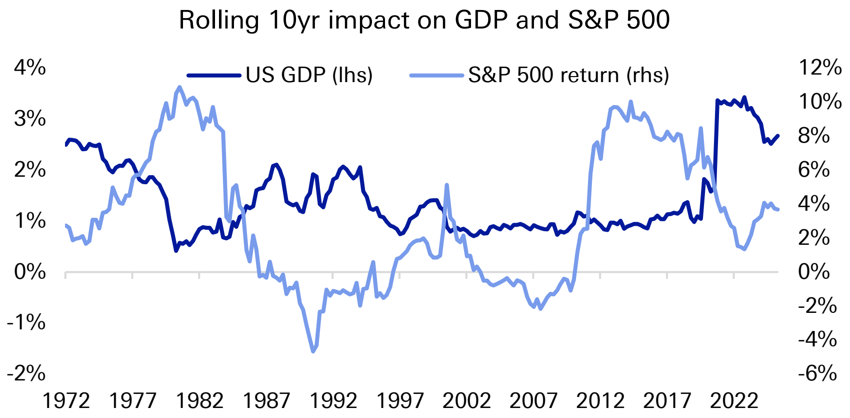
Source: Finaeon, Deutsche Bank.



5. Individual megatrends: The Q1 impact

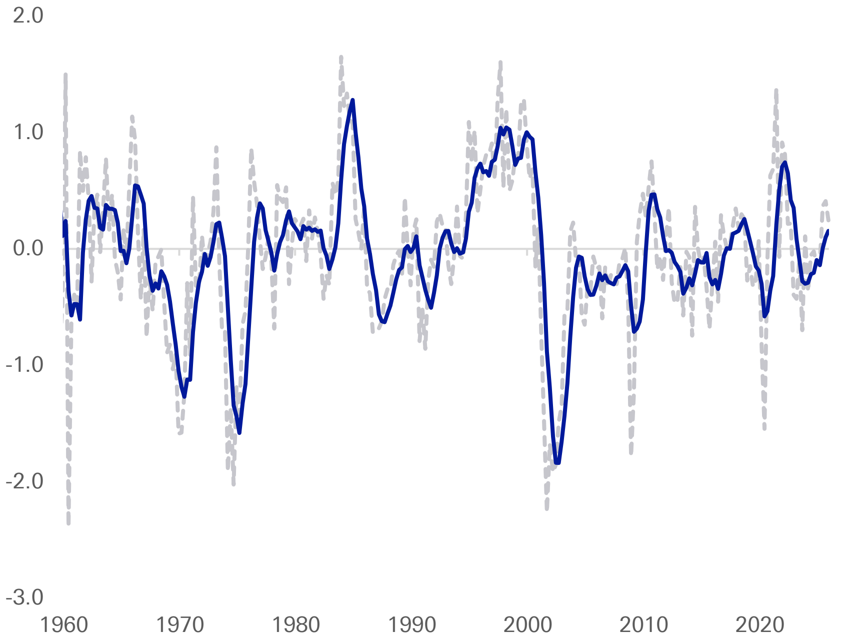
Rising tech spending lacks productivity growth for now

The -13% QoQ fall in tech stocks will be a drag on our indicator in Q1 that has been uplifted by increased tech spending in GDP accounts in recent quarters. Despite the increases in tech investment, business productivity remains somewhat tepid.



Economy-wide technology growth indicator

Higher = more spending, higher productivity

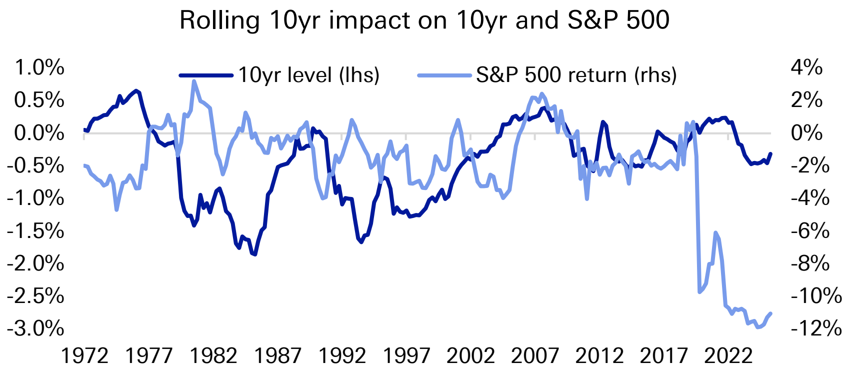


Source: Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Sovereign debt and the government's economic role

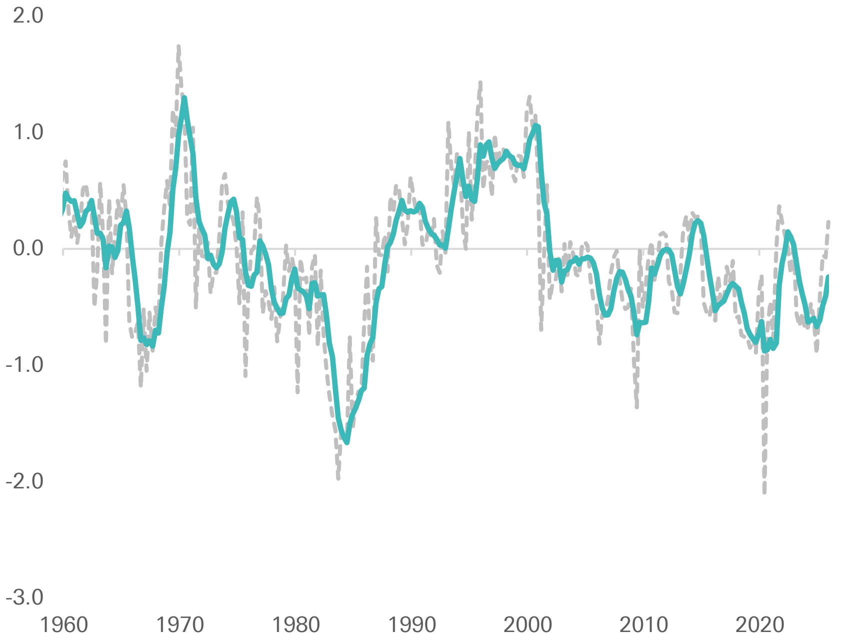
Stock gains have become dependent on fiscal expansion this decade

Our indicator remains below average for the last 10 years, amidst high debt-to-GDP, rising interest costs and elevated fiscal policy uncertainty. But a narrowing deficit, the still robust demand for treasuries and comparatively contained yields have been propping up our composite in recent quarters.



US sovereign debt and the government's role indicator

Higher = improving fiscal position, lower govt spending

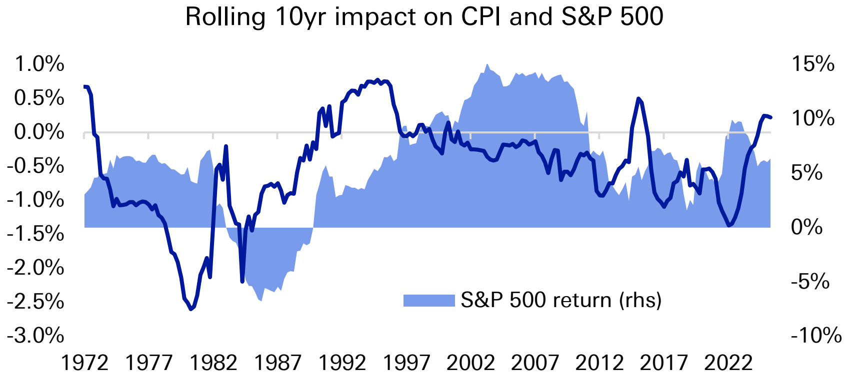


Source: Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Geopolitics & globalisation

Our gauge is at a 10-year neutral level as trade and activity hold up despite deteriorating policy

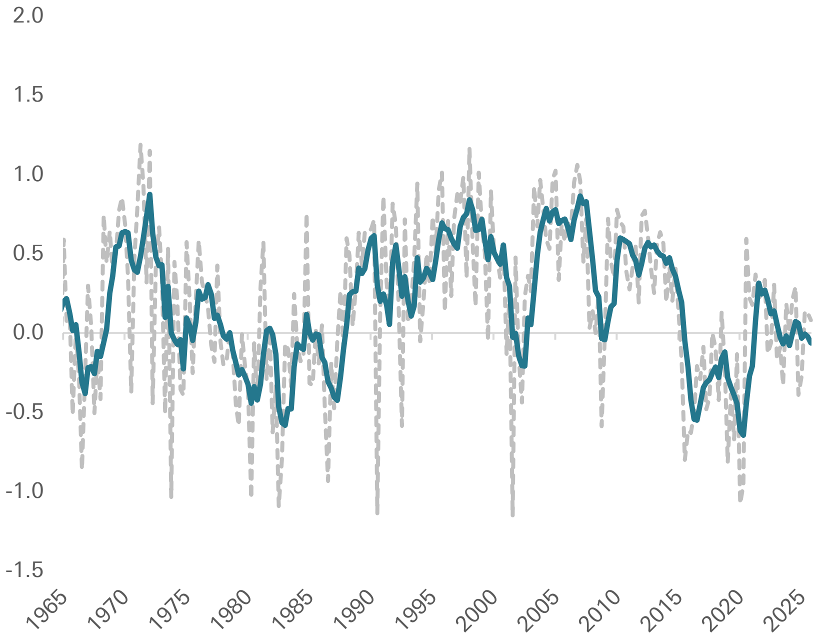
Deteriorating geopolitical backdrop and policy uncertainty have been outweighed by still robust trade and global activity backdrop, especially since the post-covid disruptions in recent quarters. Overall, however, our index has shown significant deterioration since 2010 and relative to 1990s-2000s prints.



Source: Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Geopolitics & globalisation index

Higher = more stable geopolitical backdrop, more trade



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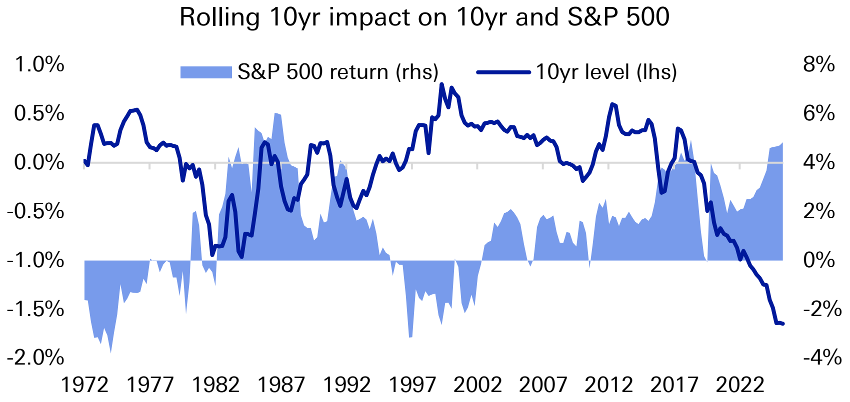
The energy transition

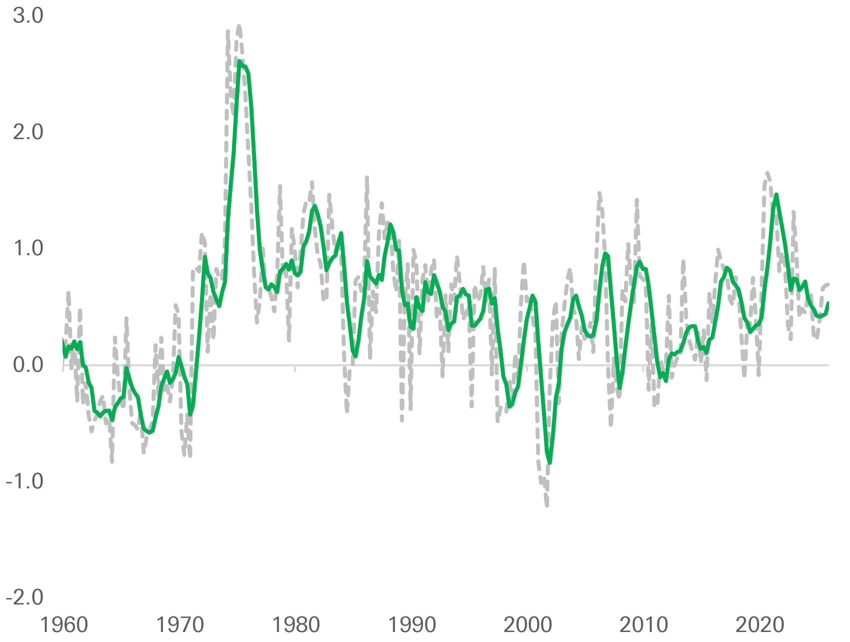
A net rise in factors helping the transition has been associated with lower yields and higher stock returns

Rising share of renewables and falling CO2 relative to GDP have been the upward drivers of our index, offsetting the impact of higher policy uncertainty and growth in absolute fossil fuels production. Metals key to the transition have also been rallying.



Energy transition index

Higher = more renewables, greater transition incentives

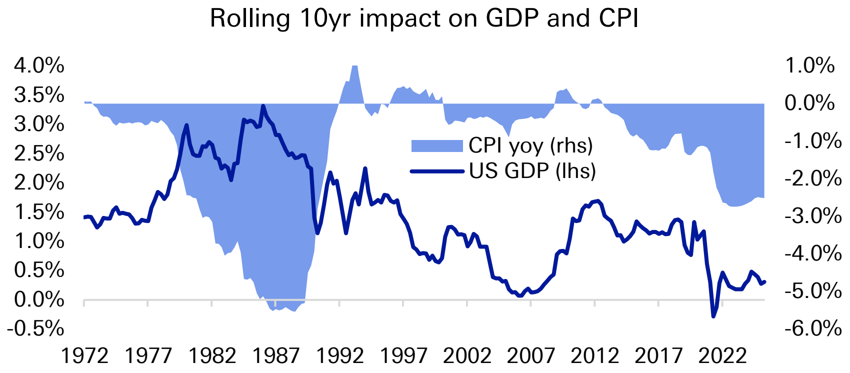


Source: Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

Domestic politics & social tensions

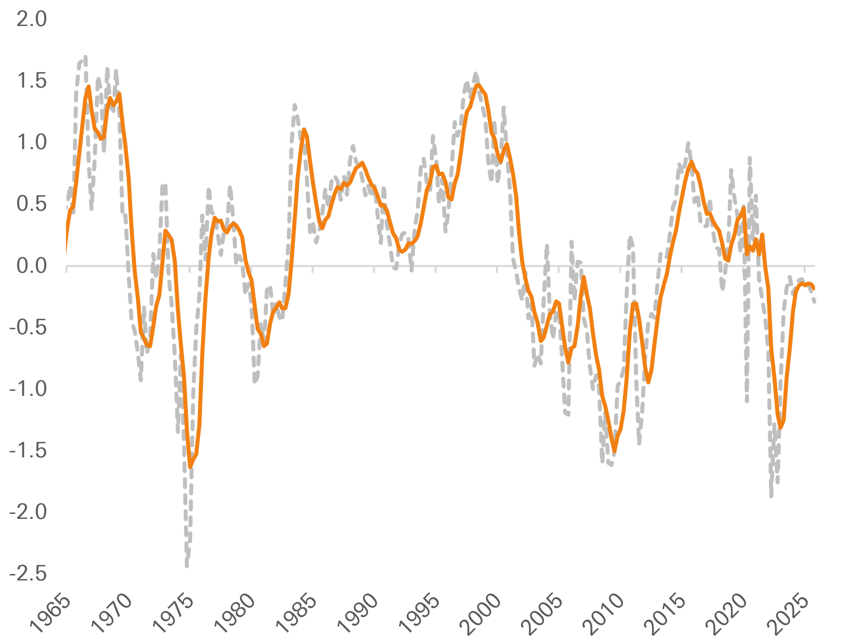
The US economy has become gradually less exposed to some of the key factors affecting consumer mood

Affordability and consumer sentiment remain a drag on our indicator, with some relative deceleration in labour market indicators also starting to weigh in. While off the lows seen in early 2020s, our gauge is yet to signal material improvement and a few indicators may deteriorate soon, including inflation.



US domestic politics and social tensions index

Higher = better sentiment, higher affordability, better labour market



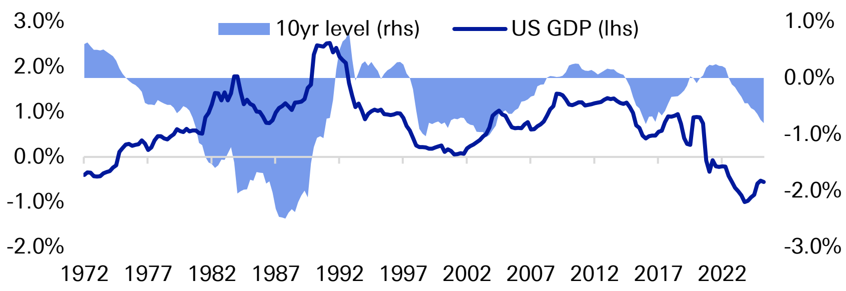
Source: Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Demographic shifts

A long-term fiscal headwind, but adaptation in some countries will likely mitigate some of the downside

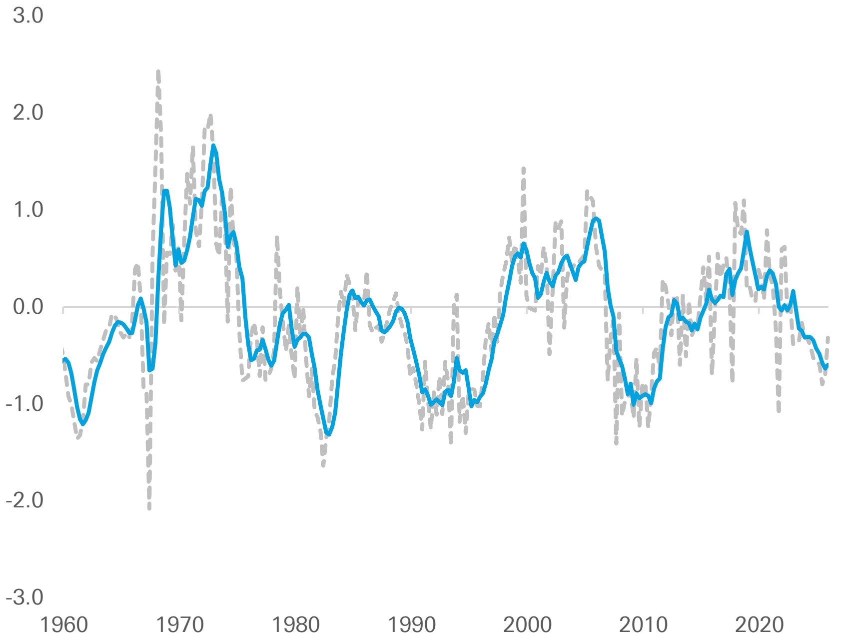
Deteriorating trends in working-age population ratio, births and immigration have been exacerbated by negative housing market dynamics recently, keeping our index c -0.5 standard deviations below the 10-yr mean. In the 1980s, the relationship between GDP (higher), CPI (lower) and better demographics was especially pronounced. This effect is much weaker now.

Rolling 10yr impact on GDP and 10yr



US demographic shifts index

Higher = younger working age population, more households and immigration



Source: Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Appendix 1

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Research

How the war has disrupted markets and economies

April 2026

Luke Templeman | Galina Pozdnyakova

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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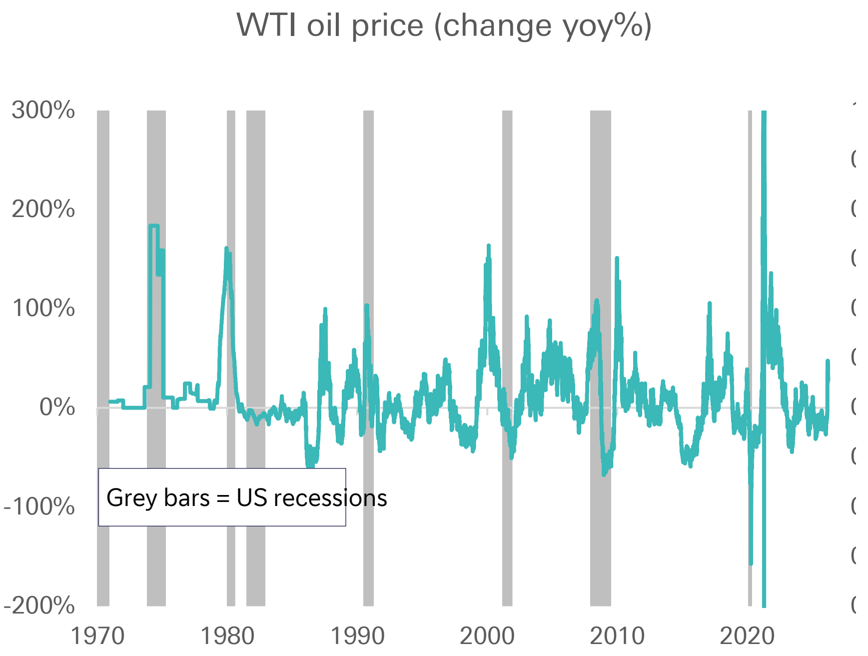
Summary: it may be too early to call the end of the conflict with Iran despite the encouraging signs. Thus, we are watching for how resilient the disruption to the world's economies and markets will be for the rest of 2026.

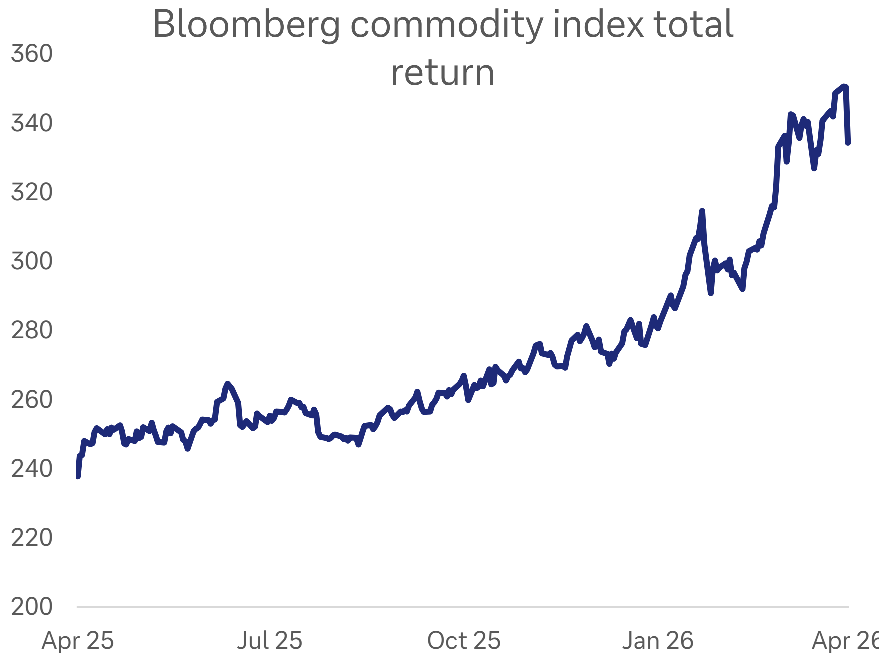
A reset? Or the return to a dangerous status quo?

- Following the announcement of the two-week conditional ceasefire between the US and Iran overnight, attention is now turning to exactly how oil tankers might begin to transit the Strait of Hormuz.
- There are inconsistencies in the logistics with Iran claiming the US has accepted its control over the strait and that transit will only be allowed under the "supervision" of its armed forces and subject to "technical limitations".
- There is also a real fear that the broader situation may return to the status quo of US sanctions and fruitless talks while a reinvigorated Iran restarts its nuclear programme. That will be a key discussion point ahead of likely US-Iran negotiations on Friday in Pakistan.
- On a day-by-day basis, there are deep uncertainties as to how long the ceasefire can hold.
- Upcoming negotiations will be difficult given Iran has claimed that the US will offer sanctions relief and security guarantees, things that may not eventuate in part or full. The most important demand in Iran's ten-point plan, its right to enrich uranium, also crosses a US red line.

It wasn't just oil – other commodities have surged and stayed high

The spike in oil prices (yoy) was less pronounced than some prior episodes that coincided with US recessions

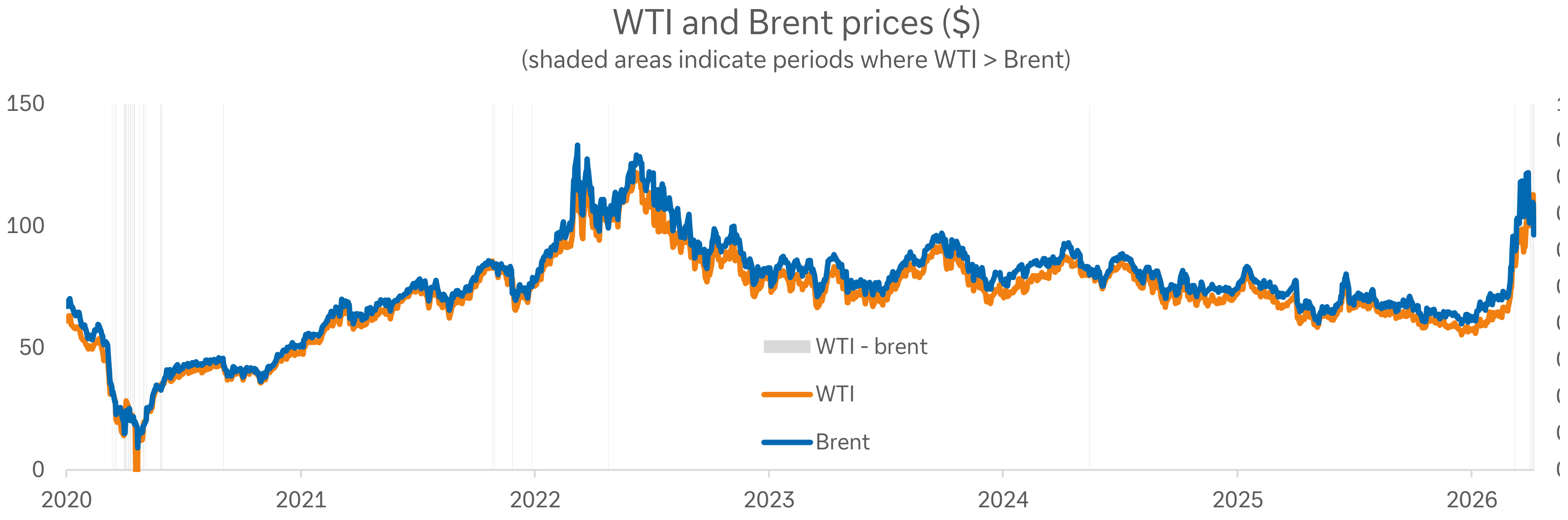




Source: Bloomberg Finance LP, Deutsche Bank.

WTI oil prices jumped above brent – that is extremely unusual this decade

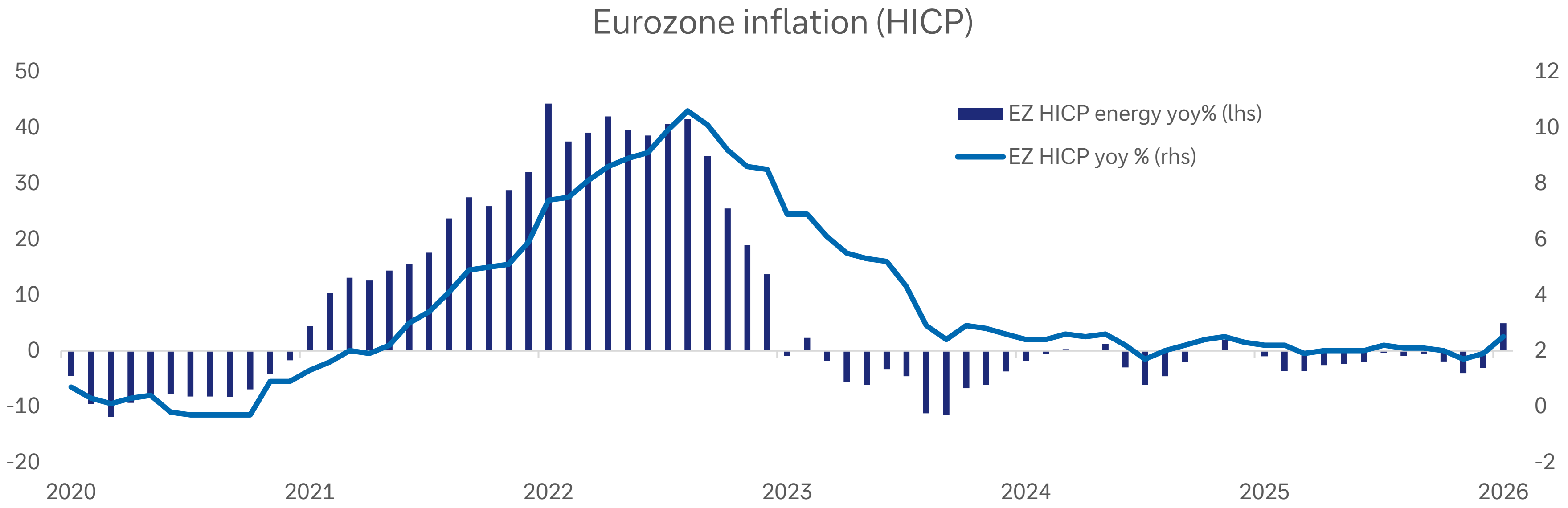
WTI's one-month delivery schedule (compared with two months for brent) was a key reason why traders moved to WTI contracts



Source: Bloomberg Finance LP, Deutsche Bank.

European inflation has climbed, however, the energy shock is so far nothing like that around the time Russia invaded Ukraine.

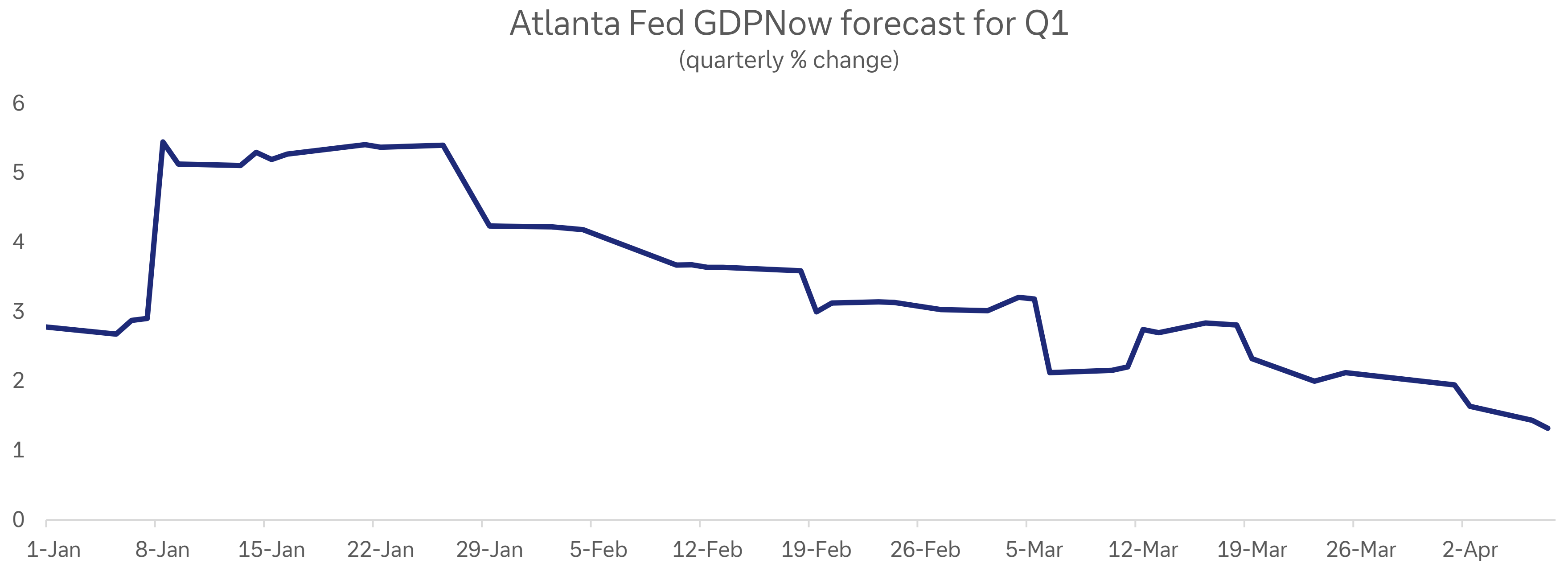
Once April's figures come in, they seem on track to show a continuing upwards trend in energy inflation. But Europe's dependence on the Middle East does not compare with its dependence on Russia five years ago.



Source: Bloomberg Finance LP, Deutsche Bank.

The war has exacerbated the falling trend for US growth forecasts

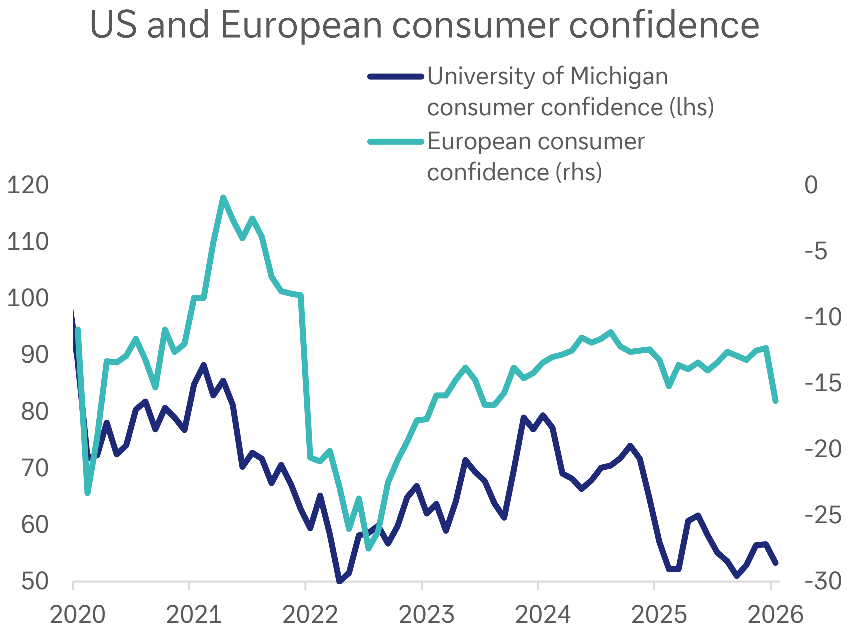
Economic growth expectations were falling before the war in Iran started. However, are more data points are released this month some will certainly put downwards pressure on growth estimates.

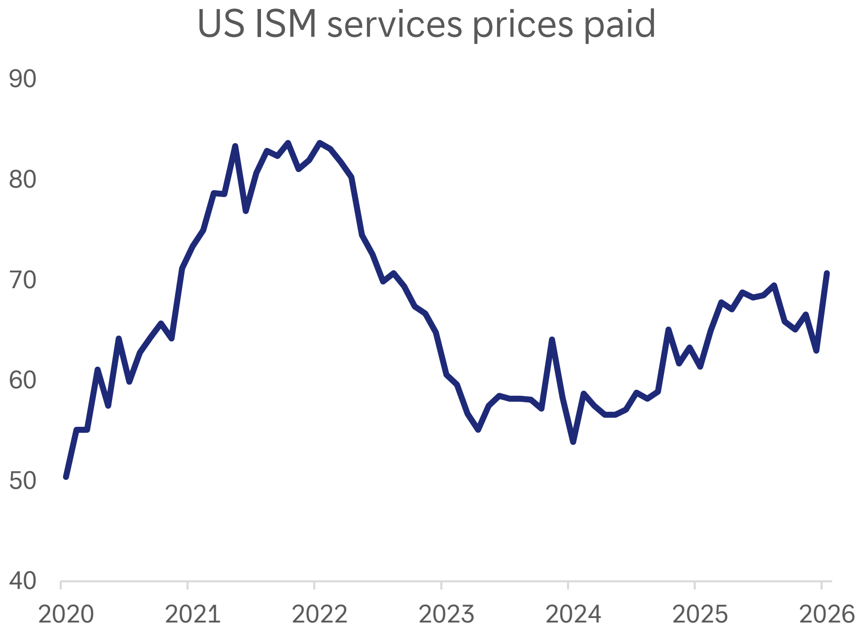


Source: Bloomberg Finance LP, Deutsche Bank.

The drop in consumer confidence might be expected. How long it lasts will be influenced by the resulting inflation.

Consumer confidence is, at least partially, dependent on whether the uptick in input cost inflation remains robust. That will also be influenced by how inflation filters through to consumers. Some corporates may decide to absorb the costs themselves.

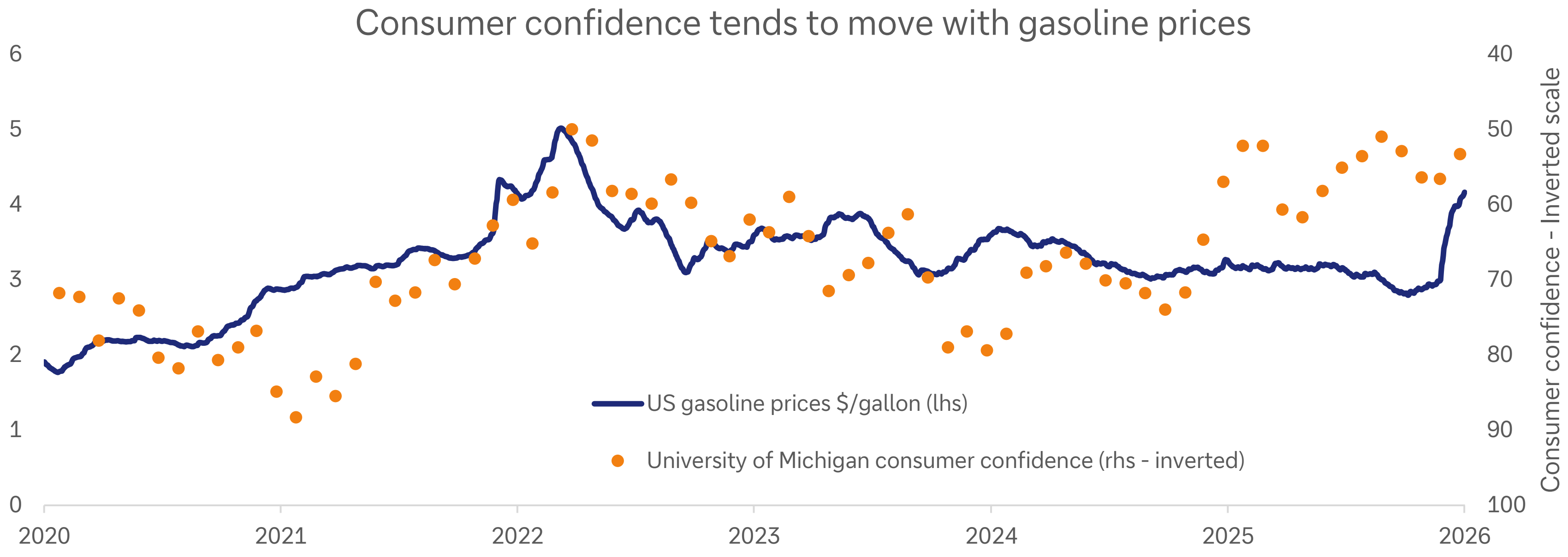




Source: Bloomberg Finance LP, Deutsche Bank.

Petrol prices

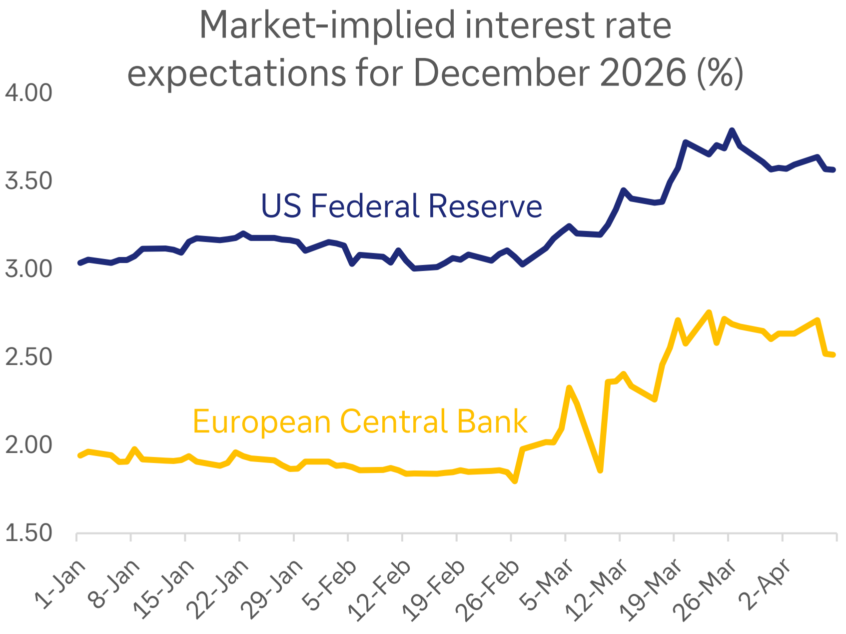
It is an inescapable reality that people care about the price they pay at the pump. That has a disproportionate impact on their overall confidence in the economy and feeds through to their voting intentions.

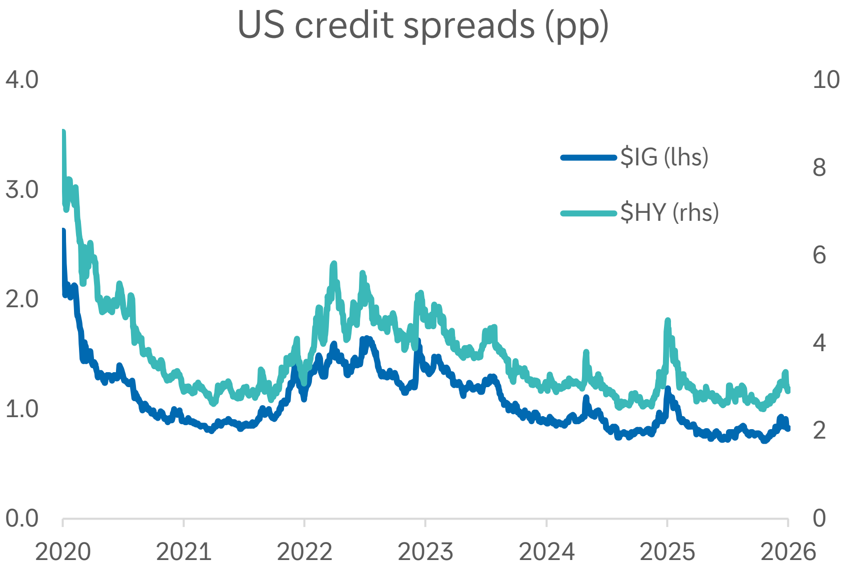


Source: Bloomberg Finance LP, Deutsche Bank.

Interest rate expectations have jumped

Credit spreads ticked upwards as the war commenced, but they weren't too concerned. Even as private credit and software worries continued to plague the market, and especially high-yield, investors seemed relatively sanguine about credit quality. The large amount of liquidity in the market certainly helps



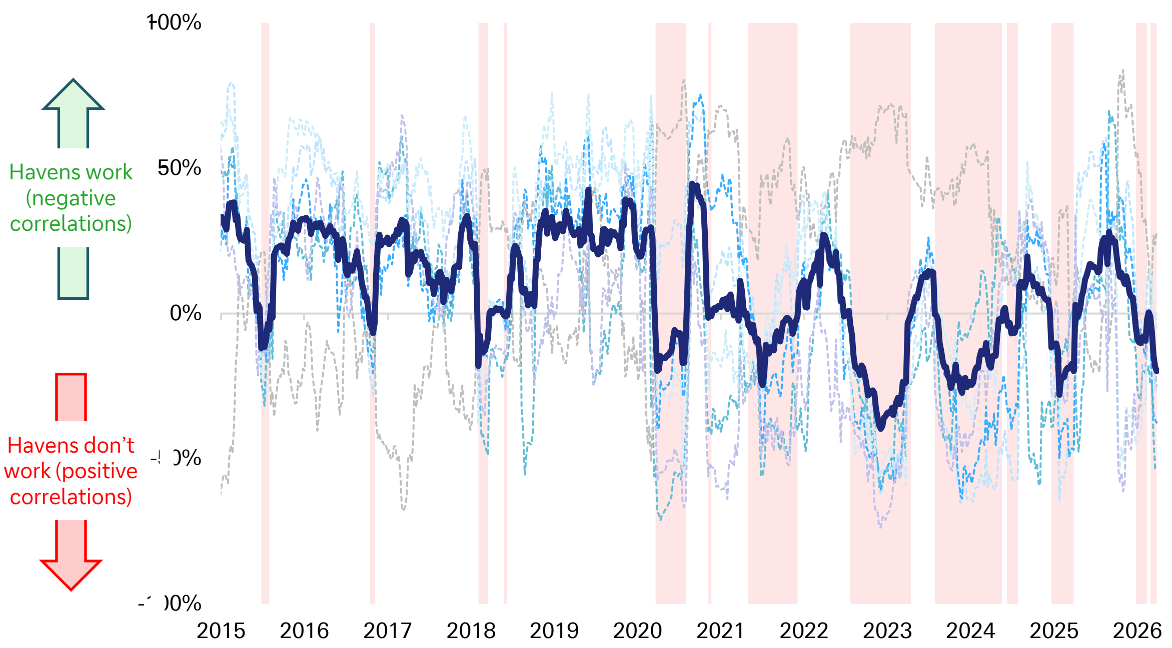


Source: Bloomberg Finance LP, Deutsche Bank.

Haven assets were little help during the Iran war – a 2020s trend

Our 'haven indicator' shows that traditional havens have been largely absent in the 2020s

Our haven indicator (20-wk rolling, avg across pairwise correlations with S&P 500)



Source: Bloomberg Finance LP, Deutsche Bank.

How our haven indicator works

We take six traditional haven assets:

1. Gold
2. US dollar
3. Japanese yen
4. Swiss franc
5. 10yr treasury
6. 10yr bund

We then analyse their individual correlation with the S&P 500 (our proxy for 'risk') over time. Finally, we take an average of the six correlations.

The red shading indicates periods where the average haven **did not work** (ie. positively correlated with the S&P 500). This includes:

1. Iran war in 2026
2. US tariff crisis in 2025
3. Peak fear about interest rate rises in 2022
4. Initial covid turmoil in 2020

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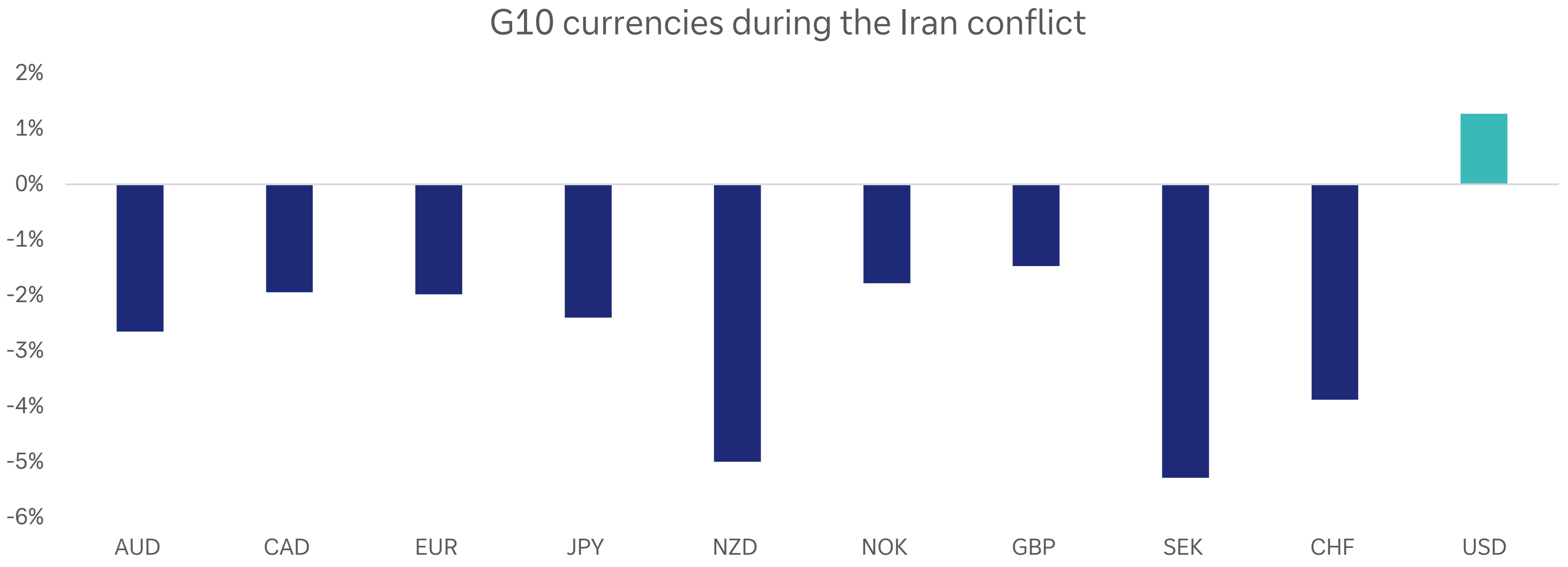
The US dollar was an exception

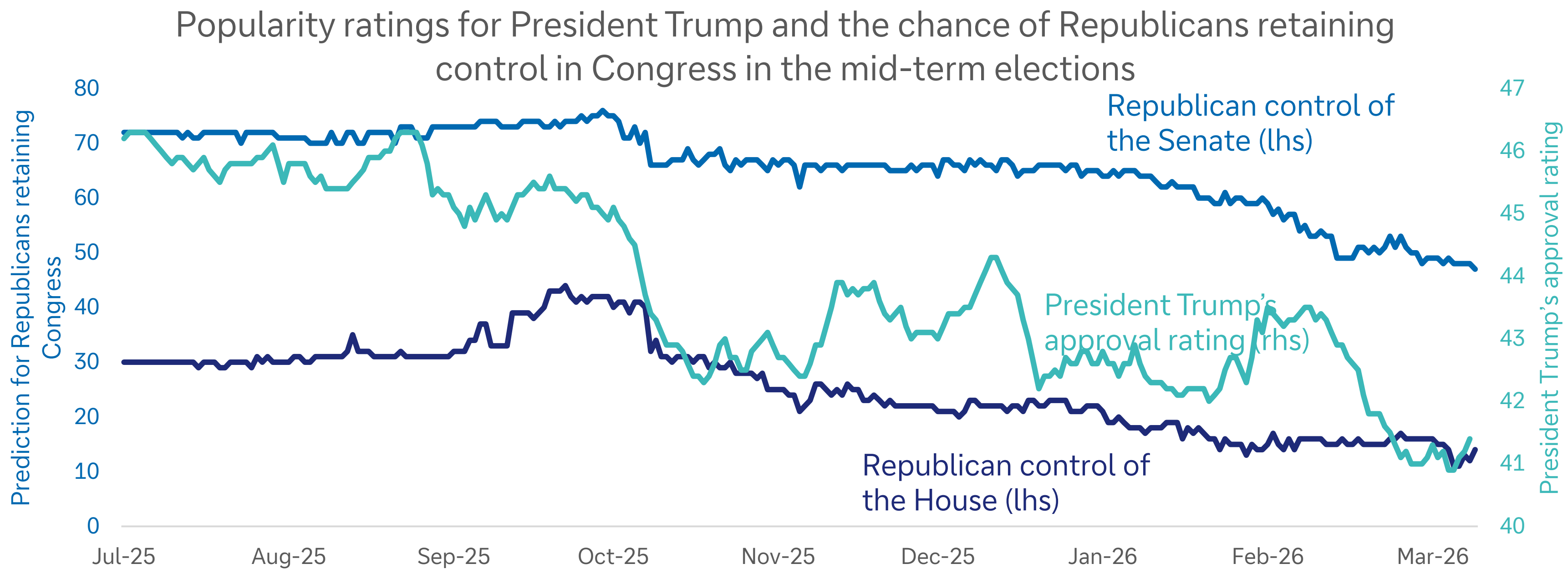
There was a sharp dichotomy between the risk-off period during the Iran war and the risk-off period during the tariff crisis in 2025. This time, investors were content with the safety of the US dollar. That was no doubt influenced by the fact that a kinetic conflict presented a different type of risk-off profile to the economic challenges last year.



Source: Bloomberg Finance LP, Deutsche Bank.

The political cost

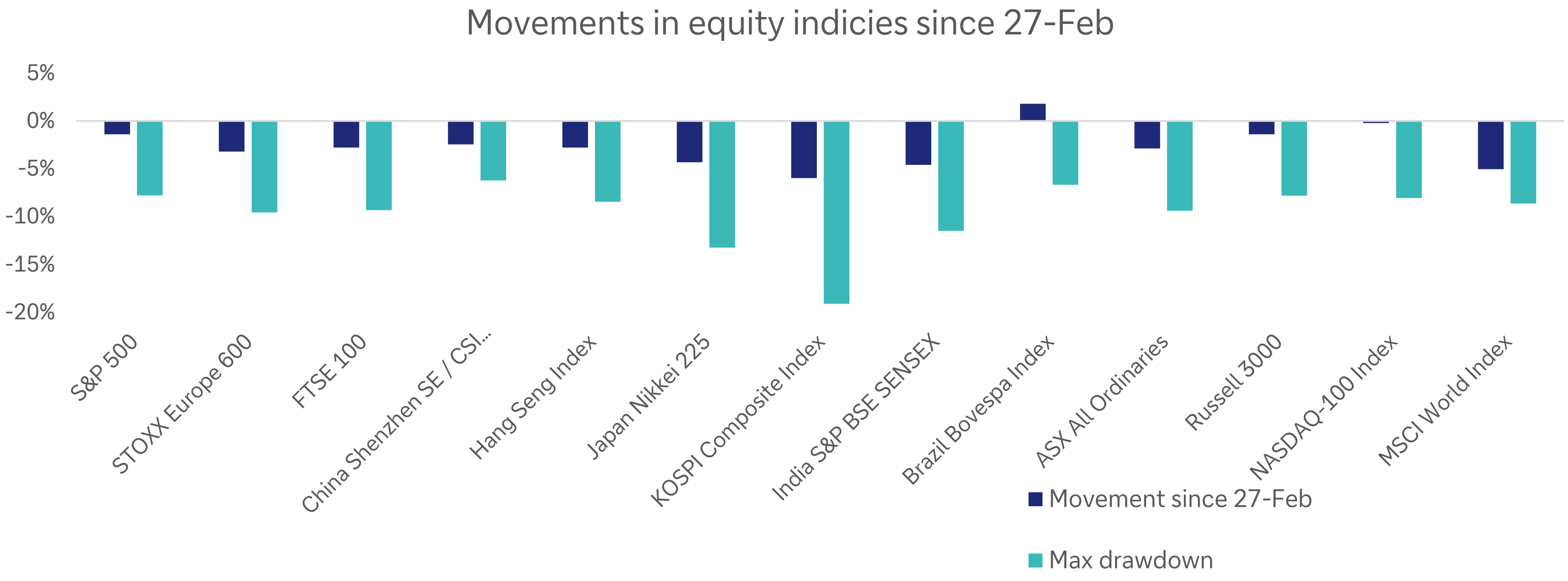
President Trump's approval rating, and the estimate of the Republicans holding the Senate, have fallen during the last five weeks.



Source: Bloomberg Finance LP, Deutsche Bank.

Global equities – most were relatively stubborn considering the severity of the closure of the Strait of Hormuz

Many investors have learned the lessons of several geopolitical shocks in the 2020s and were expecting a rapid rebound. It may be too early to be quite so optimistic.



Source: Bloomberg Finance LP, Deutsche Bank.

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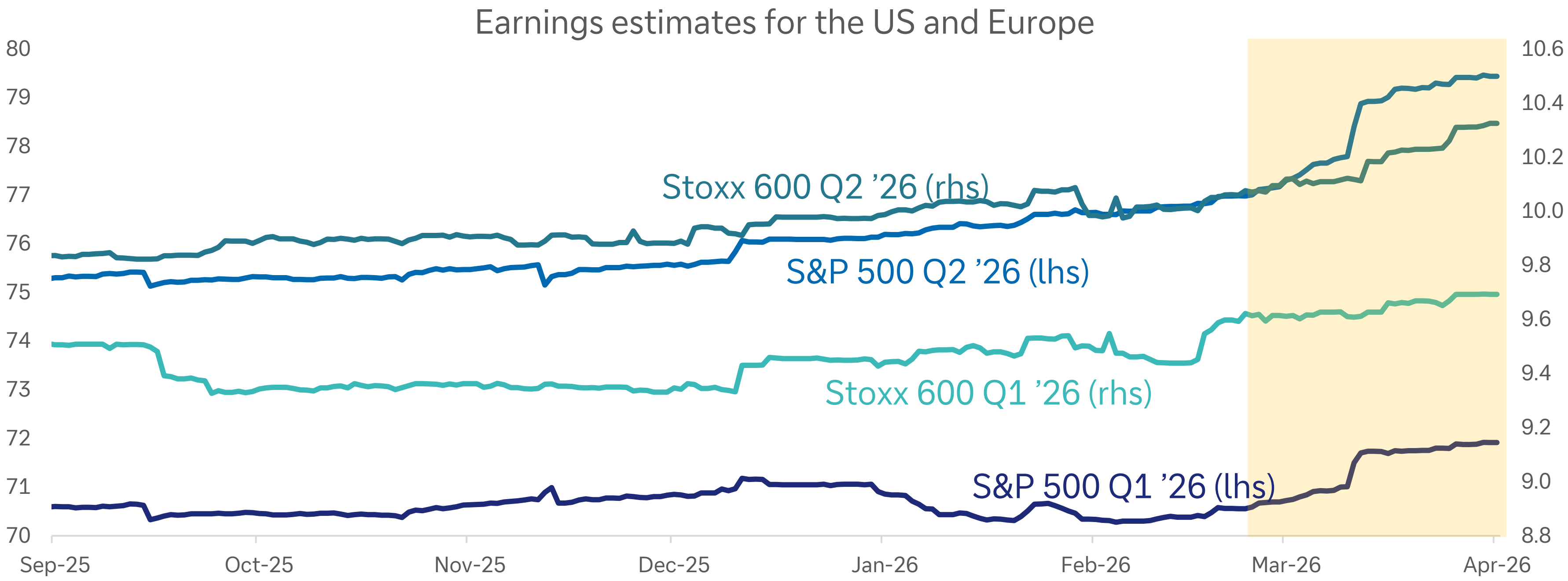
Earnings estimates have stayed strong despite the energy shock

It appears many investors have been reluctant to price in too much downside into corporate earnings until they actually see the financial impact

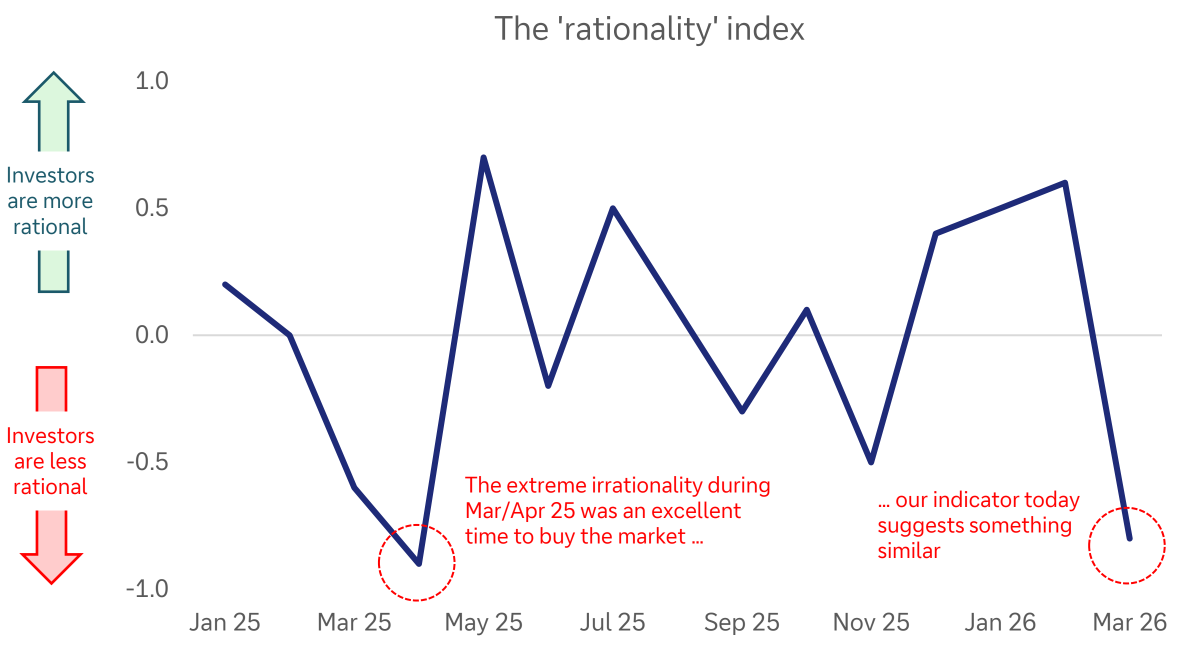


Source: Bloomberg Finance LP, Deutsche Bank.

Are investors rational? During the thick of the crisis, our indicator said “No”

Investors can be good at analysing markets during normal times and terrible during periods of uncertainty

Our dbLumina gauge indicates investors have been irrational during market drops and periods of geopolitical tensions



Source: dbLumina, Deutsche Bank.

What does it mean to be ‘rational’

March 2026 Extreme Irrationality (Panic/Fear):
The escalation of the Iran conflict led to a spike in oil prices and a broad market sell-off, characterized as a "stagflationary shock". This overwhelming fear, with investors largely selling off risk assets, indicates a highly emotional and irrational response, despite brief moments of optimism.

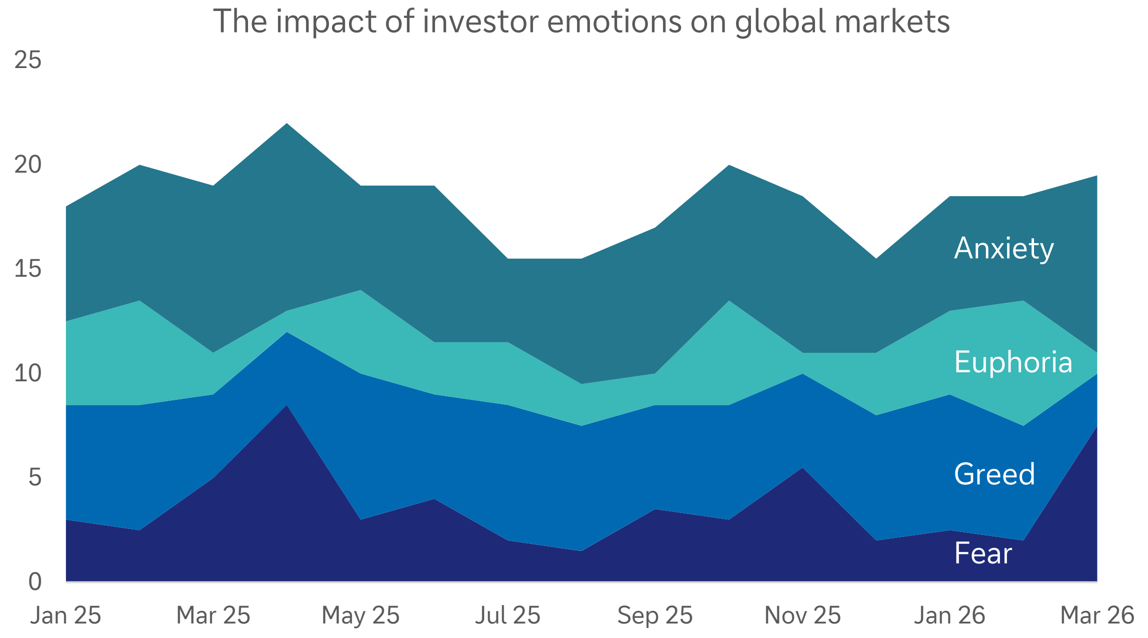
Fear Indicators:

Sharp, disproportionate reactions to threats, widespread sell-offs on announcements, panic, flight to safety beyond what might be fundamentally warranted, strong emotional language in descriptions (e.g., "roiled," "plunge," "blistering sell-off," "panic selling,“).

The impact of four market-moving emotions: fear, greed, euphoria and anxiety

dbLumina tells us that anxiety and greed can go hand-in-hand even as fear and euphoria wax and wane.

Anxiety and fear ruled Q1



Source: dbLumina, Deutsche Bank.

Status-quo bias is strong. That is creating more market volatility

Investors have been reluctant this decade to consider the upside and downside before the event. So, when something ‘bad’ happens, market drops can be more aggressive than expected.

Some examples:

1. Demography and fiscal largesse is slow moving and predictable
2. Rush to price in the downsides of tech disruption
3. Fear of considering sovereign risk
4. Fear of underperforming peers is greater than the desire to beat them
5. Unwillingness to price in the upside of iterative technology improvements
6. Narrow vision of what globalisation is and the fear that it has peaked

Appendix 1

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